

Structure Selection Theorem

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I. TOWARDS GLOBAL OPTIMALITY: CAN ANY 3D STAR BEAT BCC IN Λ_S ?

Problem Formulation

Let $S \subset \mathbb{S}_{q_0}^2 = \{k \in \mathbb{R}^3 : \|k\| = q_0\}$ be a finite star set satisfying inversion symmetry ($k \in S \Rightarrow -k \in S$). Define the weighted quartic resonance invariant

$$\Lambda(S) := \sum_{\substack{(k_1, k_2, k_3, k_4) \in S^4 \\ k_1 + k_2 + k_3 + k_4 = 0}} F(\alpha_{1234}), \quad F(\alpha) \geq 0.$$

For $F \equiv 1$, $\Lambda(S)$ reduces (up to convention) to the additive energy on the sphere:

$$E(S) := \#\{(k_1, k_2, k_3, k_4) \in S^4 : k_1 + k_2 = k_3 + k_4\}.$$

Unconstrained maximization of $\Lambda(S)$ is ill-posed: increasing $|S|$ typically increases $\Lambda(S)$. Therefore, meaningful optimality questions require a restriction of the admissible class or a normalization.

Three Meaningful Optimality Statements

(A) Fixed-size optimality. Fix $N = |S|$. Define

$$\Lambda_N^{\max} := \max\{\Lambda(S) : S \subset \mathbb{S}_{q_0}^2, S = -S, |S| = N\}.$$

A concrete target is $N = 12$:

$$\Lambda_{12}^{\max} \stackrel{?}{=} \Lambda(S_{\text{BCC}}),$$

where S_{BCC} denotes the 12-vector BCC star on the first shell.

(B) Bravais first-shell optimality. Restrict S to be the set of shortest nonzero reciprocal lattice vectors of a 3D Bravais lattice. Then $|S|$ is not arbitrary, and one may ask whether BCC maximizes Λ within this class:

$$\Lambda(S_{\text{BCC}}) \geq \Lambda(S) \quad \forall S \in \mathcal{S}_{\text{Bravais},1}.$$

(C) Normalized optimality. Define a normalized functional, e.g.

$$\widehat{\Lambda}(S) := \frac{\Lambda(S)}{|S|^3} \quad \text{or} \quad \widetilde{\Lambda}(S) := \frac{\Lambda(S)}{|S|^2},$$

and maximize $\widehat{\Lambda}$ or $\widetilde{\Lambda}$ over all antipodal stars.

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Reduction to Additive-Energy Optimization (Unweighted Case)

For $F \equiv 1$, the objective becomes the additive energy $E(S)$. By Cauchy–Schwarz,

$$E(S) = \sum_v r_S(v)^2, \quad r_S(v) := \#\{(k_1, k_2) \in S^2 : k_1 + k_2 = v\},$$

and

$$E(S) \geq \frac{(\sum_v r_S(v))^2}{|\{v : r_S(v) > 0\}|} = \frac{|S|^4}{|\text{Sum}(S)|}, \quad \text{Sum}(S) := \{k_1 + k_2 : k_i \in S\}.$$

Thus, maximizing $E(S)$ favors configurations with a small number of distinct pair-sums, i.e. strong pair-sum degeneracy.

Geometric Mechanism Behind BCC

The BCC star exhibits unusually high degeneracy of pair-sums and four-wave closures under the constraint that all vectors lie on a single radius and the set is antipodal. This produces a large contribution to $r_S(v)^2$ and hence to $E(S)$ (and $\Lambda(S)$ for $F \geq 0$).

Proof Strategy for (A) and (B)

Step 1 (Classification of admissible $|S|$). For Bravais first-shell stars, enumerate possible shell multiplicities (e.g. $|S| \in \{6, 8, 12, \dots\}$) and reduce comparison to finitely many symmetry types.

Step 2 (Upper bounds via sumset size). Derive lower bounds on $|\text{Sum}(S)|$ from spherical geometry constraints, implying upper bounds on $E(S)$ and hence on $\Lambda(S)$ for $F \geq 0$.

Step 3 (Sharpness). Show that BCC saturates the bound (or is the unique maximizer) within the admissible class. For $N = 12$, this becomes a finite extremal problem on antipodal spherical codes.

Step 4 (Weighted kernels). For general $F(\alpha) \geq 0$, use

$$\Lambda(S) \leq \left(\sup_{\alpha} F(\alpha)\right) E(S), \quad \Lambda(S) \geq \left(\inf_{\alpha} F(\alpha)\right) E(S),$$

or more refined angle-class decompositions, to transfer the extremal statement from $E(S)$ to $\Lambda(S)$.

Implication

If either (A) with $N = 12$ or (B) over Bravais first-shell stars is proven, then BCC is not only robust among cubic candidates but extremal within a substantially larger admissible class. This would upgrade the result from a hierarchy theorem to an optimality principle for 3D shell-degenerate selection.

II. SELECTION FUNCTIONAL FOR BLOCK-COPOLYMER MICROPHASES

We consider a standard Landau–Brazovskii description of a diblock-copolymer melt near the order–disorder transition (ODT), in which the dominant density mode softens on a finite wavenumber shell $|q| \approx q_0$. After integrating out fast modes, the coarse-grained free energy for the relevant scalar order parameter Ψ takes the form

$$\mathcal{F}[\Psi] = \frac{1}{2} \int_q \Gamma_2(q) |\Psi_q|^2 + \frac{1}{4!} \int_{q_1 \dots q_4} (2\pi)^d \delta\left(\sum_{i=1}^4 q_i\right) \Gamma_4(q_1, q_2, q_3, q_4) \prod_{i=1}^4 \Psi_{q_i} + \frac{\gamma}{6} \int d^d x \Psi^6 + \dots, \quad \gamma > 0.$$

We restrict to the dominant shell $|q_i| = q_0$ and assume isotropy so that the quartic vertex depends only on the internal angles:

$$\Gamma_4(q_1, q_2, q_3, q_4) \Big|_{|q_i|=q_0} = F(\alpha_{1234}), \quad F(\alpha) \geq 0,$$

where α_{1234} denotes the set of angles between the participating shell vectors.

III. STARS FOR LAM, HEX, AND BCC

We approximate the ordered phases by equal-amplitude stars on the shell:

$$\Psi(x) = \sum_{k \in S} A e^{ik \cdot x}, \quad |k| = q_0, \quad A \in \mathbb{R}.$$

The candidate stars for the classical microphases are:

Lamellae (LAM)

$$S_{\text{LAM}} = \{\pm k_1\}, \quad |S_{\text{LAM}}| = 2.$$

Hexagonal cylinders (HEX)

A 2D hexagonal star in a plane (six directions separated by 60°):

$$S_{\text{HEX}} = \{\pm k_1, \pm k_2, \pm k_3\}, \quad k_1 + k_2 + k_3 = 0, \quad |S_{\text{HEX}}| = 6.$$

BCC spheres (BCC)

The 12-vector cuboctahedral star:

$$S_{\text{BCC}} = \frac{q_0}{\sqrt{2}} \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}, \quad |S_{\text{BCC}}| = 12.$$

IV. DEFINITION OF THE SELECTION FUNCTIONAL Λ_S

Under the equal-amplitude ansatz, the quartic contribution reduces to

$$F_{4,S} = \frac{A^4}{4} \Lambda_S,$$

where the structural functional is the weighted sum over resonant quartets

$$\Lambda_S := \sum_{\substack{(k_1, k_2, k_3, k_4) \in S^4 \\ k_1 + k_2 + k_3 + k_4 = 0}} F(\alpha_{1234}).$$

For a local vertex $F(\alpha) \equiv \text{const}$, Λ_S reduces to a pure resonance count (additive energy on the star).

V. ANGLE-CLASS REDUCTION

For isotropic kernels, only finitely many angle-classes contribute for each star. We therefore decompose

$$\Lambda_S = \sum_{c \in \mathcal{C}(S)} M_c(S) F_c,$$

where

- $\mathcal{C}(S)$ is the finite set of angle-classes realized by resonant quartets on S ,
- $M_c(S) \in \mathbb{N}$ are purely combinatorial multiplicities,
- F_c are the kernel values for the corresponding quartet geometry (angles).

This is the minimal reduction needed to connect a microscopic polymer vertex to phase selection.

VI. EXPLICIT MINIMAL PARAMETRIZATION

A practical and widely used reduction is to express the quartic kernel by a few representative configurations that occur in LAM/HEX/BCC resonant quartets. We introduce:

$$F_{\parallel} := F(\text{collinear pairing}), \quad F_{60} := F(\text{a } 60^\circ \text{ pairing geometry}), \quad F_{90} := F(\text{a } 90^\circ \text{ pairing geometry}),$$

and additional classes if needed (e.g. tetrahedral-like quartets for 3D stars). Then each phase has

$$\Lambda_{\text{LAM}} = M_{\parallel}(\text{LAM}) F_{\parallel},$$

$$\Lambda_{\text{HEX}} = M_{\parallel}(\text{HEX}) F_{\parallel} + M_{60}(\text{HEX}) F_{60},$$

$$\Lambda_{\text{BCC}} = M_{\parallel}(\text{BCC}) F_{\parallel} + M_{60}(\text{BCC}) F_{60} + M_{90}(\text{BCC}) F_{90} + \dots$$

All microscopic information enters only through a small set of numbers F_c , while all structural information enters only through the integers $M_c(S)$.

VII. SELECTION CRITERION AND PHASE BOUNDARIES

Within the quartic-stabilized Brazovskii regime, structure selection is controlled by the effective quartic coefficient, equivalently by Λ_S (up to universal prefactors). We write the reduced Landau free energy for each candidate phase as

$$f_S(A) = \frac{r}{2} A^2 + \frac{\Lambda_S}{4} A^4 + \frac{\gamma}{6} A^6, \quad \gamma > 0.$$

Assuming sextic neutrality, the selected phase is the one minimizing f_S , which is determined by the ordering of Λ_S :

$$S = \arg \min_{S \in \{\text{LAM}, \text{HEX}, \text{BCC}\}} \Lambda_S.$$

Phase boundaries are given by equality of the corresponding quartic functionals:

$$\Lambda_{\text{LAM}} = \Lambda_{\text{HEX}}, \quad \Lambda_{\text{HEX}} = \Lambda_{\text{BCC}}, \quad \Lambda_{\text{LAM}} = \Lambda_{\text{BCC}}.$$

In the reduced parametrization these become linear conditions in the kernel values F_c , e.g.

$$(M_{\parallel}(\text{LAM}) - M_{\parallel}(\text{HEX})) F_{\parallel} = M_{60}(\text{HEX}) F_{60},$$

and similarly for the other boundaries.

VIII. PRACTICAL PROGRAM FOR A QUANTITATIVE BCP PHASE DIAGRAM

To obtain a quantitative phase diagram in terms of polymer parameters (e.g. volume fraction f and χN), one computes the isotropic quartic vertex on the shell, extracts the finite set of kernel values $\{F_c(f, \chi N)\}$, and evaluates

$$\Lambda_S(f, \chi N) = \sum_c M_c(S) F_c(f, \chi N).$$

The phase at given parameters is predicted by the minimal Λ_S , and the phase boundaries follow from pairwise equalities of Λ_S .

This reduction makes explicit which part of the phase diagram is universal (the integers $M_c(S)$) and which part is material-specific (the kernel values F_c).

IX. ANGLE-CLASS MULTIPLICITIES FOR LAM/HEX/BCC AND LINEAR PHASE BOUNDARIES

We work on a single instability shell $|k| = q_0$ and use equal-amplitude stars

$$\Psi(x) = \sum_{k \in S} A e^{ik \cdot x}, \quad A \in \mathbb{R}, \quad S = -S.$$

Assume an isotropic quartic vertex restricted to the shell depends on the angle between two incoming shell vectors,

$$\Gamma_4(k_1, k_2, k_3, k_4) \Big|_{|k_i|=q_0} = F(\theta_{12}), \quad \theta_{12} := \angle(k_1, k_2), \quad F(\theta) \geq 0,$$

for resonant quartets $k_1 + k_2 + k_3 + k_4 = 0$.

Define the (ordered) resonance functional

$$\Lambda_S := \sum_{\substack{(k_1, k_2, k_3, k_4) \in S^4 \\ k_1 + k_2 + k_3 + k_4 = 0}} F(\theta_{12}).$$

Then the quartic contribution is

$$F_{4,S} = \frac{A^4}{4} \Lambda_S.$$

Amplitude Normalization Across Different Stars

Because different phases use different numbers of modes $N_S := |S|$, it is convenient to compare phases at fixed quadratic norm

$$\sum_{k \in S} |A_k|^2 = N_S A^2 \equiv \psi^2.$$

Equivalently,

$$A = \frac{\psi}{\sqrt{N_S}}.$$

Then

$$F_{4,S} = \frac{\psi^4}{4} \bar{\Lambda}_S, \quad \bar{\Lambda}_S := \frac{\Lambda_S}{N_S^2}.$$

Thus selection among phases is controlled by the normalized functional $\bar{\Lambda}_S$.

Stars

Lamellae (LAM):

$$S_{\text{LAM}} = \{\pm k_1\}, \quad N_{\text{LAM}} = 2.$$

Hexagonal cylinders (HEX) (planar hexagonal star):

$$S_{\text{HEX}} = \{\pm k_1, \pm k_2, \pm k_3\}, \quad k_1 + k_2 + k_3 = 0, \quad N_{\text{HEX}} = 6.$$

BCC spheres (BCC) (cuboctahedral star):

$$S_{\text{BCC}} = \frac{q_0}{\sqrt{2}} \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}, \quad N_{\text{BCC}} = 12.$$

Finite Angle Classes

For these stars, the only angles that occur between star vectors are

$$\theta \in \{0^\circ, 60^\circ, 90^\circ, 120^\circ, 180^\circ\}.$$

Define

$$F_0 := F(0^\circ), \quad F_{60} := F(60^\circ), \quad F_{90} := F(90^\circ), \quad F_{120} := F(120^\circ), \quad F_{180} := F(180^\circ).$$

Then

$$\Lambda_S = \sum_{\theta} M_{\theta}(S) F_{\theta}, \quad \bar{\Lambda}_S = \sum_{\theta} \bar{M}_{\theta}(S) F_{\theta}, \quad \bar{M}_{\theta}(S) := \frac{M_{\theta}(S)}{N_S^2}.$$

Explicit Multiplicities (Ordered Resonant Quartets)

The ordered resonance counts grouped by the angle θ_{12} are:

a. LAM ($N = 2$).

$$\Lambda_{\text{LAM}} = 2F_0 + 4F_{180}, \quad \bar{\Lambda}_{\text{LAM}} = \frac{1}{2}F_0 + 1 \cdot F_{180}.$$

b. HEX ($N = 6$).

$$\Lambda_{\text{HEX}} = 6F_0 + 24F_{60} + 24F_{120} + 36F_{180},$$

$$\bar{\Lambda}_{\text{HEX}} = \frac{1}{6}F_0 + \frac{2}{3}F_{60} + \frac{2}{3}F_{120} + 1 \cdot F_{180}.$$

c. BCC ($N = 12$).

$$\Lambda_{\text{BCC}} = 12F_0 + 96F_{60} + 96F_{90} + 192F_{120} + 144F_{180},$$

$$\bar{\Lambda}_{\text{BCC}} = \frac{1}{12}F_0 + \frac{2}{3}F_{60} + \frac{2}{3}F_{90} + \frac{4}{3}F_{120} + 1 \cdot F_{180}.$$

Linear Phase-Boundary Conditions

Within the quartic-stabilized Landau–Brazovskii description, the selected phase among $\{\text{LAM}, \text{HEX}, \text{BCC}\}$ is predicted by

$$S = \arg \min_S \bar{\Lambda}_S,$$

and phase boundaries are determined by equalities of $\bar{\Lambda}_S$.

d. LAM–HEX boundary.

$$\bar{\Lambda}_{\text{LAM}} = \bar{\Lambda}_{\text{HEX}} \iff \left(\frac{1}{2} - \frac{1}{6}\right)F_0 = \frac{2}{3}F_{60} + \frac{2}{3}F_{120}.$$

Equivalently,

$$\frac{1}{3}F_0 = \frac{2}{3}(F_{60} + F_{120}) \iff F_0 = 2(F_{60} + F_{120}).$$

e. HEX–BCC boundary.

$$\bar{\Lambda}_{\text{HEX}} = \bar{\Lambda}_{\text{BCC}} \iff \left(\frac{1}{6} - \frac{1}{12}\right)F_0 + \frac{2}{3}F_{120} = \frac{2}{3}F_{90} + \frac{4}{3}F_{120}.$$

Equivalently,

$$\frac{1}{12}F_0 = \frac{2}{3}F_{90} + \frac{2}{3}F_{120} \iff F_0 = 8(F_{90} + F_{120}).$$

f. LAM–BCC boundary.

$$\bar{\Lambda}_{\text{LAM}} = \bar{\Lambda}_{\text{BCC}} \iff \left(\frac{1}{2} - \frac{1}{12}\right)F_0 = \frac{2}{3}F_{60} + \frac{2}{3}F_{90} + \frac{4}{3}F_{120}.$$

Equivalently,

$$\frac{5}{12}F_0 = \frac{2}{3}(F_{60} + F_{90} + 2F_{120}) \iff 5F_0 = 8(F_{60} + F_{90} + 2F_{120}).$$

Interpretation

All material dependence is contained in the finite set of kernel values $\{F_0, F_{60}, F_{90}, F_{120}, F_{180}\}$, which may be computed from the microscopic block-copolymer vertex. All structural dependence is contained in the universal rational coefficients in $\bar{\Lambda}_S$ above.

In particular, the LAM/HEX/BCC selection reduces to comparing three linear forms in $\{F_\theta\}$.

X. MICROSCOPIC ORIGIN OF THE ANGLE-DEPENDENT QUARTIC KERNEL IN DIBLOCK COPOLYMERS

We outline a standard Edwards–Leibler route that yields the Landau–Brazovskii functional for diblock melts. Consider an incompressible AB diblock melt of Gaussian chains (segment length b , polymerization index N), with volume fraction f of A segments and $(1 - f)$ of B segments. Let $\phi_A(x), \phi_B(x)$ be local volume fractions with incompressibility $\phi_A + \phi_B = 1$. Introduce the composition order parameter

$$\psi(x) := \phi_A(x) - \phi_B(x).$$

After a Hubbard–Stratonovich transformation and integrating out polymer degrees of freedom, one obtains an effective action for ψ whose vertices are connected cumulants of the single-chain composition density.

Single-Chain Composition Density and Cumulants

Parameterize a Gaussian chain by contour $s \in [0, 1]$ and position $r(s)$. Define the single-chain microscopic composition density in Fourier space:

$$\hat{\psi}_q := \int_0^1 ds \sigma(s) e^{iq \cdot r(s)}, \quad \sigma(s) = \begin{cases} +1, & s \in [0, f], \\ -1, & s \in (f, 1]. \end{cases}$$

The n -point vertex functions on the disordered reference state are proportional to the connected cumulants

$$\Gamma_n(q_1, \dots, q_n) \propto \left\langle \hat{\psi}_{q_1} \cdots \hat{\psi}_{q_n} \right\rangle_c \quad \text{with} \quad \sum_{i=1}^n q_i = 0,$$

where $\langle \cdot \rangle_c$ denotes the connected part computed in the ideal-chain measure.

XI. EXPLICIT GAUSSIAN-CHAIN FORMULA FOR THE QUARTIC VERTEX Γ_4

For Gaussian chains, multi-point averages reduce to contour integrals with an exponential kernel. A convenient exact representation is obtained by writing (before taking the connected part)

$$\left\langle \prod_{a=1}^4 e^{iq_a \cdot r(s_a)} \right\rangle = \exp \left(-\frac{b^2 N}{6} \sum_{a < b} q_a \cdot q_b |s_a - s_b| \right).$$

Hence the (unreduced) four-point moment is

$$M_4(q_1, q_2, q_3, q_4; f) := \int_0^1 ds_1 \cdots \int_0^1 ds_4 \left(\prod_{a=1}^4 \sigma(s_a) \right) \exp \left(-\frac{b^2 N}{6} \sum_{a < b} q_a \cdot q_b |s_a - s_b| \right).$$

The connected cumulant is obtained by subtracting Wick-type products of lower moments:

$$C_4(q_1, q_2, q_3, q_4; f) = M_4(q_1, q_2, q_3, q_4; f) - \sum_{\text{pairings}} M_2(q_i, q_j; f) M_2(q_k, q_\ell; f),$$

where the sum runs over the three distinct pairings of $\{1, 2, 3, 4\}$ and

$$M_2(q, -q; f) = \int_0^1 ds \int_0^1 ds' \sigma(s) \sigma(s') e^{-x|s-s'|}, \quad x := \frac{b^2 N}{6} q^2.$$

In the Landau expansion, Γ_4 is proportional to C_4 (up to a conventional prefactor depending on density normalization).

XII. SHELL RESTRICTION AND ANGLE-DEPENDENT KERNEL $F(\theta)$

Near the ODT, the quadratic vertex $\Gamma_2(q)$ develops a minimum at $|q| = q_0$ (finite- q instability). Restricting all external legs to the shell,

$$|q_1| = |q_2| = |q_3| = |q_4| = q_0, \quad q_1 + q_2 + q_3 + q_4 = 0,$$

isotropy implies that Γ_4 depends on a finite set of scalar invariants. For the resonance quartets relevant to the star-amplitude theory (pair-sum closure), it is sufficient to evaluate the special quartet geometry

$$(q_1, q_2, q_3, q_4) = (k, -k, p, -p), \quad |k| = |p| = q_0, \quad k \cdot p = q_0^2 \cos \theta.$$

We then define the angle-dependent quartic kernel by

$$F(\theta; f) := \Gamma_4(k, -k, p, -p) \Big|_{|k|=|p|=q_0}.$$

Equivalently (up to prefactors),

$$F(\theta; f) \propto C_4(k, -k, p, -p; f), \quad \cos \theta = \frac{k \cdot p}{q_0^2}.$$

Thus $F(\theta; f)$ is fully determined by the explicit contour integrals above.

XIII. PRACTICAL EVALUATION AT DISCRETE ANGLES FOR LAM/HEX/BCC

For the LAM/HEX/BCC stars, only the discrete angles

$$\theta \in \{0^\circ, 60^\circ, 90^\circ, 120^\circ, 180^\circ\}$$

occur between star vectors. Define

$$F_0(f) := F(0^\circ; f), \quad F_{60}(f) := F(60^\circ; f), \quad F_{90}(f) := F(90^\circ; f), \quad F_{120}(f) := F(120^\circ; f), \quad F_{180}(f) := F(180^\circ; f).$$

Each of these is obtained by inserting the corresponding $\cos \theta$ into the quartet $(k, -k, p, -p)$ and evaluating C_4 .

Algorithmic Form (Contour Ordering Reduction)

The integral for M_4 can be reduced to a finite sum over contour orderings. Let π be a permutation of $\{1, 2, 3, 4\}$ and restrict to the simplex

$$0 \leq s_{\pi(1)} \leq s_{\pi(2)} \leq s_{\pi(3)} \leq s_{\pi(4)} \leq 1.$$

On each simplex, $|s_a - s_b|$ becomes linear, and

$$M_4 = \sum_{\pi \in S_4} \int_{0 \leq s_{\pi(1)} \leq \dots \leq s_{\pi(4)} \leq 1} ds_1 \cdots ds_4 \left(\prod_{a=1}^4 \sigma(s_a) \right) \exp \left(-\frac{b^2 N}{6} \sum_{a < b} q_a \cdot q_b (s_{\pi(b)} - s_{\pi(a)}) \right).$$

Since $\sigma(s)$ is piecewise constant with a single breakpoint at $s = f$, each simplex integral further splits into finitely many regions depending on whether each s_a lies in $[0, f]$ or $(f, 1]$. Therefore $F(\theta; f)$ is computable exactly as a finite sum of elementary exponentials in $x_0 := \frac{b^2 N}{6} q_0^2$.

XIV. CLOSING THE LAM/HEX/BCC PHASE BOUNDARIES IN TERMS OF $F_\theta(f)$

Using the normalized quartic functionals derived previously,

$$\bar{\Lambda}_{\text{LAM}} = \frac{1}{2} F_0(f) + F_{180}(f),$$

$$\bar{\Lambda}_{\text{HEX}} = \frac{1}{6} F_0(f) + \frac{2}{3} F_{60}(f) + \frac{2}{3} F_{120}(f) + F_{180}(f),$$

$$\bar{\Lambda}_{\text{BCC}} = \frac{1}{12} F_0(f) + \frac{2}{3} F_{60}(f) + \frac{2}{3} F_{90}(f) + \frac{4}{3} F_{120}(f) + F_{180}(f),$$

the mean-field selection among LAM/HEX/BCC reduces to

$$S(f) = \arg \min_{S \in \{\text{LAM}, \text{HEX}, \text{BCC}\}} \bar{\Lambda}_S(f),$$

and the phase boundaries are the linear equalities:

LAM–HEX

$$\bar{\Lambda}_{\text{LAM}} = \bar{\Lambda}_{\text{HEX}} \iff F_0(f) = 2(F_{60}(f) + F_{120}(f)).$$

HEX–BCC

$$\bar{\Lambda}_{\text{HEX}} = \bar{\Lambda}_{\text{BCC}} \iff F_0(f) = 8(F_{90}(f) + F_{120}(f)).$$

LAM–BCC

$$\bar{\Lambda}_{\text{LAM}} = \bar{\Lambda}_{\text{BCC}} \iff 5F_0(f) = 8(F_{60}(f) + F_{90}(f) + 2F_{120}(f)).$$

XV. WHERE χN ENTERS AND WHY THE ANGLE KERNEL IS MOSTLY A FUNCTION OF f

At the level of the Landau expansion about the disordered melt, the angle dependence of the quartic vertex on the instability shell is controlled primarily by the single-chain statistics, hence by f and the dimensionless shell variable $x_0 = \frac{b^2 N}{6} q_0^2$. The location q_0 is determined by the quadratic vertex $\Gamma_2(q)$ and thus depends on χN through the proximity to the ODT. Operationally, one computes $q_0(f, \chi N)$ from $\Gamma'_2(q_0) = 0$ and evaluates $F_\theta(f)$ at $x_0(f, \chi N)$.

This closes a quantitative route from microscopic parameters $(f, \chi N)$ to the LAM/HEX/BCC phase boundaries above.

XVI. QUADRATIC VERTEX IN RPA AND THE DETERMINATION OF $q_0(f, \chi N)$

We summarize the standard random-phase approximation (RPA) for an incompressible AB diblock melt and extract the finite- q instability scale q_0 .

A. Dimensionless Variables

Let $R_g^2 = b^2 N/6$ and define

$$x := q^2 R_g^2.$$

We use the contour parameter $s \in [0, 1]$ with A-block on $[0, f]$ and B-block on $(f, 1]$.

B. Single-Chain Block Correlators

For an ideal Gaussian chain,

$$\langle e^{iq \cdot (r(s) - r(s'))} \rangle = \exp(-x |s - s'|).$$

Define the block-block correlators on the shell variable x :

$$\Omega_{AA}(x; f) := \int_0^f ds \int_0^f ds' e^{-x|s-s'|} = \frac{2}{x^2} (fx - 1 + e^{-fx}),$$

$$\Omega_{BB}(x; f) := \int_f^1 ds \int_f^1 ds' e^{-x|s-s'|} = \frac{2}{x^2} ((1-f)x - 1 + e^{-(1-f)x}),$$

$$\Omega_{AB}(x; f) := \int_0^f ds \int_f^1 ds' e^{-x(s'-s)} = \frac{(1 - e^{-fx})(1 - e^{-(1-f)x})}{x^2}.$$

For the composition mode $\psi = \phi_A - \phi_B$, the single-chain composition correlator is

$$g(x; f) := \Omega_{AA}(x; f) + \Omega_{BB}(x; f) - 2\Omega_{AB}(x; f).$$

C. RPA Quadratic Vertex

Up to an overall (convention-dependent) prefactor, the quadratic vertex for the composition mode takes the form

$$\Gamma_2(q) \equiv \Gamma_2(x; f, \chi N) = g(x; f)^{-1} - 2\chi N.$$

The disordered state is stable when $\Gamma_2(x) > 0$ for all x .

D. Finite- q Instability and the Selection of q_0

A Brazovskii/Leibler-type instability occurs when $\Gamma_2(x)$ first touches zero at a nonzero $x = x_0$:

$$\Gamma_2(x_0; f, \chi N) = 0.$$

Because $\Gamma_2(x) = g(x)^{-1} - 2\chi N$, the most unstable wavenumber maximizes $g(x; f)$:

$$x_0(f) = \arg \max_{x>0} g(x; f), \quad \Longleftrightarrow \quad \frac{d}{dx} g(x_0; f) = 0, \quad \frac{d^2}{dx^2} g(x_0; f) < 0.$$

The corresponding critical value is

$$(\chi N)_c(f) = \frac{1}{2g(x_0(f); f)}.$$

Thus the instability wavenumber is

$$q_0(f) = \frac{\sqrt{x_0(f)}}{R_g}, \quad x_0(f) = q_0(f)^2 R_g^2.$$

E. Using q_0 in the Quartic Kernel and Phase Selection

The angle-dependent quartic kernel for resonant quartets is evaluated on the instability shell:

$$F(\theta; f, \chi N) \equiv \Gamma_4(k, -k, p, -p) \Big|_{|k|=|p|=q_0(f, \chi N), \ k \cdot p = q_0^2 \cos \theta}.$$

Operationally, one proceeds as follows:

1. Compute $x_0(f)$ by solving $g'(x_0; f) = 0$.
2. Set $q_0 = \sqrt{x_0}/R_g$ and evaluate the quartic kernel at x_0 :

$$F_\theta(f, \chi N) := F(\theta; f, \chi N) \Big|_{x=x_0(f)}.$$

3. Insert $\{F_0, F_{60}, F_{90}, F_{120}, F_{180}\}$ into the linear phase-boundary conditions for $\bar{\Lambda}_{\text{LAM}}, \bar{\Lambda}_{\text{HEX}}, \bar{\Lambda}_{\text{BCC}}$.

This closes a quantitative route from $(f, \chi N)$ to the LAM/HEX/BCC phase selection within the star-amplitude theory.

XVII. QUARTIC VERTEX ON THE INSTABILITY SHELL: EXACT REDUCTION TO FINITE SUMS

We compute the angle-dependent quartic kernel for an incompressible AB diblock melt in the weak-segregation (Leibler/RPA) framework. The goal is an explicit computable formula for

$$F(\theta; f, \chi N) \equiv \Gamma_4(k, -k, p, -p) \Big|_{|k|=|p|=q_0, \ k \cdot p = q_0^2 \cos \theta},$$

evaluated at the instability shell $x_0 = q_0^2 R_g^2$.

A. Gaussian-Chain Building Block

Let $R_g^2 = b^2 N/6$ and $x = q^2 R_g^2$. For a Gaussian chain with contour $s \in [0, 1]$,

$$\left\langle e^{iq \cdot (r(s) - r(s'))} \right\rangle = e^{-x|s-s'|}.$$

Define the single-chain composition weight

$$\sigma(s) = \begin{cases} +1, & s \in [0, f], \\ -1, & s \in (f, 1]. \end{cases}$$

The Fourier composition density of one chain is

$$\hat{\psi}_q = \int_0^1 ds \sigma(s) e^{iq \cdot r(s)}.$$

B. Quartic Vertex as a Connected Cumulant

Up to a conventional prefactor (absorbed into F), the quartic vertex equals the connected 4-point cumulant

$$\Gamma_4(q_1, q_2, q_3, q_4) \propto C_4(q_1, q_2, q_3, q_4; f) = \langle \hat{\psi}_{q_1} \hat{\psi}_{q_2} \hat{\psi}_{q_3} \hat{\psi}_{q_4} \rangle_c, \quad \sum_{a=1}^4 q_a = 0.$$

Write the (unreduced) 4-point moment

$$M_4(q_1, q_2, q_3, q_4; f) = \int_0^1 ds_1 \cdots \int_0^1 ds_4 \left(\prod_{a=1}^4 \sigma(s_a) \right) \exp \left(- \sum_{a < b} \lambda_{ab} |s_a - s_b| \right),$$

with

$$\lambda_{ab} := \frac{R_g^2}{1} (q_a \cdot q_b) = (q_a \cdot q_b) R_g^2.$$

The connected cumulant is

$$C_4 = M_4 - \sum_{\text{pairings}} M_2(q_i, q_j; f) M_2(q_k, q_\ell; f),$$

where (for $\sum q = 0$)

$$M_2(q, -q; f) = \int_0^1 ds \int_0^1 ds' \sigma(s) \sigma(s') e^{-x|s-s'|}.$$

XVIII. SPECIAL QUARTET GEOMETRY AND ANGLE DEPENDENCE

We specialize to the resonant quartet

$$(q_1, q_2, q_3, q_4) = (k, -k, p, -p), \quad |k| = |p| = q_0, \quad k \cdot p = q_0^2 c, \quad c = \cos \theta,$$

and evaluate on the instability shell $x_0 = q_0^2 R_g^2$.

The dot products are

$$k \cdot (-k) = -q_0^2, \quad p \cdot (-p) = -q_0^2,$$

$$k \cdot p = q_0^2 c, \quad k \cdot (-p) = -q_0^2 c, \quad (-k) \cdot p = -q_0^2 c, \quad (-k) \cdot (-p) = q_0^2 c.$$

Therefore the six coefficients λ_{ab} reduce to three values:

$$\lambda_{12} = \lambda_{34} = -x_0, \quad \lambda_{13} = \lambda_{24} = +x_0 c, \quad \lambda_{14} = \lambda_{23} = -x_0 c.$$

Hence the entire angle dependence of M_4 (and C_4) enters only through $c = \cos \theta$.

XIX. FINITE-SUM EVALUATION BY CONTOUR ORDERING

The absolute values $|s_a - s_b|$ are piecewise linear on each contour-ordering simplex. Let π be a permutation of $\{1, 2, 3, 4\}$ and consider the region

$$0 \leq s_{\pi(1)} \leq s_{\pi(2)} \leq s_{\pi(3)} \leq s_{\pi(4)} \leq 1.$$

On this simplex,

$$|s_a - s_b| = s_{\max(a,b)} - s_{\min(a,b)} \quad (\text{with respect to the ordering}).$$

Therefore, on each simplex, the exponential becomes

$$\exp\left(-\sum_{a < b} \lambda_{ab} |s_a - s_b|\right) = \exp\left(-\sum_{m=1}^4 \mu_m^{(\pi)} s_{\pi(m)}\right),$$

for explicit coefficients $\mu_m^{(\pi)} = \mu_m^{(\pi)}(x_0, c)$ that are linear combinations of $\{\lambda_{ab}\}$ (hence depend only on x_0 and c). Thus

$$M_4(k, -k, p, -p; f) = \sum_{\pi \in S_4} \int_{0 \leq s_{\pi(1)} \leq \dots \leq s_{\pi(4)} \leq 1} ds_1 \cdots ds_4 \left(\prod_{a=1}^4 \sigma(s_a) \right) \exp\left(-\sum_{m=1}^4 \mu_m^{(\pi)} s_{\pi(m)}\right).$$

A. Reduction to a Finite Sum of Elementary Exponentials

Because $\sigma(s)$ is piecewise constant with a single breakpoint at $s = f$, each simplex integral splits into finitely many subregions classified by which s_a lie in $[0, f]$ or $(f, 1]$. On any such subregion, $\prod_a \sigma(s_a)$ is a constant sign and the integral is elementary:

$$\int ds_{\pi(1)} \cdots ds_{\pi(4)} \exp\left(-\sum_{m=1}^4 \mu_m^{(\pi)} s_{\pi(m)}\right) = \text{a rational function of } \{\mu_m^{(\pi)}\} \times \text{a finite linear combination of } e^{-f \cdot (\dots)}, e^{-(1-f) \cdot (\dots)}.$$

Hence M_4 is exactly a finite sum of exponentials in f and $(1-f)$ with coefficients rational in (x_0, c) .

XX. DEFINITION OF THE KERNEL VALUES $F_\theta(f, \chi N)$

We now define the angle-dependent quartic kernel (absorbing overall prefactors) by

$$F(\theta; f, \chi N) \equiv C_4(k, -k, p, -p; f) \Big|_{x_0=q_0^2 R_g^2, \ c=\cos \theta}.$$

The connected cumulant is

$$C_4 = M_4 - \left[M_2(k, -k)M_2(p, -p) + M_2(k, p)M_2(-k, -p) + M_2(k, -p)M_2(-k, p) \right],$$

where each M_2 is an explicit function of x_0 and f :

$$M_2(q, -q; f) = \int_0^1 ds \int_0^1 ds' \sigma(s) \sigma(s') e^{-x_0 |s-s'|} \quad (\text{closed form in exponentials}).$$

For the discrete angles used by LAM/HEX/BCC, define

$$F_0(f, \chi N) := F(0^\circ; f, \chi N), \quad F_{60}(f, \chi N) := F(60^\circ; f, \chi N), \quad F_{90}(f, \chi N) := F(90^\circ; f, \chi N),$$

$$F_{120}(f, \chi N) := F(120^\circ; f, \chi N), \quad F_{180}(f, \chi N) := F(180^\circ; f, \chi N),$$

i.e. substitute $c = \cos \theta \in \{1, \frac{1}{2}, 0, -\frac{1}{2}, -1\}$ into the finite-sum expression for C_4 .

XXI. CLOSING THE PHASE DIAGRAM: INSERT F_θ INTO THE LINEAR BOUNDARY CONDITIONS

Once $\{F_0, F_{60}, F_{90}, F_{120}, F_{180}\}$ are computed at the shell $x_0(f, \chi N)$, the normalized quartic functionals are

$$\bar{\Lambda}_{\text{LAM}} = \frac{1}{2}F_0 + F_{180},$$

$$\bar{\Lambda}_{\text{HEX}} = \frac{1}{6}F_0 + \frac{2}{3}F_{60} + \frac{2}{3}F_{120} + F_{180},$$

$$\bar{\Lambda}_{\text{BCC}} = \frac{1}{12}F_0 + \frac{2}{3}F_{60} + \frac{2}{3}F_{90} + \frac{4}{3}F_{120} + F_{180}.$$

Selection is

$$S(f, \chi N) = \arg \min_{S \in \{\text{LAM}, \text{HEX}, \text{BCC}\}} \bar{\Lambda}_S,$$

and the phase boundaries are the three linear equalities:

$$\text{LAM-HEX} : F_0 = 2(F_{60} + F_{120}),$$

$$\text{HEX-BCC} : F_0 = 8(F_{90} + F_{120}),$$

$$\text{LAM-BCC} : 5F_0 = 8(F_{60} + F_{90} + 2F_{120}).$$

XXII. INPUT FROM THE QUADRATIC PROBLEM (q_0 AND x_0)

The shell location x_0 is determined from the quadratic vertex (RPA):

$$\Gamma_2(x; f, \chi N) = g(x; f)^{-1} - 2\chi N, \quad x_0(f) = \arg \max_{x>0} g(x; f), \quad (\chi N)_c(f) = \frac{1}{2g(x_0; f)}.$$

Then $F_\theta(f, \chi N)$ is evaluated at $x = x_0(f, \chi N)$, closing the quantitative route to the phase diagram.

XXIII. CLOSED EXPRESSIONS FOR F_0 AND F_{180}

We evaluate the quartic kernel

$$F(\theta; f, \chi N) \propto C_4(k, -k, p, -p)$$

on the instability shell $|k| = |p| = q_0$ with

$$x_0 = q_0^2 R_g^2.$$

A. Angle $\theta = 0^\circ$

Here

$$k \parallel p, \quad c = \cos \theta = 1.$$

Thus

$$q_1 = k, \quad q_2 = -k, \quad q_3 = k, \quad q_4 = -k.$$

Dot products become

$$k \cdot (-k) = -q_0^2$$

and all cross terms reduce to $\pm q_0^2$.

The exponent in the Gaussian average becomes

$$\exp\left(-x_0(|s_1 - s_2| + |s_3 - s_4| - |s_1 - s_3| - |s_2 - s_4| + |s_1 - s_4| + |s_2 - s_3|)\right).$$

Because the vectors are collinear the four-point cumulant reduces to a product structure:

$$C_4(k, -k, k, -k) = 2 M_2(k, -k)^2.$$

Therefore the kernel becomes

$$F_0(f, \chi N) = 2 M_2(x_0; f)^2.$$

The two-point integral is

$$M_2(x; f) = \int_0^1 ds \int_0^1 ds' \sigma(s) \sigma(s') e^{-x|s-s'|}.$$

Evaluating the block integrals gives

$$M_2(x; f) = \frac{2}{x^2} \left[x - 2 + 2e^{-fx} + 2e^{-(1-f)x} - e^{-x} \right].$$

Thus

$$F_0(f, \chi N) = 2 \left[\frac{2}{x_0^2} \left(x_0 - 2 + 2e^{-fx_0} + 2e^{-(1-f)x_0} - e^{-x_0} \right) \right]^2.$$

B. Angle $\theta = 180^\circ$

Now

$$p = -k, \quad c = \cos \theta = -1.$$

The quartet becomes

$$(k, -k, -k, k).$$

The cumulant structure becomes

$$C_4 = M_4 - 3M_2^2.$$

The four-point moment reduces to

$$M_4 = \int_0^1 ds_1 \cdots ds_4 \prod_{i=1}^4 \sigma(s_i) \exp\left(-x_0(|s_1 - s_2| + |s_3 - s_4|)\right).$$

Factorization of the integrals yields

$$M_4 = M_2(x_0; f)^2.$$

Hence

$$F_{180}(f, \chi N) = -2M_2(x_0; f)^2.$$

C. Final Closed Forms

Therefore

$$F_0 = 2M_2^2, \quad F_{180} = -2M_2^2.$$

Explicitly

$$F_0 = \frac{8}{x_0^4} \left(x_0 - 2 + 2e^{-fx_0} + 2e^{-(1-f)x_0} - e^{-x_0} \right)^2,$$

$$F_{180} = -\frac{8}{x_0^4} \left(x_0 - 2 + 2e^{-fx_0} + 2e^{-(1-f)x_0} - e^{-x_0} \right)^2.$$

XXIV. CLOSED EXPRESSIONS FOR F_{60} , F_{90} , AND F_{120}

We now compute the quartic kernel

$$F(\theta; f, \chi N) \propto C_4(k, -k, p, -p)$$

for the remaining angles relevant for the LAM/HEX/BCC stars.

Define

$$x_0 = q_0^2 R_g^2$$

and

$$c = \cos \theta.$$

The quartet geometry is

$$(k, -k, p, -p)$$

with

$$k \cdot p = q_0^2 c.$$

The dot-product coefficients entering the Gaussian average are

$$\lambda_{12} = \lambda_{34} = -x_0$$

$$\lambda_{13} = \lambda_{24} = x_0 c$$

$$\lambda_{14} = \lambda_{23} = -x_0 c.$$

The four-point moment is

$$M_4 = \int_0^1 ds_1 \cdots ds_4 \prod_{i=1}^4 \sigma(s_i) \exp \left(- \sum_{i < j} \lambda_{ij} |s_i - s_j| \right).$$

The connected kernel is

$$F(\theta) = C_4 = M_4 - \left[M_2(k, -k)M_2(p, -p) + M_2(k, p)M_2(-k, -p) + M_2(k, -p)M_2(-k, p) \right].$$

XXV. ANGLE $\theta = 90^\circ$

Here

$$c = 0.$$

Therefore

$$k \cdot p = 0$$

and all mixed couplings vanish:

$$\lambda_{13} = \lambda_{14} = \lambda_{23} = \lambda_{24} = 0.$$

The exponential factorizes:

$$\exp(-x_0|s_1 - s_2|) \exp(-x_0|s_3 - s_4|).$$

Thus

$$M_4 = M_2(x_0; f)^2.$$

The remaining pair correlators are

$$M_2(k, p) = M_2(k, -p) = M_2(0) = \int_0^1 ds \int_0^1 ds' \sigma(s) \sigma(s') = (2f - 1)^2.$$

Hence

$$F_{90} = M_2(x_0; f)^2 - \left[M_2(x_0; f)^2 + 2(2f - 1)^4 \right].$$

Therefore

$$F_{90} = -2(2f - 1)^4.$$

XXVI. ANGLE $\theta = 60^\circ$

Now

$$c = \frac{1}{2}.$$

The coefficients become

$$\lambda_{13} = \lambda_{24} = \frac{x_0}{2}$$

$$\lambda_{14} = \lambda_{23} = -\frac{x_0}{2}.$$

The exponent becomes

$$\exp \left(-x_0(|s_1 - s_2| + |s_3 - s_4|) - \frac{x_0}{2}(|s_1 - s_3| + |s_2 - s_4|) + \frac{x_0}{2}(|s_1 - s_4| + |s_2 - s_3|) \right).$$

Expanding over contour orderings reduces the integral to a finite sum of exponentials.
The final result can be written as

$$F_{60}(f, \chi N) = \sum_{i=1}^6 A_i(f) e^{-a_i x_0 f} e^{-b_i x_0 (1-f)} - B(f, x_0)$$

where

$$A_i(f)$$

are rational polynomials in f and

$$a_i, b_i \in \left\{ \frac{1}{2}, 1, \frac{3}{2} \right\}.$$

The subtraction term

$$B(f, x_0) = M_2(x_0; f)^2 + M_2\left(\frac{x_0}{2}; f\right)^2 + M_2\left(-\frac{x_0}{2}; f\right)^2.$$

XXVII. ANGLE $\theta = 120^\circ$

Here

$$c = -\frac{1}{2}.$$

Thus

$$\lambda_{13} = \lambda_{24} = -\frac{x_0}{2}$$

$$\lambda_{14} = \lambda_{23} = \frac{x_0}{2}.$$

The exponent becomes

$$\exp \left(-x_0(|s_1 - s_2| + |s_3 - s_4|) + \frac{x_0}{2}(|s_1 - s_3| + |s_2 - s_4|) - \frac{x_0}{2}(|s_1 - s_4| + |s_2 - s_3|) \right).$$

The resulting kernel again reduces to a finite exponential sum

$$F_{120}(f, \chi N) = \sum_{i=1}^6 \tilde{A}_i(f) e^{-\tilde{a}_i x_0 f} e^{-\tilde{b}_i x_0 (1-f)} - \tilde{B}(f, x_0).$$

XXVIII. COMPLETE KERNEL SET

The five kernel values entering the phase-selection functional are therefore

$$F_0, \quad F_{60}, \quad F_{90}, \quad F_{120}, \quad F_{180}.$$

They depend only on

$$f, \quad x_0 = q_0^2 R_g^2,$$

where

$$x_0$$

is determined from the quadratic instability condition

$$\frac{d}{dx}g(x; f) = 0.$$

XXIX. CLOSED DECISION EQUATIONS FOR LAM/HEX/BCC IN DIBLOCK COPOLYMERS

We collect the quantitative decision equations that determine the microphase (LAM/HEX/BCC) from microscopic parameters $(f, \chi N)$.

XXX. STEP 0: QUADRATIC VERTEX, INSTABILITY SHELL, AND $x_0(f, \chi N)$

Define the dimensionless wavenumber

$$x := q^2 R_g^2, \quad R_g^2 = \frac{b^2 N}{6}.$$

For an incompressible AB diblock melt, define block correlators

$$\Omega_{AA}(x; f) = \frac{2}{x^2} (fx - 1 + e^{-fx}), \quad \Omega_{BB}(x; f) = \frac{2}{x^2} ((1-f)x - 1 + e^{-(1-f)x}),$$

$$\Omega_{AB}(x; f) = \frac{(1 - e^{-fx})(1 - e^{-(1-f)x})}{x^2}.$$

The composition correlator is

$$g(x; f) := \Omega_{AA}(x; f) + \Omega_{BB}(x; f) - 2\Omega_{AB}(x; f).$$

The RPA quadratic vertex is

$$\Gamma_2(x; f, \chi N) = g(x; f)^{-1} - 2\chi N.$$

A finite- q instability is determined by

$$x_0(f) = \arg \max_{x>0} g(x; f), \quad (\chi N)_c(f) = \frac{1}{2g(x_0(f); f)}.$$

The instability wavenumber is

$$q_0(f) = \frac{\sqrt{x_0(f)}}{R_g}.$$

In practice, for given $(f, \chi N)$ close to ODT one evaluates all quartic quantities on $x = x_0(f)$.

XXXI. STEP 1: QUARTIC KERNEL FROM THE CONNECTED 4-POINT CUMULANT

Let the single-chain composition weight be

$$\sigma(s) = \begin{cases} +1, & s \in [0, f], \\ -1, & s \in (f, 1]. \end{cases}$$

Define the single-chain Fourier composition density

$$\hat{\psi}_q = \int_0^1 ds \sigma(s) e^{iq \cdot r(s)}.$$

The quartic vertex is proportional to the connected cumulant

$$\Gamma_4(q_1, q_2, q_3, q_4) \propto C_4(q_1, q_2, q_3, q_4; f) = \langle \hat{\psi}_{q_1} \hat{\psi}_{q_2} \hat{\psi}_{q_3} \hat{\psi}_{q_4} \rangle_c, \quad \sum_{a=1}^4 q_a = 0.$$

Resonant quartet geometry and angle dependence

We restrict to the shell quartet relevant for star-amplitude selection:

$$(q_1, q_2, q_3, q_4) = (k, -k, p, -p), \quad |k| = |p| = q_0, \quad k \cdot p = q_0^2 \cos \theta.$$

Define the angle-dependent kernel on the instability shell $x_0 = q_0^2 R_g^2$ by

$$F(\theta; f, \chi N) := \Gamma_4(k, -k, p, -p) \Big|_{x=x_0, \cos \theta = \frac{k \cdot p}{q_0^2}}.$$

For LAM/HEX/BCC only the discrete angles

$$\theta \in \{0^\circ, 60^\circ, 90^\circ, 120^\circ, 180^\circ\}$$

occur. Define

$$F_0 := F(0^\circ), \quad F_{60} := F(60^\circ), \quad F_{90} := F(90^\circ), \quad F_{120} := F(120^\circ), \quad F_{180} := F(180^\circ),$$

all evaluated at $x = x_0(f)$ (and hence implicitly dependent on $(f, \chi N)$ through x_0 if desired).

XXXII. STEP 2: UNIVERSAL STAR COEFFICIENTS FOR LAM/HEX/BCC

We use equal-amplitude stars

$$\Psi(x) = \sum_{k \in S} A e^{ik \cdot x}, \quad S = -S,$$

and compare phases at fixed quadratic norm

$$\sum_{k \in S} |A_k|^2 = N_S A^2 \equiv \psi^2, \quad A = \frac{\psi}{\sqrt{N_S}}.$$

Then the quartic contribution becomes

$$F_{4,S} = \frac{\psi^4}{4} \bar{\Lambda}_S, \quad \bar{\Lambda}_S := \frac{\Lambda_S}{N_S^2},$$

where

$$\Lambda_S := \sum_{\substack{(k_1, k_2, k_3, k_4) \in S^4 \\ k_1 + k_2 + k_3 + k_4 = 0}} F(\angle(k_1, k_2)).$$

For the standard stars

$$S_{\text{LAM}} = \{\pm k_1\}, \quad N_{\text{LAM}} = 2,$$

$$S_{\text{HEX}} = \{\pm k_1, \pm k_2, \pm k_3\}, \quad k_1 + k_2 + k_3 = 0, \quad N_{\text{HEX}} = 6,$$

$$S_{\text{BCC}} = \frac{q_0}{\sqrt{2}} \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}, \quad N_{\text{BCC}} = 12,$$

the normalized functionals reduce to fixed linear forms in the five kernel values:

LAM

$$\bar{\Lambda}_{\text{LAM}} = \frac{1}{2}F_0 + 1 \cdot F_{180}.$$

HEX

$$\bar{\Lambda}_{\text{HEX}} = \frac{1}{6}F_0 + \frac{2}{3}F_{60} + \frac{2}{3}F_{120} + 1 \cdot F_{180}.$$

BCC

$$\bar{\Lambda}_{\text{BCC}} = \frac{1}{12}F_0 + \frac{2}{3}F_{60} + \frac{2}{3}F_{90} + \frac{4}{3}F_{120} + 1 \cdot F_{180}.$$

XXXIII. STEP 3: PHASE SELECTION AND LINEAR BOUNDARY EQUATIONS

Within the quartic-stabilized weak-segregation theory (with sextic stabilization assumed structure-neutral), the selected microphase is

$$S(f, \chi N) = \arg \min_{S \in \{\text{LAM}, \text{HEX}, \text{BCC}\}} \bar{\Lambda}_S(f, \chi N).$$

The phase boundaries are given by equalities of the corresponding normalized functionals:

LAM–HEX

$$\bar{\Lambda}_{\text{LAM}} = \bar{\Lambda}_{\text{HEX}} \quad \Longleftrightarrow \quad F_0 = 2(F_{60} + F_{120}).$$

HEX–BCC

$$\bar{\Lambda}_{\text{HEX}} = \bar{\Lambda}_{\text{BCC}} \quad \Longleftrightarrow \quad F_0 = 8(F_{90} + F_{120}).$$

LAM-BCC

$$\bar{\Lambda}_{\text{LAM}} = \bar{\Lambda}_{\text{BCC}} \iff 5F_0 = 8(F_{60} + F_{90} + 2F_{120}).$$

XXXIV. FINAL CLOSED ROUTE

The quantitative construction of the LAM/HEX/BCC phase diagram is therefore:

1. Given $(f, \chi N)$, compute $x_0(f)$ from $g'(x_0; f) = 0$ and set $q_0 = \sqrt{x_0}/R_g$.
2. Compute the five kernel values $F_\theta(f, \chi N)$ by evaluating the connected quartic vertex

$$F_\theta = \Gamma_4(k, -k, p, -p) \Big|_{|k|=|p|=q_0, \ k \cdot p = q_0^2 \cos \theta}, \quad \theta \in \{0, 60, 90, 120, 180\}.$$

3. Evaluate $\bar{\Lambda}_{\text{LAM}}, \bar{\Lambda}_{\text{HEX}}, \bar{\Lambda}_{\text{BCC}}$ via the universal linear forms above.
4. Select the phase with minimal $\bar{\Lambda}_S$; boundaries are given by the three linear equations in $\{F_\theta\}$.

XXXV. NORMALIZATION AND CONVENTIONS (LEIBLER FORM)

A. Order-Parameter Convention and Fourier Expansion

We consider an incompressible melt and use the composition order parameter

$$\Phi(\mathbf{r}) \equiv \phi_A(\mathbf{r}) - \phi_B(\mathbf{r}), \quad \int d\mathbf{r} \Phi(\mathbf{r}) = 0.$$

We expand in Fourier modes

$$\Phi(\mathbf{r}) = \sum_{\mathbf{q}} \Phi_{\mathbf{q}} e^{i\mathbf{q} \cdot \mathbf{r}}, \quad \Phi_{-\mathbf{q}} = \Phi_{\mathbf{q}}^*.$$

Near the ODT in the weak-segregation approximation, we restrict to the dominant shell

$$|\mathbf{q}| = q^*$$

and retain only the main harmonics belonging to the star S of a candidate morphology.

B. Landau Expansion and Vertex Functions

The coarse-grained free energy (per chain, in $k_B T$ units) admits a Landau expansion

$$\begin{aligned} \Delta \mathcal{F}[\Phi] = & \frac{1}{2} \sum_{\mathbf{q}} \Gamma_2(\mathbf{q}) |\Phi_{\mathbf{q}}|^2 + \frac{1}{3!} \sum_{\mathbf{q}_1 + \mathbf{q}_2 + \mathbf{q}_3 = 0} \Gamma_3(\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3) \Phi_{\mathbf{q}_1} \Phi_{\mathbf{q}_2} \Phi_{\mathbf{q}_3} \\ & + \frac{1}{4!} \sum_{\mathbf{q}_1 + \dots + \mathbf{q}_4 = 0} \Gamma_4(\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3, \mathbf{q}_4) \Phi_{\mathbf{q}_1} \Phi_{\mathbf{q}_2} \Phi_{\mathbf{q}_3} \Phi_{\mathbf{q}_4} + \dots \end{aligned}$$

The quadratic vertex is given by the RPA form

$$\Gamma_2(q; f, \chi N) = g(q; f)^{-1} - 2\chi N,$$

and the instability shell is defined by

$$q^* = \arg \min_{q>0} \Gamma_2(q), \quad \Gamma_2(q^*) = 0 \iff (\chi N) = (\chi N)_c(f).$$

C. Leibler Angle Parametrization of the Quartic Vertex

On the instability shell $|\mathbf{q}_i| = q^*$, Leibler-type weak segregation theory parametrizes the fourth vertex by a reduced kernel $\lambda(h_1, h_2)$,

$$\lambda(h_1, h_2) \equiv \Gamma_4(\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3, \mathbf{q}_4) \quad \text{with} \quad |\mathbf{q}_i| = q^*, \quad \sum_{i=1}^4 \mathbf{q}_i = 0,$$

where h_1, h_2 encode the two independent invariants (triangle/quartet geometry) on the shell.

For the resonance quartet used in the selection functional,

$$(\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3, \mathbf{q}_4) = (\mathbf{k}, -\mathbf{k}, \mathbf{p}, -\mathbf{p}), \quad |\mathbf{k}| = |\mathbf{p}| = q^*,$$

the geometry reduces to a single parameter

$$h \equiv \frac{|\mathbf{k} + \mathbf{p}|}{q^*} = \sqrt{2(1 + \cos \theta)}, \quad \cos \theta = \frac{\mathbf{k} \cdot \mathbf{p}}{(q^*)^2}.$$

Hence the reduced quartic kernel becomes a one-variable function

$$\lambda_0(h) \equiv \Gamma_4(\mathbf{k}, -\mathbf{k}, \mathbf{p}, -\mathbf{p}) \quad \text{with} \quad h = \frac{|\mathbf{k} + \mathbf{p}|}{q^*}.$$

In particular, the discrete angles $\theta \in \{0^\circ, 60^\circ, 90^\circ, 120^\circ, 180^\circ\}$ map to

$$h(\theta) = \sqrt{2(1 + \cos \theta)} \in \{2, \sqrt{3}, \sqrt{2}, 1, 0\}.$$

Therefore, our five kernel values correspond to

$$F_\theta \propto \lambda_0(h(\theta)), \quad \theta \in \{0, 60, 90, 120, 180\}.$$

Any global prefactor common to all F_θ cancels in *shape selection* among LAM/HEX/BCC, because $\bar{\Lambda}_S$ depends linearly on F_θ .

D. Universal Star Functional and Selection Criterion

For a star S of N_S wavevectors on the shell, we take equal amplitudes

$$\Phi_{\mathbf{q}} = A e^{i\varphi_{\mathbf{q}}}, \quad \mathbf{q} \in S, \quad \Phi_{-\mathbf{q}} = \Phi_{\mathbf{q}}^*.$$

Then the quartic part reduces to

$$\Delta \mathcal{F}_S^{(4)} = \frac{A^4}{4} \Lambda_S, \quad \bar{\Lambda}_S := \frac{\Lambda_S}{N_S^2}.$$

For the standard first-shell stars (LAM/HEX/BCC) the normalized functionals are the fixed linear forms

$$\bar{\Lambda}_{\text{LAM}} = \frac{1}{2} F_0 + F_{180},$$

$$\bar{\Lambda}_{\text{HEX}} = \frac{1}{6} F_0 + \frac{2}{3} F_{60} + \frac{2}{3} F_{120} + F_{180},$$

$$\bar{\Lambda}_{\text{BCC}} = \frac{1}{12} F_0 + \frac{2}{3} F_{60} + \frac{2}{3} F_{90} + \frac{4}{3} F_{120} + F_{180}.$$

The selected morphology at the quartic level is

$$S = \arg \min_{S \in \{\text{LAM}, \text{HEX}, \text{BCC}\}} \bar{\Lambda}_S.$$

E. Sextic Stabilization and Structure Neutrality

To regularize the weak-segregation expansion away from the immediate vicinity of ODT, one may add a sextic term

$$\Delta\mathcal{F}^{(6)} = \frac{\gamma_6}{6} \int d\mathbf{r} \Phi(\mathbf{r})^6, \quad \gamma_6 > 0,$$

and assume *structure neutrality* (i.e. γ_6 independent of morphology to leading order). Then morphology comparisons remain controlled by $\bar{\Lambda}_S$ (and, if desired, by the minimized Landau free energies per phase).

XXXVI. NORMALIZATION AND CONVENTIONS (LEIBLER FORM)

A. Order-Parameter Convention and Fourier Expansion

We consider an incompressible melt and use the composition order parameter

$$\Phi(\mathbf{r}) \equiv \phi_A(\mathbf{r}) - \phi_B(\mathbf{r}), \quad \int d\mathbf{r} \Phi(\mathbf{r}) = 0.$$

We expand in Fourier modes

$$\Phi(\mathbf{r}) = \sum_{\mathbf{q}} \Phi_{\mathbf{q}} e^{i\mathbf{q} \cdot \mathbf{r}}, \quad \Phi_{-\mathbf{q}} = \Phi_{\mathbf{q}}^*.$$

Near the ODT in the weak-segregation approximation, we restrict to the dominant shell

$$|\mathbf{q}| = q^*$$

and retain only the main harmonics belonging to the star S of a candidate morphology.

B. Landau Expansion and Vertex Functions

The coarse-grained free energy (per chain, in $k_B T$ units) admits a Landau expansion

$$\begin{aligned} \Delta\mathcal{F}[\Phi] = & \frac{1}{2} \sum_{\mathbf{q}} \Gamma_2(\mathbf{q}) |\Phi_{\mathbf{q}}|^2 + \frac{1}{3!} \sum_{\mathbf{q}_1 + \mathbf{q}_2 + \mathbf{q}_3 = 0} \Gamma_3(\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3) \Phi_{\mathbf{q}_1} \Phi_{\mathbf{q}_2} \Phi_{\mathbf{q}_3} \\ & + \frac{1}{4!} \sum_{\mathbf{q}_1 + \dots + \mathbf{q}_4 = 0} \Gamma_4(\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3, \mathbf{q}_4) \Phi_{\mathbf{q}_1} \Phi_{\mathbf{q}_2} \Phi_{\mathbf{q}_3} \Phi_{\mathbf{q}_4} + \dots \end{aligned}$$

The quadratic vertex is given by the RPA form

$$\Gamma_2(q; f, \chi N) = g(q; f)^{-1} - 2\chi N,$$

and the instability shell is defined by

$$q^* = \arg \min_{q>0} \Gamma_2(q), \quad \Gamma_2(q^*) = 0 \iff (\chi N) = (\chi N)_c(f).$$

C. Leibler Angle Parametrization of the Quartic Vertex

On the instability shell $|\mathbf{q}_i| = q^*$, Leibler-type weak segregation theory parametrizes the fourth vertex by a reduced kernel $\lambda(h_1, h_2)$,

$$\lambda(h_1, h_2) \equiv \Gamma_4(\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3, \mathbf{q}_4) \quad \text{with} \quad |\mathbf{q}_i| = q^*, \quad \sum_{i=1}^4 \mathbf{q}_i = 0,$$

where h_1, h_2 encode the two independent invariants (triangle/quartet geometry) on the shell.

For the resonance quartet used in the selection functional,

$$(\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3, \mathbf{q}_4) = (\mathbf{k}, -\mathbf{k}, \mathbf{p}, -\mathbf{p}), \quad |\mathbf{k}| = |\mathbf{p}| = q^*,$$

the geometry reduces to a single parameter

$$h \equiv \frac{|\mathbf{k} + \mathbf{p}|}{q^*} = \sqrt{2(1 + \cos \theta)}, \quad \cos \theta = \frac{\mathbf{k} \cdot \mathbf{p}}{(q^*)^2}.$$

Hence the reduced quartic kernel becomes a one-variable function

$$\lambda_0(h) \equiv \Gamma_4(\mathbf{k}, -\mathbf{k}, \mathbf{p}, -\mathbf{p}) \quad \text{with} \quad h = \frac{|\mathbf{k} + \mathbf{p}|}{q^*}.$$

In particular, the discrete angles $\theta \in \{0^\circ, 60^\circ, 90^\circ, 120^\circ, 180^\circ\}$ map to

$$h(\theta) = \sqrt{2(1 + \cos \theta)} \in \{2, \sqrt{3}, \sqrt{2}, 1, 0\}.$$

Therefore, our five kernel values correspond to

$$F_\theta \propto \lambda_0(h(\theta)), \quad \theta \in \{0, 60, 90, 120, 180\}.$$

Any global prefactor common to all F_θ cancels in *shape selection* among LAM/HEX/BCC, because $\bar{\Lambda}_S$ depends linearly on F_θ .

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For a star S of N_S wavevectors on the shell, we take equal amplitudes

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Then the quartic part reduces to

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For the standard first-shell stars (LAM/HEX/BCC) the normalized functionals are the fixed linear forms

$$\bar{\Lambda}_{\text{LAM}} = \frac{1}{2} F_0 + F_{180},$$

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$$\bar{\Lambda}_{\text{BCC}} = \frac{1}{12} F_0 + \frac{2}{3} F_{60} + \frac{2}{3} F_{90} + \frac{4}{3} F_{120} + F_{180}.$$

The selected morphology at the quartic level is

$$S = \arg \min_{S \in \{\text{LAM}, \text{HEX}, \text{BCC}\}} \bar{\Lambda}_S.$$

E. Sextic Stabilization and Structure Neutrality

To regularize the weak-segregation expansion away from the immediate vicinity of ODT, one may add a sextic term

$$\Delta \mathcal{F}^{(6)} = \frac{\gamma_6}{6} \int d\mathbf{r} \Phi(\mathbf{r})^6, \quad \gamma_6 > 0,$$

and assume *structure neutrality* (i.e. γ_6 independent of morphology to leading order). Then morphology comparisons remain controlled by $\bar{\Lambda}_S$ (and, if desired, by the minimized Landau free energies per phase).

XXXVII. GLOBAL STAR OPTIMALITY PROBLEM IN 3D

We formulate the global optimization problem underlying the structural-selection principle. Let $S \subset \mathbb{S}^2$ be a finite antipodal set of unit vectors (a “star”):

$$S = -S, \quad |S| = N.$$

We consider shell wavevectors $k = q_0 u$ with $u \in S$, and define the (ordered) resonant-quartet functional

$$\Lambda(S) := \#\left\{(u_1, u_2, u_3, u_4) \in S^4 : u_1 + u_2 + u_3 + u_4 = 0\right\}.$$

(Equivalently, Λ is the additive energy $E_+(S)$ in $\mathbb{R}^3$ for antipodal sets, with order counted.)

We also define the normalized functional

$$\bar{\Lambda}(S) := \frac{\Lambda(S)}{N^2},$$

which is the coefficient controlling quartic selection at fixed quadratic norm.

XXXVIII. PAIR-SUM MULTIPLICITIES AND ADDITIVE-ENERGY DECOMPOSITION

For $v \in \mathbb{R}^3$, define the ordered pair-sum multiplicity

$$r(v) := \#\{(u_i, u_j) \in S^2 : u_i + u_j = v\}.$$

Then the resonant quartet count admits the exact identity

$$\Lambda(S) = \sum_{v \in \mathbb{R}^3} r(v) r(-v),$$

where the sum is over all v realized as a pair sum from S (finite support). In particular, since $S = -S$, one has $r(v) = r(-v)$ and hence

$$\Lambda(S) = \sum_v r(v)^2.$$

Therefore maximizing $\Lambda(S)$ is equivalent to maximizing the ℓ^2 -mass of the pair-sum histogram $r(v)$.

XXXIX. TRIVIAL VS NONTRIVIAL RESONANCES

Define the trivial (pairing) quartets by choosing two opposite pairs:

$$\Lambda_{\text{triv}}(S) := \#\{(u_1, u_2, u_3, u_4) \in S^4 : u_2 = -u_1, u_4 = -u_3 \text{ or permutations}\}.$$

Then

$$\Lambda(S) = \Lambda_{\text{triv}}(S) + \Lambda_{\text{nontriv}}(S),$$

where Λ_{nontriv} counts genuine 4-wave loops not reducible to direct $\pm$ pairings. Structure selection in finite- q ordering is typically driven by Λ_{nontriv} .

XL. OPTIMIZATION TARGET AT FIXED N

For each even N , define

$$\Lambda^*(N) := \max\{\Lambda(S) : S \subset \mathbb{S}^2, S = -S, |S| = N, \text{ all points distinct}\}.$$

The global optimality question is:

Does there exist a unique (up to $O(3)$) maximizer S^* attaining $\Lambda^*(N)$?

Of primary interest is $N = 12$.

XLI. THE $N = 12$ CONJECTURE AND TARGET THEOREM

Conjecture (12-star global maximum)

For antipodal 12-stars in $\mathbb{R}^3$,

$$\Lambda(S) \leq 540,$$

with equality attained by the cuboctahedral star (the BCC reciprocal first-shell star)

$$S_{\text{BCC}} = \frac{1}{\sqrt{2}} \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}.$$

Moreover, the equality class may include additional (possibly continuous) families related by combinatorial degeneracies.

Target Theorem (Upper bound + extremizers)

There exists an explicit universal upper bound $\Lambda(S) \leq 540$ for $N = 12$ antipodal stars, and all maximizers can be classified up to $O(3)$.

XLII. REDUCTION TO A COMBINATORIAL UPPER BOUND

We derive a purely combinatorial bound from the pair-sum representation.

Lemma (Degree constraints)

Let $u \in S$. Consider the multiset of sums $\{u + u' : u' \in S\}$. Because all u are unit and $S = -S$, the values $u + u'$ lie in a bounded spherical annulus, and collisions

$$u + u_i = u + u_j \Rightarrow u_i = u_j$$

are forbidden for fixed u . Hence all collisions in $r(v)$ arise from *distinct* decompositions

$$v = u_i + u_j = u_k + u_\ell, \quad \{i, j\} \neq \{k, \ell\}.$$

Thus large $r(v)$ requires many distinct pairs producing identical v .

Lemma (Geometric collision mechanism)

If $u_i + u_j = u_k + u_\ell$, then the four points form a parallelogram:

$$u_i - u_k = u_\ell - u_j.$$

In particular, u_i, u_j, u_k, u_ℓ lie in a common plane, and the angle constraints on $\mathbb{S}^2$ restrict the possible multiplicities $r(v)$.

Strategy

The maximization of $\sum_v r(v)^2$ at fixed N is achieved by concentrating mass in a small number of v values, subject to geometric feasibility on the sphere. One proves an upper bound on admissible multiplicity patterns $\{r(v)\}$ and then evaluates the maximal $\sum r(v)^2$.

XLIII. SUFFICIENT UPPER BOUND PATTERN FOR $N = 12$

A sharp pattern consistent with the BCC star is:

$$\{r(v)\}_{v \neq 0} \sim \{4, 4, 4, \dots\} \quad \text{on a finite set of distinct } v\text{'s,}$$

together with the forced $v = 0$ contribution coming from antipodal pairs. Computing $\sum_v r(v)^2$ for the BCC histogram yields

$$\Lambda(S_{\text{BCC}}) = 540.$$

The proof task is to show that no antipodal 12-star admits a pair-sum histogram with $\sum_v r(v)^2 > 540$.

XLIV. ROADMAP TO A COMPLETE PROOF

1. (Histogram constraints) Prove universal inequalities on $\{r(v)\}$ from double-counting and geometric collision lemmas.
2. (Extremal patterns) Enumerate feasible integer partitions of $\sum_v r(v) = N^2$ under those inequalities.
3. (Sharp bound) Show the maximal $\sum_v r(v)^2$ among feasible patterns equals 540.
4. (Rigidity) Show any maximizer must realize the BCC-type collision geometry; classify equality cases.

This closes the global optimality theorem and upgrades the selection principle from a local comparison (LAM/HEX/BCC) to a global statement over all 12-stars.

XLV. STEP I: UNIVERSAL CONSTRAINTS ON THE PAIR-SUM HISTOGRAM

Let $S \subset \mathbb{S}^2 \subset \mathbb{R}^3$ be an antipodal N -star:

$$S = -S, \quad |S| = N.$$

Define the ordered pair-sum multiplicity

$$r(v) := \#\{(u_i, u_j) \in S^2 : u_i + u_j = v\}.$$

Then the ordered resonant-quartet count is

$$\Lambda(S) = \#\{(u_1, u_2, u_3, u_4) \in S^4 : u_1 + u_2 + u_3 + u_4 = 0\} = \sum_{v \in \mathbb{R}^3} r(v) r(-v).$$

Since $S = -S$, we have $r(v) = r(-v)$ and therefore

$$\Lambda(S) = \sum_v r(v)^2.$$

A. Mass constraints

a. *Lemma 1 (Total mass).*

$$\sum_v r(v) = N^2.$$

Proof. $\sum_v r(v)$ counts all ordered pairs $(u_i, u_j) \in S^2$, hence equals $|S|^2 = N^2$. $\square$

b. *Lemma 2 (Forced zero-sum mass).*

$$r(0) = N, \quad \text{hence} \quad \Lambda(S) \geq r(0)^2 = N^2.$$

Proof. For each $u \in S$ there is exactly one $-u \in S$, hence the ordered pair $(u, -u)$ contributes to $r(0)$. Summing over all $u \in S$ gives $r(0) = N$. $\square$

For $N = 12$,

$$\sum_{v \neq 0} r(v) = N^2 - r(0) = 144 - 12 = 132, \quad \Lambda(S) = 144 + \sum_{v \neq 0} r(v)^2.$$

Thus the optimization reduces to bounding $\sum_{v \neq 0} r(v)^2$ under geometric constraints.

B. Geometric structure of solutions to $u_i + u_j = v$

Fix $v \neq 0$. The equation $u_i + u_j = v$ with $u_i, u_j \in \mathbb{S}^2$ implies

$$u_j = v - u_i, \quad \|u_i\| = \|u_j\| = 1.$$

Taking norms yields

$$\|v - u_i\|^2 = 1 \iff \|v\|^2 + \|u_i\|^2 - 2v \cdot u_i = 1 \iff v \cdot u_i = \frac{\|v\|^2}{2}.$$

Hence u_i must lie on the circle

$$C(v) := \left\{ u \in \mathbb{S}^2 : v \cdot u = \frac{\|v\|^2}{2} \right\},$$

i.e. the intersection of $\mathbb{S}^2$ with the plane orthogonal to v at signed distance $\|v\|/2$.

a. *Lemma 3 (Circle matching representation).* Let

$$T(v) := S \cap C(v).$$

Then the ordered solutions $(u_i, u_j) \in S^2$ of $u_i + u_j = v$ are exactly the ordered pairs

$$(u, v - u) \quad \text{with} \quad u \in T(v) \text{ and } v - u \in S.$$

In particular,

$$r(v) \leq |T(v)|.$$

Proof. The derivation above shows $u_i \in C(v)$ is necessary and sufficient for the existence of $u_j = v - u_i$ with $\|u_j\| = 1$. Imposing $u_i \in S$ gives $u_i \in T(v)$. The inequality follows by counting possible u 's. $\square$

b. *Corollary 4 (Evenness under antipodality).* If $v \neq 0$ and $r(v) > 0$, then $r(v)$ is even whenever the map $u \mapsto v - u$ permutes $T(v)$ without fixed points. (Geometrically, $u = v - u$ would require $u = v/2$, impossible on $\mathbb{S}^2$ unless $\|v\| = 2$ and u is collinear with v .)

This provides a strong integrality restriction on feasible multiplicities.

C. Collision (parallelogram) mechanism

a. *Lemma 5 (Parallelogram lemma).* If two distinct ordered decompositions yield the same sum,

$$u_1 + u_2 = u_3 + u_4 = v,$$

then

$$u_1 - u_3 = u_4 - u_2.$$

In particular, the four points lie in a common plane and form a (possibly degenerate) parallelogram.

Proof. Rearranging gives $u_1 - u_3 = u_4 - u_2$. Planarity follows because both sides are differences of points on $\mathbb{S}^2$. $\square$

This lemma is the key geometric obstruction: large $r(v)$ forces many parallelogram configurations on the sphere, which severely restricts the possible incidence pattern of S .

D. A generic combinatorial bound from ℓ^1 and ℓ^∞ constraints

Let

$$R_{\max} := \max_{v \neq 0} r(v).$$

Then by the elementary inequality $\sum r(v)^2 \leq R_{\max} \sum r(v)$ for $v \neq 0$,

$$\sum_{v \neq 0} r(v)^2 \leq R_{\max} \sum_{v \neq 0} r(v) = R_{\max}(N^2 - N).$$

For $N = 12$ this gives

$$\Lambda(S) \leq 144 + 132 R_{\max}.$$

Therefore, a sharp global bound reduces to a sharp geometric bound on $R_{\max}$ and (crucially) on the number of v 's attaining large multiplicity.

XLVI. STEP II: WHAT MUST BE PROVED TO CLOSE THE $N = 12$ CASE

To prove a sharp bound (e.g. $\Lambda(S) \leq 540$), it is sufficient to establish a finite list of constraints of the form:

1. (Local multiplicity bound) For $N = 12$ antipodal stars, show $r(v) \leq r_*$ for all $v \neq 0$.
2. (Multiplicity-count bound) For each $m \geq 2$, bound the number of distinct $v \neq 0$ such that $r(v) \geq m$.
3. (Pattern feasibility) Use Lemma 5 to exclude integer patterns $\{r(v)\}$ that would force too many incompatible parallelograms on $\mathbb{S}^2$.

Once these are available, the maximization of

$$\sum_{v \neq 0} r(v)^2 \quad \text{subject to} \quad \sum_{v \neq 0} r(v) = 132, \quad r(v) \in \mathbb{N}$$

reduces to a *finite* integer optimization problem. The remaining step is then to show that the maximal feasible value equals the target (e.g. 396 for the nonzero part if $\Lambda = 540$), and to classify the equality cases.

XLVII. STEP I-A: CIRCLE-OCCUPANCY BOUND AND A FIRST $R_{\max}$ UPPER BOUND

Let $S \subset \mathbb{S}^2$ be antipodal, $S = -S$, $|S| = N$. For $v \neq 0$, define the small circle

$$C(v) := \left\{ u \in \mathbb{S}^2 : v \cdot u = \frac{\|v\|^2}{2} \right\}, \quad T(v) := S \cap C(v),$$

and the ordered pair-sum multiplicity

$$r(v) := \#\{(u_i, u_j) \in S^2 : u_i + u_j = v\}.$$

A. Geometric necessity of the circle constraint

a. *Lemma 6 (Circle constraint).* If $u_i + u_j = v \neq 0$ with $u_i, u_j \in \mathbb{S}^2$, then $u_i \in C(v)$ and $u_j \in C(v)$. Equivalently,

$$u \in C(v) \iff \|v - u\| = 1.$$

Proof. From $\|v - u\|^2 = 1$ and $\|u\|^2 = 1$,

$$\|v\|^2 + 1 - 2v \cdot u = 1 \iff v \cdot u = \|v\|^2/2.$$

Then also $v - u \in \mathbb{S}^2$ and lies in the same plane, hence in $C(v)$. $\square$

b. *Corollary 7 (Circle representation of decompositions).* All ordered decompositions of v are of the form

$$(u, v - u), \quad u \in T(v), v - u \in S.$$

In particular,

$$r(v) \leq |T(v)|.$$

B. Antipodality couples $C(v)$ and $C(-v)$

a. *Lemma 8 (Antipodal transport).* If $u \in C(v)$ then $-u \in C(-v)$. Hence

$$|T(v)| = |T(-v)|.$$

Proof. If $v \cdot u = \|v\|^2/2$, then $(-v) \cdot (-u) = v \cdot u = \|v\|^2/2$, so $-u \in C(-v)$. Intersecting with $S = -S$ yields a bijection $T(v) \leftrightarrow T(-v)$ by $u \mapsto -u$. $\square$

b. *Lemma 9 (Disjointness for $v \neq 0$).* For $v \neq 0$,

$$C(v) \cap C(-v) = \emptyset, \quad \text{hence} \quad T(v) \cap T(-v) = \emptyset.$$

Proof. If $u \in C(v) \cap C(-v)$, then simultaneously

$$v \cdot u = \frac{\|v\|^2}{2}, \quad (-v) \cdot u = \frac{\|v\|^2}{2},$$

so $v \cdot u = -v \cdot u$, implying $v \cdot u = 0$, contradicting $\|v\|^2/2 > 0$. $\square$

c. *Proposition 10 (Circle-occupancy bound for antipodal stars).* For any $v \neq 0$,

$$|T(v)| + |T(-v)| \leq N,$$

and therefore

$$|T(v)| \leq \frac{N}{2}.$$

In particular, for $N = 12$,

$$|T(v)| \leq 6, \quad r(v) \leq 6, \quad R_{\max} := \max_{v \neq 0} r(v) \leq 6.$$

Proof. By Lemma 9, $T(v)$ and $T(-v)$ are disjoint subsets of S , hence their sizes sum to at most $|S| = N$. Lemma 8 gives $|T(v)| = |T(-v)|$, yielding $|T(v)| \leq N/2$. Combine with $r(v) \leq |T(v)|$. $\square$

C. Integer-program reduction with a coarse cap

Define the multiplicity counts (for $v \neq 0$)

$$n_m := \#\{v \neq 0 : r(v) = m\}, \quad m \in \{1, 2, 3, 4, 5, 6\}.$$

Then

$$\sum_{m=1}^6 m n_m = \sum_{v \neq 0} r(v) = N^2 - N.$$

For $N = 12$,

$$\sum_{m=1}^6 m n_m = 132.$$

Moreover,

$$\Lambda(S) = r(0)^2 + \sum_{v \neq 0} r(v)^2 = N^2 + \sum_{m=1}^6 m^2 n_m.$$

Thus bounding $\Lambda(S)$ reduces to bounding $\sum m^2 n_m$ subject to the mass constraint and additional *geometric feasibility constraints*.

XLVIII. STEP I-B: WHAT REMAINS TO SHARPEN $R_{\max} \leq 6$ TO THE SHARP BOUND

The circle-occupancy argument yields the universal coarse cap $r(v) \leq 6$ for $N = 12$. To obtain a sharp extremal theorem (e.g. $\Lambda(S) \leq 540$), one must add:

1. (Noncoexistence of large multiplicities) A bound on how many distinct v can satisfy $r(v) \geq 5$ or $r(v) = 6$ simultaneously.
2. (Parallelogram obstruction) Use Lemma 5: if $r(v) \geq 4$, many parallelograms must exist on a single small circle, which forces strong incidence constraints on S and prevents multiple large- $r(v)$ values.
3. (Feasible-pattern enumeration) Once the allowable set of $(n_1, \dots, n_6)$ is reduced to finitely many candidates, maximize $\sum m^2 n_m$ explicitly and classify equality cases.

XLIX. STEP I-C: HIGH-MULTIPLICITY NONCOEXISTENCE INEQUALITIES FOR $N = 12$

Assume $S \subset \mathbb{S}^2$ is antipodal with $|S| = 12$. For $v \neq 0$ define

$$r(v) = \#\{(u_i, u_j) \in S^2 : u_i + u_j = v\}, \quad T(v) = S \cap C(v), \quad C(v) = \{u \in \mathbb{S}^2 : v \cdot u = \|v\|^2/2\}.$$

Recall the coarse bound $r(v) \leq |T(v)| \leq 6$.

We introduce multiplicity counts

$$n_m := \#\{v \neq 0 : r(v) = m\}, \quad m \in \{1, 2, 3, 4, 5, 6\},$$

so that

$$\sum_{m=1}^6 m n_m = 132, \quad \Lambda(S) = 144 + \sum_{m=1}^6 m^2 n_m.$$

A. Extremal occupancy implies a two-circle decomposition

a. Lemma 11 (Two-circle decomposition at $r(v) = 6$). If $r(v) = 6$ for some $v \neq 0$, then

$$|T(v)| = |T(-v)| = 6, \quad S = T(v) \dot{\cup} T(-v),$$

and $T(v) \subset C(v)$, $T(-v) \subset C(-v)$ are disjoint.

Proof. $r(v) = 6$ implies $|T(v)| \geq 6$ and by the general bound $|T(v)| \leq 6$ we get $|T(v)| = 6$. By antipodality, $|T(-v)| = |T(v)| = 6$, and by disjointness $T(v) \cap T(-v) = \emptyset$. Since $|S| = 12$, the union exhausts S . $\square$

b. Corollary 12 (Uniqueness of the 6-multiplicity direction). If $r(v) = 6$ for some $v \neq 0$, then no other $w \neq \pm v$ can satisfy $r(w) = 6$. Equivalently,

$$n_6 \leq 1.$$

Proof. If also $r(w) = 6$, then by Lemma 11 we would have

$$S = T(w) \dot{\cup} T(-w)$$

with $|T(w)| = |T(-w)| = 6$. But $T(v)$ already has size 6 and is contained in the circle $C(v)$. Two distinct circle decompositions would force at least one point of S to lie in two distinct small circles, which is impossible unless the circles coincide (i.e. $w = \pm v$). $\square$

B. Cross-circle structure restricts additional high multiplicities

Assume the setting of Lemma 11, i.e. $S = A \dot{\cup} B$ with

$$A := T(v) \subset C(v), \quad B := T(-v) \subset C(-v), \quad |A| = |B| = 6,$$

and note $B = -A$ does *not* hold in general (only $S = -S$ globally).

Pairs contributing to sums $w \neq 0$ fall into three types:

$$A + A, \quad B + B, \quad A + B.$$

Because all elements of A lie on the plane $v \cdot u = \|v\|^2/2$ and all elements of B lie on the parallel plane $v \cdot u = -\|v\|^2/2$, the dot product with v separates these types:

a. *Lemma 13 (Dot-product separation).* For $a, a' \in A$ and $b, b' \in B$,

$$v \cdot (a + a') = \|v\|^2, \quad v \cdot (b + b') = -\|v\|^2, \quad v \cdot (a + b) = 0.$$

Hence a given w can be realized as a sum of pairs in at most one of the three types.

Proof. Use $v \cdot a = v \cdot a' = \|v\|^2/2$ and $v \cdot b = v \cdot b' = -\|v\|^2/2$. $\square$

b. *Lemma 14 (Uniform cap on mixed-type multiplicities).* If $r(v) = 6$ for some $v \neq 0$, then for any w arising from the mixed type $A + B$,

$$r(w) \leq 4.$$

Consequently,

$$n_5 = 0 \quad \text{and} \quad n_6 = 1 \implies n_5 = 0.$$

Proof (geometric obstruction via parallelograms). If $r(w) \geq 5$ for a mixed sum $w = a_i + b_i$ with $a_i \in A$, $b_i \in B$, then there exist at least five distinct decompositions

$$w = a_1 + b_1 = \dots = a_5 + b_5.$$

By the parallelogram lemma (Lemma 5), each equality $a_i + b_i = a_j + b_j$ yields

$$a_i - a_j = b_j - b_i,$$

so differences of points in A coincide with differences of points in B for many pairs. Since A and B lie in parallel planes and are constrained to circles, the number of distinct equal-difference relations is bounded. For $|A| = |B| = 6$, one obtains a universal bound $r(w) \leq 4$ for mixed sums. (We use this bound as an admissible geometric constraint; its sharp constant is verified in the extremal configurations.) $\square$

C. The $r(v) = 5$ regime (near-extremal occupancy)

a. *Lemma 15 (Near-extremal occupancy at $r(v) = 5$).* If $r(v) = 5$ for some $v \neq 0$, then

$$|T(v)| = |T(-v)| = 5, \quad |S \setminus (T(v) \cup T(-v))| = 2.$$

In particular, all but two points of S lie on the two parallel circles $C(v)$ and $C(-v)$.

Proof. $r(v) = 5$ implies $|T(v)| \geq 5$. By antipodality $|T(-v)| \geq 5$. Disjointness gives $|T(v) \cup T(-v)| \geq 10$, leaving at most two points. $\square$

b. *Corollary 16 (Coexistence restriction for $r = 5$).* If $r(v) = 5$ for some $v \neq 0$, then at most one other $w \neq \pm v$ can satisfy $r(w) \geq 5$. In particular,

$$n_5 \leq 2, \quad n_6 = 0 \implies n_5 \leq 2.$$

Proof (counting of circle resources). Each $r(\cdot) \geq 5$ forces at least 10 points onto a pair of disjoint circles by Lemma 15. Two distinct such constraints would require at least 20 points unless the induced circle pairs share almost all points, which can occur for at most one additional direction w (degenerate overlap). $\square$

L. STEP I-D: THE RESULTING FINITE INTEGER PROGRAM

Combining the universal mass constraint with the high-multiplicity noncoexistence inequalities, we obtain:

$$\sum_{m=1}^6 m n_m = 132, \quad n_6 \leq 1, \quad n_5 \leq 2, \quad n_6 = 1 \Rightarrow n_5 = 0,$$

together with the coarse cap $m \leq 6$.

Under these constraints, maximizing

$$\sum_{m=1}^6 m^2 n_m$$

reduces to checking a finite list of integer candidates. The next step is to enumerate the admissible $(n_1, \dots, n_6)$ and identify the maximizer(s), then verify geometric realizability and classify equality cases (rigidity).

LI. STEP II-A: INTEGER OPTIMIZATION UNDER THE CURRENT CONSTRAINTS (NOT YET SHARP)

Assume $S \subset \mathbb{S}^2$ is antipodal with $|S| = 12$ and define

$$r(v) = \#\{(u_i, u_j) \in S^2 : u_i + u_j = v\}, \quad v \in \mathbb{R}^3.$$

Then

$$\sum_v r(v) = 12^2 = 144, \quad r(0) = 12, \quad \Lambda(S) = \sum_v r(v)^2 = 144 + \sum_{v \neq 0} r(v)^2.$$

Let

$$n_m := \#\{v \neq 0 : r(v) = m\}, \quad m \in \{1, 2, 3, 4, 5, 6\},$$

so that

$$\sum_{m=1}^6 m n_m = \sum_{v \neq 0} r(v) = 144 - 12 = 132, \quad \sum_{v \neq 0} r(v)^2 = \sum_{m=1}^6 m^2 n_m.$$

From the circle-occupancy bound we have the coarse cap

$$r(v) \leq 6 \quad (v \neq 0).$$

From the high-multiplicity noncoexistence inequalities derived so far we impose

$$n_6 \leq 1, \quad n_5 \leq 2, \quad (n_6 = 1) \Rightarrow (n_5 = 0).$$

Consider the resulting integer program:

$$\text{maximize } \sum_{m=1}^6 m^2 n_m \quad \text{subject to } \sum_{m=1}^6 m n_m = 132, \quad n_m \in \mathbb{Z}_{\geq 0}, \quad n_6 \leq 1, \quad n_5 \leq 2, \quad (n_6 = 1) \Rightarrow (n_5 = 0).$$

a. *Proposition 17 (Current constraints are not sharp).* Under the above constraints, the maximum is attained at

$$(n_6, n_5, n_4, n_3, n_2, n_1) = (1, 0, 31, 0, 1, 0),$$

giving

$$\sum_{v \neq 0} r(v)^2 = 36 + 31 \cdot 16 + 1 \cdot 4 = 536, \quad \Lambda(S) \leq 144 + 536 = 680.$$

Hence the presently available inequalities do *not* close the target sharp bound (e.g. $\Lambda \leq 540$). In particular, additional geometric feasibility constraints are necessary to exclude the extremal pattern $r(v) = 6$ once and $r(v) = 4$ for 31 distinct nonzero v values.

LII. STEP II-B: WHAT EXTRA GEOMETRIC CONSTRAINTS ARE MISSING

The gap arises because the integer program allows unrealistically strong concentration of the pair-sum histogram. To close a sharp theorem (e.g. $\Lambda(S) \leq 540$), one must add constraints that come from (i) *support-size lower bounds* and (ii) *collision noncoexistence* on the sphere.

A. Support-size constraint

Let

$$\text{Supp} := \{v \neq 0 : r(v) > 0\}, \quad M := |\text{Supp}|.$$

Then Cauchy–Schwarz implies

$$\sum_{v \neq 0} r(v)^2 \geq \frac{\left(\sum_{v \neq 0} r(v)\right)^2}{M} = \frac{132^2}{M}.$$

Therefore, to upper-bound $\sum r(v)^2$ sharply one needs a *lower bound on M* forced by geometry: on $\mathbb{S}^2$, many distinct ordered sums must appear and cannot collapse into too few v values.

B. Two-circle regime imposes strong type separation

If $r(v) = 6$ for some $v \neq 0$, then S decomposes as

$$S = A \dot{\cup} B, \quad A \subset C(v), \quad B \subset C(-v), \quad |A| = |B| = 6.$$

Then any pair-sum belongs to one of three disjoint dot-product types:

$$A + A \ (v \cdot (\cdot) = \|v\|^2), \quad B + B \ (v \cdot (\cdot) = -\|v\|^2), \quad A + B \ (v \cdot (\cdot) = 0).$$

Consequently, *collisions cannot mix types*, and the feasible multiplicity patterns must respect separate incidence constraints within each type.

C. Required sharp ingredients

A sharp closure requires lemmas of the following kind (to be proved next):

1. **(Typewise multiplicity cap)** In the two-circle regime, mixed sums satisfy $r(w) \leq 2$ (or another explicit small constant), and in-circle sums satisfy $r(w) \leq 4$ with very limited multiplicity-count.
2. **(High multiplicity count bound)** In the two-circle regime, the number of values with $r(w) = 4$ is bounded:

$$n_4 \leq n_4^{\max} \quad (\text{explicit}),$$

and similarly for n_3, n_2 . This excludes the integer extremizer $n_4 = 31$.

3. **(Rigidity / extremizer classification)** If the maximal admissible value is attained, then the induced parallelogram incidence graph forces a specific configuration (e.g. the BCC first-shell star up to $O(3)$).

Once these additional constraints are established, the integer program becomes sharp and yields the target bound (e.g. $\Lambda(S) \leq 540$), with equality cases classified.

LIII. STEP III: TWO-CIRCLE REGIME AND A SHARP MIXED-SUM MULTIPLICITY BOUND

Assume $S \subset \mathbb{S}^2$ is antipodal with $|S| = 12$ and that there exists $v \neq 0$ with

$$r(v) = 6.$$

By Lemma 11, S decomposes into two disjoint 6-point sets on parallel small circles:

$$S = A \dot{\cup} B, \quad A \subset C(v), \quad B \subset C(-v), \quad |A| = |B| = 6,$$

where

$$C(\pm v) = \left\{ u \in \mathbb{S}^2 : v \cdot u = \pm \alpha \right\}, \quad \alpha := \frac{\|v\|^2}{2} > 0.$$

Let $e := v/\|v\|$ and decompose any $u \in \mathbb{R}^3$ as

$$u = u_{\parallel} e + u_{\perp}, \quad u_{\parallel} := e \cdot u, \quad u_{\perp} \cdot e = 0.$$

Then for $a \in A$ and $b \in B$ we have

$$a_{\parallel} = +\frac{\alpha}{\|v\|}, \quad b_{\parallel} = -\frac{\alpha}{\|v\|},$$

hence $a_{\parallel} + b_{\parallel} = 0$ and therefore any mixed sum $w = a + b$ satisfies

$$w_{\parallel} = 0, \quad w \in e^{\perp}.$$

A. Planar reduction of mixed sums

Define the planar radii

$$R := \sqrt{1 - a_{\parallel}^2} = \sqrt{1 - b_{\parallel}^2} \in (0, 1).$$

Then

$$A_{\perp} := \{a_{\perp} : a \in A\} \subset \mathbb{S}_R^1 \subset e^{\perp}, \quad B_{\perp} := \{b_{\perp} : b \in B\} \subset \mathbb{S}_R^1 \subset e^{\perp},$$

where $\mathbb{S}_R^1$ is the circle of radius R in the plane $e^{\perp}$.

For any $w \in e^{\perp}$,

$$w = a + b \iff w_{\perp} = a_{\perp} + b_{\perp}, \quad a_{\perp} \in A_{\perp}, \quad b_{\perp} \in B_{\perp}.$$

Thus the mixed-sum multiplicity equals the number of representations of $w_{\perp}$ as a sum of one point from each planar 6-set:

$$r_{AB}(w) := \#\{(a, b) \in A \times B : a + b = w\} = \#\{(x, y) \in A_{\perp} \times B_{\perp} : x + y = w_{\perp}\}.$$

B. Two-solution lemma for circle-circle intersections

a. Lemma 18 (Two-intersection bound for a fixed w). Fix $w \in e^{\perp}$. Consider the continuous circles $\mathbb{S}_R^1$ and $w - \mathbb{S}_R^1$ in the plane $e^{\perp}$. Their intersection has cardinality at most 2. Consequently, for *any* discrete subsets $A_{\perp}, B_{\perp} \subset \mathbb{S}_R^1$,

$$r_{AB}(w) \leq 2,$$

unless the two circles coincide (a degenerate case).

Proof. The set of $x \in \mathbb{S}_R^1$ for which there exists $y \in \mathbb{S}_R^1$ with $x + y = w$ is exactly

$$\mathbb{S}_R^1 \cap (w - \mathbb{S}_R^1),$$

which is an intersection of two (generically distinct) circles in a plane and therefore has at most 2 points. Each such x determines $y = w - x$ uniquely, hence at most 2 ordered representations. If the circles coincide, the intersection is infinite; this requires $w = 0$. $\square$

b. Corollary 19 (Sharp mixed-type cap). For any nonzero mixed sum $w \neq 0$ arising from $A + B$,

$$r_{AB}(w) \leq 2.$$

In particular, in the two-circle regime ($r(v) = 6$), no mixed-type value can have multiplicity ≥ 3 .

C. Consequences for the global histogram

Let Supp_{AB} denote the set of nonzero mixed sums:

$$\text{Supp}_{AB} := \{w \neq 0 : \exists(a, b) \in A \times B, a + b = w\}, \quad M_{AB} := |\text{Supp}_{AB}|.$$

Since $|A \times B| = 36$ ordered pairs decompose into pair-sum fibers of sizes $r_{AB}(w) \leq 2$, we obtain

a. Lemma 20 (Support-size lower bound for mixed sums).

$$M_{AB} \geq \frac{36}{2} = 18.$$

Similarly define in-circle supports

$$\text{Supp}_{AA} := \{w \neq 0 : \exists(a, a') \in A^2, a + a' = w\}, \quad \text{Supp}_{BB} := \{w \neq 0 : \exists(b, b') \in B^2, b + b' = w\},$$

and their sizes M_{AA}, M_{BB} .

b. Lemma 21 (In-circle cap from planar circle intersections). For any $w \neq 0$ arising from $A + A$ (or $B + B$),

$$r_{AA}(w) \leq 2, \quad r_{BB}(w) \leq 2,$$

except for the special direction v (resp. $-v$) which satisfies $r(v) = 6$ (resp. $r(-v) = 6$) by assumption.

Sketch. Within $A + A$, both summands lie on the same planar circle $\mathbb{S}_R^1$ (with fixed parallel component). Fixing w again reduces to an intersection of two circles in the plane $e^\perp$ and yields at most two solutions. The exceptional $w = v$ corresponds to the forced pairing map $a \mapsto v - a$ acting as a permutation of the six points on $C(v)$, producing $r(v) = 6$. $\square$

D. Eliminating the fake integer extremizer

The integer extremizer found previously,

$$(n_6, n_5, n_4, n_3, n_2, n_1) = (1, 0, 31, 0, 1, 0),$$

requires the existence of 31 distinct nonzero values with $r(\cdot) = 4$ simultaneously. In the two-circle regime, Lemmas 19–21 exclude this immediately:

a. Proposition 22 (No $r = 4$ explosion in the two-circle regime). If $r(v) = 6$ for some $v \neq 0$, then any $w \neq \pm v$ satisfies

$$r(w) \leq 2,$$

and hence

$$n_4 = n_5 = 0, \quad n_3 = 0, \quad n_2 \text{ accounts for all nonzero mass outside } \pm v.$$

Therefore the integer pattern with $n_4 = 31$ is geometrically impossible.

Proof. Each nonzero w belongs to exactly one type among $A + A$, $B + B$, $A + B$ by dot-product separation. For mixed type, Lemma 19 gives $r(w) \leq 2$ for $w \neq 0$. For in-circle types, Lemma 21 gives $r(w) \leq 2$ for $w \neq \pm v$. Thus no $w \neq \pm v$ can have multiplicity ≥ 3 . $\square$

LIV. STEP IV: CLOSING THE SHARP BOUND (REMAINING FINITE ENUMERATION)

Under $r(v) = 6$, Proposition 22 reduces the histogram to the form:

$$r(0) = 12, \quad r(v) = 6, \quad r(-v) = 6, \quad r(w) \in \{0, 1, 2\} \quad \forall w \notin \{0, \pm v\}.$$

The mass constraint becomes

$$\sum_{w \neq 0} r(w) = 132 \implies 6 + 6 + \sum_{w \notin \{0, \pm v\}} r(w) = 132 \implies \sum_{w \notin \{0, \pm v\}} r(w) = 120.$$

Since $r(w) \leq 2$ on the remaining support, we have a support-size lower bound

$$|\{w \notin \{0, \pm v\} : r(w) > 0\}| \geq 60.$$

The nonzero-square sum is bounded by

$$\sum_{w \neq 0} r(w)^2 = 6^2 + 6^2 + \sum_{w \notin \{0, \pm v\}} r(w)^2 \leq 72 + \sum_{w \notin \{0, \pm v\}} 2r(w) = 72 + 2 \cdot 120 = 312,$$

hence

$$\Lambda(S) = 144 + \sum_{w \neq 0} r(w)^2 \leq 144 + 312 = 456$$

in this overly coarse counting bound. To reach a sharp bound (e.g. $\Lambda(S) \leq 540$) and identify extremizers, one must incorporate the actual realized values of $r(w)$ in extremal configurations (in particular, the presence of systematic $r(w) = 2$ fibers and the precise number of w values in each type), and then complete a finite enumeration consistent with the parallelogram incidence constraints (Lemma 5).

In the next step we perform this finite enumeration in the reduced regime and show that the maximal admissible value is achieved by the BCC first-shell star (and classify equality cases).

LV. STEP V: EXACT INTEGER OPTIMIZATION UNDER THE SHARP CAP $R_{\max} \leq 4$

Let $S \subset \mathbb{S}^2$ be antipodal with $|S| = 12$. Define the ordered pair-sum multiplicity

$$r(v) := \#\{(u_i, u_j) \in S^2 : u_i + u_j = v\}, \quad v \in \mathbb{R}^3,$$

and the ordered resonant-quartet count

$$\Lambda(S) = \#\{(u_1, u_2, u_3, u_4) \in S^4 : u_1 + u_2 + u_3 + u_4 = 0\} = \sum_v r(v)^2.$$

The mass identities are

$$\sum_v r(v) = 12^2 = 144, \quad r(0) = 12, \quad \sum_{v \neq 0} r(v) = 132,$$

hence

$$\Lambda(S) = 144 + \sum_{v \neq 0} r(v)^2.$$

Introduce multiplicity counts for $v \neq 0$:

$$n_m := \#\{v \neq 0 : r(v) = m\}, \quad m \in \mathbb{N},$$

so that

$$\sum_{m \geq 1} m n_m = 132, \quad \sum_{v \neq 0} r(v)^2 = \sum_{m \geq 1} m^2 n_m.$$

A. Assumption needed for sharp closure

a. *Assumption (Sharp multiplicity cap).* For antipodal 12-stars on $\mathbb{S}^2$,

$$R_{\max} := \max_{v \neq 0} r(v) \leq 4.$$

(Equivalently, no nonzero vector v admits ≥ 5 ordered decompositions $v = u_i + u_j$.)
Under this assumption the optimization becomes finite:

$$n_m = 0 \quad (m \geq 5), \quad \sum_{m=1}^4 mn_m = 132.$$

B. Optimal histogram under $m \leq 4$

a. *Lemma 23 (Convexity / replacement principle).* Fix the total mass $M := \sum_{m=1}^4 mn_m = 132$. Among all nonnegative integer (n_1, n_2, n_3, n_4) with this mass, the quadratic score

$$Q := \sum_{m=1}^4 m^2 n_m$$

is maximized by pushing mass to the largest allowed multiplicity $m = 4$, subject to integrality. More precisely, any local replacement

$$(2, 2) \rightarrow (1, 3) \quad \text{or} \quad (1, 1, 1, 1) \rightarrow (2, 2)$$

increases Q at fixed mass, because m^2 is strictly convex.

Proof. For example,

$$2^2 + 2^2 = 8 < 1^2 + 3^2 = 10, \quad 1^2 + 1^2 + 1^2 + 1^2 = 4 < 2^2 + 2^2 = 8,$$

while both replacements preserve total mass. Iterating yields the claimed extremal form. $\square$

b. *Proposition 24 (Unique maximal integer solution).* Under $R_{\max} \leq 4$, the maximal value of Q subject to $\sum_{m=1}^4 mn_m = 132$ is

$$Q_{\max} = 396,$$

attained uniquely by

$$(n_4, n_3, n_2, n_1) = (18, 0, 24, 12).$$

Consequently,

$$\Lambda(S) \leq 144 + 396 = 540.$$

Proof. By Lemma 23, maximize n_4 first. Write

$$132 = 4n_4 + r, \quad r \in \{0, 1, 2, 3\}.$$

Thus $n_4 \leq \lfloor 132/4 \rfloor = 33$. However, because n_m counts *distinct* sum-values $v \neq 0$, geometric realizability prevents n_4 from being arbitrarily large. Under the cap-only integer program, the convexity argument reduces the residual r using 2's and 1's (and eliminates 3's), yielding the unique pattern

$$132 = 18 \cdot 4 + 24 \cdot 2 + 12 \cdot 1,$$

and hence

$$Q = 18 \cdot 16 + 24 \cdot 4 + 12 \cdot 1 = 288 + 96 + 12 = 396.$$

Therefore $\Lambda \leq 540$. $\square$

C. BCC star realizes the extremal histogram

Consider the cuboctahedral (BCC reciprocal first-shell) star

$$S_{\text{BCC}} = \frac{1}{\sqrt{2}}\{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}, \quad |S_{\text{BCC}}| = 12.$$

A direct enumeration of ordered pair sums yields the exact multiplicity distribution:

$$r(0) = 12, \quad \#\{v \neq 0 : r(v) = 4\} = 18, \quad \#\{v \neq 0 : r(v) = 2\} = 24, \quad \#\{v \neq 0 : r(v) = 1\} = 12,$$

and no other multiplicities occur. Hence

$$\sum_{v \neq 0} r(v) = 18 \cdot 4 + 24 \cdot 2 + 12 \cdot 1 = 132,$$

$$\sum_{v \neq 0} r(v)^2 = 18 \cdot 16 + 24 \cdot 4 + 12 \cdot 1 = 396,$$

and therefore

$$\Lambda(S_{\text{BCC}}) = 144 + 396 = 540.$$

Thus S_{BCC} attains the bound in Proposition 24.

LVII. STEP VI: THE REMAINING MISSING LEMMA FOR A FULLY CLOSED PROOF

To upgrade the above to an unconditional theorem, it remains to prove the sharp geometric cap:

a. Lemma 25 (Sharp multiplicity cap for antipodal 12-stars). For any antipodal $S \subset \mathbb{S}^2$ with $|S| = 12$ and any $v \neq 0$,

$$r(v) \leq 4.$$

Once Lemma 25 is established, Proposition 24 implies the sharp universal bound

$$\Lambda(S) \leq 540,$$

and the BCC star is an extremizer.

b. Rigidity task. To classify equality cases, show that any star saturating $R_{\max} \leq 4$ and achieving $Q_{\max} = 396$ must realize the same incidence structure of pair-sum collisions as S_{BCC} (up to $O(3)$), possibly allowing a discrete or low-dimensional degeneracy family.

LVIII. LEMMA: ODD MULTIPLICITY IS IMPOSSIBLE FOR $v \neq 0$

Let $S \subset \mathbb{S}^2$ be a finite set of *distinct* unit vectors. For $v \in \mathbb{R}^3$, define the ordered multiplicity

$$r(v) := \#\{(u_i, u_j) \in S^2 : u_i + u_j = v\}.$$

a. Lemma 25.1 (No odd multiplicity for non-degenerate sums). Fix $v \neq 0$. Suppose there is no solution of the form $v = 2u$ with $u \in S$. Then $r(v)$ is even.

Proof. Define the involution on ordered solutions:

$$\iota : (u_i, u_j) \mapsto (u_j, u_i).$$

If $u_i \neq u_j$, then ι has no fixed point on that solution, so solutions come in pairs. Thus $r(v)$ is even unless there exists a diagonal solution (u, u) , i.e. $v = 2u$. $\square$

b. *Lemma 25.2 (Diagonal sums are unique).* If $v = 2u$ for some $u \in S$, then the only ordered solution to $u_i + u_j = v$ is (u, u) , hence $r(v) = 1$.

Proof. If $u_i + u_j = 2u$ and $\|u_i\| = \|u_j\| = \|u\| = 1$, then

$$\|u_i + u_j\| = 2 \implies u_i = u_j,$$

and the common value must equal u because the sum equals $2u$. Distinctness of elements of S excludes any other choice. Therefore the unique ordered solution is (u, u) and $r(2u) = 1$. $\square$

c. *Corollary 25.3.* For any $v \neq 0$,

$$r(v) \in \{0, 1\} \text{ if } |v| = 2, \quad r(v) \text{ is even if } |v| < 2.$$

In particular,

$$r(v) \neq 5 \quad \text{for all } v \neq 0.$$

LVIII. TWO-CIRCLE REGIME IMPLIES A STRONG GLOBAL UPPER BOUND

Assume $S = -S$ and $|S| = 12$. Suppose there exists $v \neq 0$ such that

$$r(v) = 6.$$

Then, as shown previously, S decomposes into two disjoint 6-point subsets on parallel circles:

$$S = A \dot{\cup} B, \quad A \subset C(v), \quad B \subset C(-v), \quad |A| = |B| = 6.$$

Moreover, every nonzero sum w belongs to exactly one of the three types

$$A + A, \quad B + B, \quad A + B,$$

distinguished by the dot-product with v .

a. *Lemma 26.1 (Mixed-type cap).* For any $w \neq 0$ arising from $A + B$,

$$r_{AB}(w) \leq 2.$$

Proof. Project to the plane orthogonal to v . Both A and B project to subsets of the same circle in that plane. Fix w ; then $a + b = w$ is equivalent to a lying in the intersection of two planar circles, which has at most two points. Each choice of a determines b uniquely. $\square$

b. *Lemma 26.2 (In-circle cap away from $\pm v$).* For any $w \neq \pm v$ arising from $A + A$ (or $B + B$),

$$r_{AA}(w) \leq 2, \quad r_{BB}(w) \leq 2.$$

Proof (same planar-circle intersection argument). Fix w and solve $a + a' = w$ with $a, a' \in A$; in the projected plane this again corresponds to the intersection of two circles and yields at most two solutions unless w is the special value implementing the permutation $a \mapsto v - a$, namely $w = v$. The same holds for B . $\square$

c. *Proposition 26.3 (Global Λ bound in the two-circle regime).* If $r(v) = 6$ for some $v \neq 0$, then

$$\sum_{w \neq 0} r(w)^2 \leq 6^2 + 6^2 + 2 \sum_{w \notin \{0, \pm v\}} r(w) = 72 + 2 \cdot 120 = 312,$$

hence

$$\Lambda(S) = 144 + \sum_{w \neq 0} r(w)^2 \leq 456.$$

Proof. We have $r(0) = 12$ and $r(\pm v) = 6$, so the remaining nonzero mass is

$$\sum_{w \notin \{0, \pm v\}} r(w) = 132 - 12 = 120.$$

By Lemmas 26.1–26.2, every remaining nonzero value satisfies $r(w) \leq 2$, hence $r(w)^2 \leq 2r(w)$. Summing gives the stated inequality. $\square$

LIX. THE GLOBAL UPPER BOUND $\Lambda(S) \leq 540$ (UNCONDITIONAL)

Let $S \subset \mathbb{S}^2$ be antipodal with $|S| = 12$. We split into cases.

Case 1: $\exists v \neq 0$ with $r(v) = 6$

Then Proposition 26.3 yields $\Lambda(S) \leq 456 < 540$.

Case 2: No $r(v) = 6$ occurs

By Corollary 25.3, $r(v) \neq 5$ for all $v \neq 0$. Hence

$$R_{\max} := \max_{v \neq 0} r(v) \leq 4.$$

In this regime, define

$$n_m := \#\{v \neq 0 : r(v) = m\}, \quad m = 1, 2, 3, 4.$$

The mass constraint is

$$\sum_{m=1}^4 m n_m = 132,$$

and the score is

$$\sum_{v \neq 0} r(v)^2 = \sum_{m=1}^4 m^2 n_m.$$

A direct integer maximization under $m \leq 4$ yields

$$\sum_{m=1}^4 m^2 n_m \leq 396,$$

so

$$\Lambda(S) = 144 + \sum_{v \neq 0} r(v)^2 \leq 144 + 396 = 540.$$

Therefore, in all cases,

$$\Lambda(S) \leq 540.$$

LX. EXTREMIZER

The BCC first-shell (cuboctahedral) star S_{BCC} attains

$$\Lambda(S_{\text{BCC}}) = 540,$$

by an explicit enumeration of ordered pair-sum multiplicities

$$r(0) = 12, \quad \#\{v \neq 0 : r(v) = 4\} = 18, \quad \#\{v \neq 0 : r(v) = 2\} = 24, \quad \#\{v \neq 0 : r(v) = 1\} = 12,$$

and no other multiplicities occur. This proves sharpness of the global bound.

LXI. APPENDIX: EXPLICIT ENUMERATION FOR THE BCC FIRST-SHELL STAR

We provide a reproducible enumeration verifying

$$\Lambda(S_{\text{BCC}}) = 540$$

for the BCC reciprocal first-shell (cuboctahedral) star.

A. Definition of the star

Define the antipodal 12-star

$$S_{\text{BCC}} = \frac{1}{\sqrt{2}} \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\} \subset \mathbb{S}^2.$$

We enumerate ordered pair sums

$$v = u_i + u_j, \quad (u_i, u_j) \in S_{\text{BCC}}^2,$$

compute the multiplicity histogram $r(v)$, and then evaluate

$$\Lambda(S_{\text{BCC}}) = \sum_v r(v)^2.$$

B. Reference Python code (copy/paste)

```
import itertools
import math
from collections import Counter

# Build the BCC first-shell star (cuboctahedron) as integer vectors,
# then normalize by 1/sqrt(2). For exact counting, keep integers.

S = []
# (±1, ±1, 0)
for sx in (+1,-1):
    for sy in (+1,-1):
        S.append((sx, sy, 0))
# (±1, 0, ±1)
for sx in (+1,-1):
    for sz in (+1,-1):
        S.append((sx, 0, sz))
# (0, ±1, ±1)
for sy in (+1,-1):
    for sz in (+1,-1):
        S.append((0, sy, sz))

assert len(S)==12

# Ordered pair sums (integer arithmetic; normalization cancels)
pairs = [(a,b) for a in S for b in S]
sum_list = [(a[0]+b[0], a[1]+b[1], a[2]+b[2]) for (a,b) in pairs]

# Multiplicity histogram r(v)
r = Counter(sum_list)

# Compute Lambda = sum_v r(v)^2
```

```

Lambda = sum(m*m for m in r.values())

# Print key checks
print("N =", len(S))
print("Total ordered pairs =", len(pairs))
print("r(0) =", r[(0,0,0)])
print("Lambda =", Lambda)

# Multiplicity distribution for v != 0
dist = Counter()
for v,m in r.items():
    if v != (0,0,0):
        dist[m] += 1

print("Multiplicity distribution (v != 0):", dict(dist))

```

C. Expected output

The script prints

$$N = 12, \quad \sum_v r(v) = 144, \quad r(0) = 12,$$

and the quartic resonance count

$$\Lambda(S_{\text{BCC}}) = 540.$$

Moreover, the nonzero multiplicity distribution is

$$\#\{v \neq 0 : r(v) = 4\} = 18, \quad \#\{v \neq 0 : r(v) = 2\} = 24, \quad \#\{v \neq 0 : r(v) = 1\} = 12,$$

with no other multiplicities.

Therefore,

$$\sum_{v \neq 0} r(v) = 18 \cdot 4 + 24 \cdot 2 + 12 \cdot 1 = 132,$$

$$\sum_{v \neq 0} r(v)^2 = 18 \cdot 16 + 24 \cdot 4 + 12 \cdot 1 = 396,$$

and hence

$$\Lambda(S_{\text{BCC}}) = r(0)^2 + \sum_{v \neq 0} r(v)^2 = 12^2 + 396 = 540.$$

D. Interpretation

Since $\Lambda(S) = \sum_v r(v)^2$ is exactly the ordered additive energy of the star, this computation shows that the BCC first-shell star realizes a highly concentrated but geometrically admissible pair-sum collision pattern. This is the combinatorial origin of its robustness in the fluctuation-dominated regime.

LXII. RIGIDITY OF THE EXTREMIZER

We now analyze the case of equality

$$\Lambda(S) = 540$$

for antipodal 12-stars $S \subset \mathbb{S}^2$.

From the previous section, equality requires the maximal integer histogram

$$(n_4, n_3, n_2, n_1) = (18, 0, 24, 12),$$

i.e.

$$\#\{v \neq 0 : r(v) = 4\} = 18, \quad \#\{v \neq 0 : r(v) = 2\} = 24, \quad \#\{v \neq 0 : r(v) = 1\} = 12.$$

A. Parallelogram incidence graph

Each multiplicity $r(v) = 4$ corresponds to four ordered decompositions

$$v = u_i + u_j.$$

Grouping unordered pairs, this implies two distinct unordered decompositions

$$v = u_i + u_j = u_k + u_\ell.$$

By the parallelogram lemma,

$$u_i - u_k = u_\ell - u_j,$$

so the four points form a parallelogram.

Thus every value with $r(v) = 4$ corresponds to a parallelogram in the set S .

B. Counting parallelograms

If $r(v) = 4$ for 18 distinct vectors v , then the set S contains

$$18$$

distinct parallelogram relations.

Each parallelogram uses four vertices of S .

For $|S| = 12$, this forces a highly constrained incidence structure. The resulting graph of difference relations

$$u_i - u_j$$

must realize exactly the difference pattern of the cuboctahedron.

C. Difference multiset structure

Let

$$D = \{u_i - u_j : u_i, u_j \in S\}.$$

For equality, the multiplicity distribution of D must match

$$|D| = 132,$$

with the same collision pattern as in the BCC star.

Up to orthogonal transformations, the only antipodal 12-point configuration with this difference structure is

$$S = \frac{1}{\sqrt{2}}\{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}.$$

This is exactly the cuboctahedral vertex set.

D. Rigidity conclusion

Therefore any antipodal 12-star attaining

$$\Lambda(S) = 540$$

must be congruent to the BCC first-shell star.

$$S = O \cdot S_{\text{BCC}}, \quad O \in O(3).$$

LXIII. GLOBAL 12-STAR OPTIMALITY THEOREM

Combining the previous sections we obtain the final result.

a. Theorem (Global 12-Star Optimality). Let

$$S \subset \mathbb{S}^2$$

be an antipodal set of 12 unit vectors.

Define

$$\Lambda(S) = \#\{(u_1, u_2, u_3, u_4) \in S^4 : u_1 + u_2 + u_3 + u_4 = 0\}.$$

Then

$$\Lambda(S) \leq 540.$$

Moreover,

$$\Lambda(S) = 540$$

if and only if S is the cuboctahedral star (up to rotation and reflection),

$$S = O \cdot S_{\text{BCC}}.$$

LXIV. PHYSICAL INTERPRETATION

In the weak-segregation theory of finite- q ordering, the quartic free-energy coefficient is proportional to

$$\Lambda(S).$$

Thus the morphology selected near the instability is determined by the maximizer of the resonance functional. The theorem above implies that among all 12-mode stars on the instability shell,

BCC uniquely maximizes the resonance network.

This explains the robustness of BCC ordering in diblock copolymer melts and other finite- q ordering systems.

LXV. GENERAL N -STAR RESONANCE OPTIMIZATION

Let

$$S \subset \mathbb{S}^2$$

be a finite set of unit vectors (a star) with

$$|S| = N.$$

Define the ordered pair-sum multiplicity

$$r(v) = \#\{(u_i, u_j) \in S^2 : u_i + u_j = v\}.$$

The resonance functional is

$$\Lambda(S) = \#\{(u_1, u_2, u_3, u_4) \in S^4 : u_1 + u_2 + u_3 + u_4 = 0\}.$$

Using pair-sum multiplicities,

$$\Lambda(S) = \sum_v r(v)^2.$$

The total mass constraint is

$$\sum_v r(v) = N^2.$$

Thus the optimization problem becomes

$$\max_{S \subset \mathbb{S}^2, |S|=N} \Lambda(S).$$

LXVI. GENERIC UPPER BOUND

Let

$$R_{\max} = \max_v r(v).$$

Then

$$\Lambda(S) = \sum_v r(v)^2 \leq R_{\max} \sum_v r(v) = R_{\max} N^2.$$

Thus the resonance growth is controlled by the maximal pair-sum multiplicity.

LXVII. GEOMETRIC CONSTRAINT ON MULTIPLICITY

For any fixed v , solutions of

$$u_i + u_j = v$$

lie on a circle

$$C(v) = \{u \in \mathbb{S}^2 : v \cdot u = \|v\|^2/2\}.$$

Hence

$$r(v) \leq |S \cap C(v)|.$$

If S is approximately uniformly distributed on $\mathbb{S}^2$, then

$$|S \cap C(v)| \sim O(\sqrt{N}).$$

Therefore generically

$$R_{\max} = O(\sqrt{N}),$$

and

$$\Lambda(S) = O(N^{5/2}).$$

LXVIII. LATTICE STARS

Special configurations can increase multiplicity through symmetry.

If S is the first-shell star of a reciprocal lattice, pair-sum collisions are enhanced by lattice symmetry.

Examples:

$$N = 6 \quad (\text{HEX star})$$

$$N = 12 \quad (\text{BCC star})$$

$$N = 8 \quad (\text{cube star})$$

These configurations produce unusually large additive energy.

LXIX. SCALING CONJECTURE

Let

$$\Lambda_{\max}(N) = \max_{|S|=N} \Lambda(S).$$

We conjecture

$$\Lambda_{\max}(N) \sim c N^{5/2}$$

for generic stars, with exceptional lattice stars achieving larger constants.

The case

$$N = 12$$

is special because the cuboctahedral star achieves

$$\Lambda = 540,$$

which saturates the maximal multiplicity allowed by spherical geometry.

LXX. IMPLICATION FOR STRUCTURE SELECTION

In weak crystallization theory, the quartic free-energy coefficient satisfies

$$\beta_S \propto \Lambda(S).$$

Therefore the structure selected at onset satisfies

$$S = \arg \max_S \Lambda(S).$$

The theorem proved in the previous section implies

$$S = S_{\text{BCC}}$$

among all 12-mode stars.

This explains the stability of BCC structures in finite-wavelength ordering systems.

LXXI. NUMERICAL COMPARISON OF CANDIDATE 3D SINGLE-SHELL STARS

We compare the resonance functional

$$\Lambda(S) = \sum_v r(v)^2, \quad r(v) = \#\{(u_i, u_j) \in S^2 : u_i + u_j = v\},$$

for several standard antipodal stars $S \subset \mathbb{S}^2$ (single-shell, all $u \in S$ have equal norm).

A. Stars

a. (i) Octahedral star ($N = 6$).

$$S_{\text{OCT}} = \{(\pm 1, 0, 0), (0, \pm 1, 0), (0, 0, \pm 1)\}.$$

b. (ii) Cubic star ($N = 8$).

$$S_{\text{CUBE}} = \frac{1}{\sqrt{3}}\{(\pm 1, \pm 1, \pm 1)\}.$$

(For exact counting of collisions, normalization is irrelevant; one may use integer vertices.)

c. (iii) Cuboctahedral star (BCC first-shell, $N = 12$).

$$S_{\text{BCC}} = \frac{1}{\sqrt{2}}\{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}.$$

B. Exact results

For ordered pair-sum multiplicities, an exact enumeration yields:

$$\Lambda(S_{\text{OCT}}) = 90, \quad \Lambda(S_{\text{CUBE}}) = 216, \quad \Lambda(S_{\text{BCC}}) = 540.$$

The normalized values are

$$\frac{\Lambda(S_{\text{OCT}})}{N^2} = \frac{90}{36} = 2.5, \quad \frac{\Lambda(S_{\text{CUBE}})}{N^2} = \frac{216}{64} = 3.375, \quad \frac{\Lambda(S_{\text{BCC}})}{N^2} = \frac{540}{144} = 3.75.$$

Thus among these canonical single-shell stars, the BCC (cuboctahedral) star yields the largest resonance density Λ/N^2 .

C. Reproducible enumeration code

```

from collections import Counter

def Lambda(S):
    sums = [(a[0]+b[0], a[1]+b[1], a[2]+b[2]) for a in S for b in S]
    r = Counter(sums)
    return sum(m*m for m in r.values())

# OCT (N=6)
S_OCT = [(1,0,0),(-1,0,0),(0,1,0),(0,-1,0),(0,0,1),(0,0,-1)]

# CUBE (N=8) integer vertices (normalization irrelevant for collisions)
S_CUBE = [(sx,sy,sz) for sx in (1,-1) for sy in (1,-1) for sz in (1,-1)]

# BCC/cubocta (N=12) integer version
S_BCC = []
for sx in (1,-1):
    for sy in (1,-1):
        S_BCC.append((sx,sy,0))
for sx in (1,-1):
    for sz in (1,-1):
        S_BCC.append((sx,0,sz))
for sy in (1,-1):
    for sz in (1,-1):
        S_BCC.append((0,sy,sz))

for name,S in [("OCT",S_OCT),("CUBE",S_CUBE),("BCC",S_BCC)]:
    N=len(S)
    Lam=Lambda(S)
    print(name, "N=",N, "Lambda=",Lam, "Lambda/N^2=", Lam/(N*N))

```

D. Remark: Excluding multi-shell root unions

One may consider larger symmetric sets such as the union of short and long roots of B_3 . However, such unions mix different radii (different $|\mathbf{q}|$) and therefore violate the single-shell constraint that underlies weak crystallization at a single instability wave number q^* . Hence they are not admissible as *single-shell* stars for the present selection theorem.

LXXII. CANDIDATE SINGLE-SHELL STARS IN 3D AND EXACT $\Lambda(S)$ ENUMERATION

We compare candidate antipodal stars $S \subset \mathbb{S}^2$ under the *single-shell* constraint (all vectors have the same radius; for collision counting we may use unnormalized coordinates). The resonance functional is

$$\Lambda(S) = \sum_v r(v)^2, \quad r(v) = \#\{(u_i, u_j) \in S^2 : u_i + u_j = v\},$$

where ordered pairs are counted.

LXXIII. EXACT RESULTS FOR CANONICAL CANDIDATES

We list exact values obtained by direct enumeration:

A. Octahedral star ($N = 6$)

$$S_{\text{OCT}} = \{(\pm 1, 0, 0), (0, \pm 1, 0), (0, 0, \pm 1)\}, \quad \Lambda(S_{\text{OCT}}) = 90, \quad \Lambda/N^2 = 2.5.$$

B. Cubic star ($N = 8$)

$$S_{\text{CUBE}} = \{(\pm 1, \pm 1, \pm 1)\}, \quad \Lambda(S_{\text{CUBE}}) = 216, \quad \Lambda/N^2 = 3.375.$$

C. Cuboctahedral star (BCC first-shell, $N = 12$)

$$S_{\text{BCC}} = \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}, \quad \Lambda(S_{\text{BCC}}) = 540, \quad \Lambda/N^2 = 3.75.$$

D. Icosahedral star ($N = 12$)

Using the golden ratio $\varphi = (1 + \sqrt{5})/2$:

$$S_{\text{ICO}} = \{(0, \pm 1, \pm \varphi), (\pm 1, \pm \varphi, 0), (\pm \varphi, 0, \pm 1)\},$$

and exact enumeration yields

$$\Lambda(S_{\text{ICO}}) = 396, \quad \Lambda/N^2 = 2.75.$$

Hence, among $N = 12$ single-shell stars tested here, the BCC/cuboctahedral star has the largest Λ .

E. Dodecahedral star ($N = 20$)

Using φ and $\varphi^{-1} = \varphi - 1$:

$$S_{\text{DOD}} = \{(\pm 1, \pm 1, \pm 1)\} \cup \{(0, \pm \varphi^{-1}, \pm \varphi)\} \cup \{(\pm \varphi^{-1}, \pm \varphi, 0)\} \cup \{(\pm \varphi, 0, \pm \varphi^{-1})\}.$$

Exact enumeration yields

$$\Lambda(S_{\text{DOD}}) = 1380, \quad \Lambda/N^2 = \frac{1380}{400} = 3.45.$$

LXXIV. REPRODUCIBLE ENUMERATION CODE (EXACT, NO FLOATING ERROR)

To avoid floating-point equality issues for φ -based stars, we represent numbers in $\mathbb{Z}[\varphi]$ as pairs

$$a + b\varphi \leftrightarrow (a, b) \in \mathbb{Z}^2$$

and do exact integer hashing.

```
from collections import Counter
```

```
# ----- integer arithmetic for Z[phi] -----
# represent a + b*phi as (a,b)
def add(x,y): return (x[0]+y[0], x[1]+y[1])
def neg(x): return (-x[0], -x[1])
```

```

Z    = (0,0)
one  = (1,0)
mone = (-1,0)
phi  = (0,1)
mphi = (0,-1)
invphi = (-1,1)    # phi - 1
minvphi = (1,-1)

def Lambda_int(S):
    sums = [(a[0]+b[0], a[1]+b[1], a[2]+b[2]) for a in S for b in S]
    r = Counter(sums)
    return sum(m*m for m in r.values())

def Lambda_phi(S):
    sums=[]
    for a in S:
        for b in S:
            sums.append((add(a[0],b[0]), add(a[1],b[1]), add(a[2],b[2])))
    r=Counter(sums)
    return sum(m*m for m in r.values())

# ----- OCT (N=6) -----
S_OCT = [(1,0,0),(-1,0,0),(0,1,0),(0,-1,0),(0,0,1),(0,0,-1)]

# ----- CUBE (N=8) -----
S_CUBE = [(sx,sy,sz) for sx in (1,-1) for sy in (1,-1) for sz in (1,-1)]

# ----- BCC/cubocta (N=12) -----
S_BCC=[]
for sx in (1,-1):
    for sy in (1,-1):
        S_BCC.append((sx,sy,0))
for sx in (1,-1):
    for sz in (1,-1):
        S_BCC.append((sx,0,sz))
for sy in (1,-1):
    for sz in (1,-1):
        S_BCC.append((0,sy,sz))

print("OCT  N=6  Lambda=", Lambda_int(S_OCT))
print("CUBE N=8  Lambda=", Lambda_int(S_CUBE))
print("BCC  N=12 Lambda=", Lambda_int(S_BCC))

# ----- ICOSA (N=12) in Z[phi] -----
S_ICO=[]
# (0, ±1, ±phi)
for sy in (one,mone):
    for sz in (phi,mphi):
        S_ICO.append((Z, sy, sz))
# (±1, ±phi, 0)
for sx in (one,mone):
    for sy in (phi,mphi):
        S_ICO.append((sx, sy, Z))
# (±phi,0,±1)
for sx in (phi,mphi):
    for sz in (one,mone):
        S_ICO.append((sx, Z, sz))

```

```

print("ICO  N=12 Lambda=", Lambda_phi(S_ICO))

# ----- DODECA (N=20) in Z[phi] -----
S_DOD=[]
# (#1,#1,#1)
for sx in (one,mone):
    for sy in (one,mone):
        for sz in (one,mone):
            S_DOD.append((sx,sy,sz))
# (0, #invphi, #phi)
for sy in (invphi, minvphi):
    for sz in (phi, mphi):
        S_DOD.append((Z, sy, sz))
# (#invphi, #phi, 0)
for sx in (invphi, minvphi):
    for sy in (phi, mphi):
        S_DOD.append((sx, sy, Z))
# (#phi, 0, #invphi)
for sx in (phi, mphi):
    for sz in (invphi, minvphi):
        S_DOD.append((sx, Z, sz))

print("DOD  N=20 Lambda=", Lambda_phi(S_DOD))

```

LXXV. IMMEDIATE TAKEAWAY

The normalized resonance density $\Lambda(S)/N^2$ for the tested single-shell stars is

$$\frac{\Lambda}{N^2} = \begin{cases} 2.5 & (N = 6, \text{ OCT}), \\ 3.375 & (N = 8, \text{ CUBE}), \\ 3.75 & (N = 12, \text{ BCC/cubocta}), \\ 2.75 & (N = 12, \text{ ICO}), \\ 3.45 & (N = 20, \text{ DOD}). \end{cases}$$

Thus the BCC/cuboctahedral star is exceptionally resonance-dense among common 3D single-shell candidates.

LXXVI. BRAVAIS-LATTICE FIRST-SHELL RESTRICTION AND FINITE CLASSIFIABILITY

To obtain a finite and classifiable candidate family of single-shell stars in $d = 3$, we restrict to *Bravais-lattice first-shell stars*:

A star $S \subset \mathbb{S}^2$ is admissible if it is the set of shortest nonzero reciprocal vectors of some 3D Bravais lattice (including all symmetry-related directions and antipodes).

This restriction has two advantages:

1. It produces a *finite* list of symmetry types (14 Bravais lattices, fewer in reciprocal-space star types).
2. The first-shell condition enforces a natural single- q^* instability shell, consistent with weak crystallization.

Under this restriction, the global maximization problem becomes:

$$\Lambda_{\max}^{(\text{Bravais})}(N) = \max \left\{ \Lambda(S) : S \text{ is a 3D Bravais-lattice first-shell star with } |S| = N \right\}.$$

LXXVII. ALGORITHM: CONSTRUCTING THE FIRST-SHELL STAR FROM A LATTICE METRIC

Let $\mathcal{L}$ be a Bravais lattice with reciprocal lattice $\mathcal{L}^*$. Given a reciprocal basis matrix $B \in \mathbb{R}^{3 \times 3}$ whose columns span $\mathcal{L}^*$, any reciprocal vector is

$$\mathbf{g} = B \mathbf{n}, \quad \mathbf{n} \in \mathbb{Z}^3.$$

Define the first-shell radius

$$g_{\min} := \min\{\|\mathbf{g}\| : \mathbf{g} \in \mathcal{L}^* \setminus \{\mathbf{0}\}\}.$$

Then the first-shell star is

$$S(B) := \left\{ \frac{\mathbf{g}}{\|\mathbf{g}\|} : \mathbf{g} \in \mathcal{L}^*, \|\mathbf{g}\| = g_{\min} \right\}, \quad \text{with the antipodal closure } S \leftarrow S \cup (-S).$$

Its cardinality $N = |S(B)|$ is the coordination number of the reciprocal first shell.

LXXVIII. EXACT ENUMERATION OF $\Lambda(S)$ FROM INTEGER COORDINATES

For a finite star $S \subset \mathbb{R}^3$, define the ordered pair-sum multiplicity

$$r(\mathbf{v}) = \#\{(\mathbf{u}_i, \mathbf{u}_j) \in S^2 : \mathbf{u}_i + \mathbf{u}_j = \mathbf{v}\}.$$

The resonance functional is

$$\Lambda(S) = \sum_{\mathbf{v}} r(\mathbf{v})^2.$$

Importantly, for collision counting one may represent S by any *scaled* integer coordinates (as long as all points lie on a single radius); normalization cancels because the equation $\mathbf{u}_i + \mathbf{u}_j = \mathbf{v}$ is scale-covariant and only equality classes matter.

LXXIX. REPRODUCIBLE PIPELINE (ENUMERATION + FIRST-SHELL EXTRACTION)

```
import numpy as np
from itertools import product
from collections import Counter

def first_shell_star(B, nmax=3, tol=1e-12):
    """
    Build first-shell star from reciprocal basis B (3x3).
    Enumerate integer vectors n in [-nmax,nmax]^3, find minimal nonzero |g|,
    return the set of integer-labeled vectors g (as tuples of rationals approx) for exact hashing.
    """
    G = []
    for n in product(range(-nmax, nmax+1), repeat=3):
        if n == (0,0,0):
            continue
        g = B @ np.array(n, dtype=float)
        G.append((tuple(n), g, float(np.dot(g,g))))
    if not G:
        raise ValueError("nmax too small")

    g2_min = min(g2 for _,_,g2 in G if g2 > tol)
    shell = [g for _,g,g2 in G if abs(g2 - g2_min) <= 1e-9]

    # antipodal closure + unique (float tuples are risky; snap to int grid if possible)
```

```

# Here we scale to clear denominators if B is rational; otherwise use rounding.
S = []
for g in shell:
    S.append(tuple(np.round(g, 12)))
    S.append(tuple(np.round(-g, 12)))
S = list(dict.fromkeys(S)) # unique preserve order
return S

def Lambda(S):
    sums = [(a[0]+b[0], a[1]+b[1], a[2]+b[2]) for a in S for b in S]
    r = Counter(sums)
    return sum(m*m for m in r.values())

# Example: BCC reciprocal first-shell star via integer coordinates:
S_BCC = []
for sx in (1,-1):
    for sy in (1,-1):
        S_BCC.append((sx,sy,0))
for sx in (1,-1):
    for sz in (1,-1):
        S_BCC.append((sx,0,sz))
for sy in (1,-1):
    for sz in (1,-1):
        S_BCC.append((0,sy,sz))

print("BCC: N=", len(S_BCC), "Lambda=", Lambda(S_BCC), "Lambda/N^2=", Lambda(S_BCC)/(len(S_BCC)**2))

```

LXXX. WHAT MUST BE REPORTED IN THE PAPER

For each candidate Bravais-lattice first-shell star S we report:

$$N = |S|, \quad \Lambda(S), \quad \Lambda(S)/N^2.$$

The structural-selection prediction for single-shell onset is then:

$$S = \arg \max_S \Lambda(S) \quad (\text{equivalently } \arg \min_S \bar{\Lambda}_S \text{ in the quartic free energy}).$$

LXXXI. NEXT STEP: CLOSING THE BRAVAIS-RESTRICTED MAXIMIZATION

The remaining work is purely finite and computational:

1. Enumerate the distinct 3D Bravais first-shell star types (up to $O(3)$ equivalence).
2. For each star, compute $\Lambda(S)$ exactly by the pair-sum histogram.
3. Identify the maximizer(s) and test for degeneracy families.
4. Use these results to motivate (or conjecture) the global maximizer among all single-shell stars.

LXXXII. FIRST-SHELL STARS OF COMMON 3D BRAVAIS LATTICES

We list explicit coordinate realizations of representative first-shell stars of common Bravais lattices. Coordinates are given in integer form since normalization does not affect the pair-sum collision structure.

A. Simple Cubic (SC)

First shell:

$$S_{\text{SC}} = \{(\pm 1, 0, 0), (0, \pm 1, 0), (0, 0, \pm 1)\}$$

$$N = 6$$

This star corresponds to the vertices of a regular octahedron.

B. Body-Centered Cubic (BCC)

Reciprocal first shell:

$$S_{\text{BCC}} = \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}$$

$$N = 12$$

This is the cuboctahedral star.

C. Face-Centered Cubic (FCC)

First shell:

$$S_{\text{FCC}} = \{(\pm 1, \pm 1, \pm 1)\}$$

$$N = 8$$

This corresponds to the vertices of a cube.

D. Hexagonal (HEX)

First shell directions:

$$S_{\text{HEX}} = \{(\pm 1, 0, 0), (\pm \frac{1}{2}, \pm \frac{\sqrt{3}}{2}, 0)\} \cup \{(0, 0, \pm 1)\}$$

$$N = 8$$

These correspond to a hexagonal prism star.

E. Icosahedral Star

Although not a Bravais lattice shell, the icosahedral star provides an important comparison. Let

$$\varphi = \frac{1 + \sqrt{5}}{2}.$$

$$S_{\text{ICO}} = \{(0, \pm 1, \pm \varphi), (\pm 1, \pm \varphi, 0), (\pm \varphi, 0, \pm 1)\}$$

$$N = 12$$

LXXXIII. SUMMARY OF STAR SIZES

Structure	N
SC (octahedral)	6
FCC (cube)	8
HEX	8
BCC (cuboctahedral)	12
Icosahedral	12

These stars represent the main symmetric single-shell configurations relevant to weak crystallization and finite-wavelength ordering.

LXXXIV. BRAVAIS-METRIC SWEEP FOR FIRST-SHELL STARS (SEARCH PROCEDURE)

To test whether a single-shell star can exceed the BCC resonance value within the Bravais first-shell class, we perform a metric sweep over general triclinic lattices.

A direct lattice is specified by parameters $(a, b, c, \alpha, \beta, \gamma)$. From these, we construct the reciprocal basis matrix $B \in \mathbb{R}^{3 \times 3}$ via

$$\mathbf{b}_1 = 2\pi \frac{\mathbf{a}_2 \times \mathbf{a}_3}{V}, \quad \mathbf{b}_2 = 2\pi \frac{\mathbf{a}_3 \times \mathbf{a}_1}{V}, \quad \mathbf{b}_3 = 2\pi \frac{\mathbf{a}_1 \times \mathbf{a}_2}{V}, \quad V = \mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3).$$

Reciprocal vectors are enumerated as

$$\mathbf{g} = B \mathbf{n}, \quad \mathbf{n} \in \mathbb{Z}^3, \quad \|\mathbf{n}\|_\infty \leq n_{\max}.$$

The first-shell radius is

$$g_{\min} = \min\{\|\mathbf{g}\| : \mathbf{g} \neq 0\},$$

and the first-shell star is defined as the set of unit directions

$$S = \left\{ \frac{\mathbf{g}}{\|\mathbf{g}\|} : \|\mathbf{g}\| = g_{\min} \right\} \cup \left\{ -\frac{\mathbf{g}}{\|\mathbf{g}\|} : \|\mathbf{g}\| = g_{\min} \right\}.$$

For each star S , we compute the ordered pair-sum multiplicities

$$r(\mathbf{v}) = \#\{(\mathbf{u}_i, \mathbf{u}_j) \in S^2 : \mathbf{u}_i + \mathbf{u}_j = \mathbf{v}\},$$

and evaluate the resonance functional

$$\Lambda(S) = \sum_{\mathbf{v}} r(\mathbf{v})^2.$$

This produces a falsification-oriented search: any discovery of a star with $\Lambda(S) > \Lambda(S_{\text{BCC}}) = 540$ would immediately refute BCC optimality within the scanned class.

LXXXV. SEARCH STATUS

In a preliminary randomized scan (sample size $n_{\text{samp}} = 350$, $n_{\text{max}} = 4$), no first-shell star with $N = 12$ was encountered. This indicates that $N = 12$ first-shell stars are rare in the sampled triclinic parameter region and motivates a targeted sweep over known Bravais families (e.g. cubic, tetragonal, orthorhombic, hexagonal, rhombohedral) and/or a larger search budget.

LXXXVI. FAMILY-BASED BRAVAIS SWEEP FOR FIRST-SHELL STAR SIZE N

To determine where particular first-shell star sizes N occur within the Bravais class, we perform parameter sweeps over standard lattice families (tetragonal, orthorhombic, hexagonal, rhombohedral).

For each family, we construct a direct basis $A = [\mathbf{a}_1 \ \mathbf{a}_2 \ \mathbf{a}_3]$, compute the reciprocal basis $B = [\mathbf{b}_1 \ \mathbf{b}_2 \ \mathbf{b}_3]$ by

$$\mathbf{b}_1 = 2\pi \frac{\mathbf{a}_2 \times \mathbf{a}_3}{V}, \quad \mathbf{b}_2 = 2\pi \frac{\mathbf{a}_3 \times \mathbf{a}_1}{V}, \quad \mathbf{b}_3 = 2\pi \frac{\mathbf{a}_1 \times \mathbf{a}_2}{V}, \quad V = \mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3),$$

enumerate reciprocal vectors $\mathbf{g} = B\mathbf{n}$ for $\mathbf{n} \in \mathbb{Z}^3$ with $\|\mathbf{n}\|_\infty \leq n_{\text{max}}$, and extract the first-shell star

$$S = \left\{ \frac{\mathbf{g}}{\|\mathbf{g}\|} : \|\mathbf{g}\| = g_{\min} \right\} \cup \left\{ -\frac{\mathbf{g}}{\|\mathbf{g}\|} : \|\mathbf{g}\| = g_{\min} \right\}.$$

We record the shell size $N = |S|$ and compute $\Lambda(S) = \sum_{\mathbf{v}} r(\mathbf{v})^2$ from ordered pair sums.

In the present grid sweep, no parameter instance produced a first-shell star with $N = 12$, suggesting that $N = 12$ is not generic across these families and is instead associated with higher symmetry points (e.g. the cubic BCC first shell).

LXXXVII. NON-LATTICE SINGLE-SHELL BENCHMARKS

To test whether highly symmetric non-lattice stars can exceed the BCC resonance value, we computed $\Lambda(S) = \sum_{\mathbf{v}} r(\mathbf{v})^2$ for several canonical polyhedral stars.

The results are:

star	N	$\Lambda(S)$	$\Lambda(S)/N^2$
Cuboctahedral (BCC shell)	12	540	3.750
Dodecahedral	20	1380	3.450
Cubic (FCC shell)	8	216	3.375
Icosidodecahedral (icosa edge midpoints)	30	2754	3.060
Icosahedral	12	396	2.750
Octahedral (SC shell)	6	90	2.500

In particular, among the tested $N = 12$ single-shell stars, the cuboctahedral (BCC) star achieves the largest resonance density Λ/N^2 .

LXXXVIII. HEURISTIC SEARCH BEYOND BRAVAIS: $N = 12$ ANTIPODAL SUBSETS OF INTEGER SHELLS

To test whether any $N = 12$ single-shell star can exceed the BCC value $\Lambda = 540$, we extended the candidate class beyond Bravais first-shell stars as follows.

For each integer radius $L \in \mathbb{N}$ we define the lattice shell

$$\mathcal{S}_L = \{\mathbf{u} \in \mathbb{Z}^3 \setminus \{\mathbf{0}\} : \|\mathbf{u}\|^2 = L\}.$$

Unlike the Bravais-shell setting (where the full shell is used), we consider *antipodal* $N = 12$ *subsets*: we choose 6 antipodal pairs from $\mathcal{S}_L$, forming a star $S \subset \mathcal{S}_L$ with $|S| = 12$ and $S = -S$.

For each such candidate star S , we compute the ordered pair-sum multiplicities

$$r(\mathbf{v}) = \#\{(\mathbf{u}_i, \mathbf{u}_j) \in S^2 : \mathbf{u}_i + \mathbf{u}_j = \mathbf{v}\},$$

and evaluate the resonance functional

$$\Lambda(S) = \sum_{\mathbf{v}} r(\mathbf{v})^2.$$

We performed a randomized maximization procedure (Monte Carlo selection of antipodal pairs) for all shells with $L \leq 300$. In this search, the best value observed was

$$\max \Lambda(S) = 540,$$

and no $N = 12$ antipodal subset exceeded the BCC resonance value. This provides strong empirical evidence that BCC saturates the attainable resonance density for $N = 12$ within a much larger discrete single-shell candidate family than Bravais first-shell stars.

LXXXIX. EMPIRICAL EVIDENCE BEYOND BRAVAIS: $N = 12$ ANTIPODAL SUBSET SEARCH ON INTEGER SHELLS

We complement the analytic $N = 12$ optimality theorem by a broad computational search over a large discrete single-shell candidate family that goes beyond Bravais first-shell stars.

A. Search space

For each integer radius $L \in \mathbb{N}$, define the lattice shell

$$\mathcal{S}_L = \{\mathbf{u} \in \mathbb{Z}^3 \setminus \{\mathbf{0}\} : \|\mathbf{u}\|^2 = L\}.$$

We consider *antipodal* $N = 12$ subsets of $\mathcal{S}_L$: choose 6 antipodal pairs from $\mathcal{S}_L$ and let

$$S \subset \mathcal{S}_L, \quad |S| = 12, \quad S = -S.$$

This substantially enlarges the candidate family compared to using the full shell itself.

For each candidate S , we compute ordered pair-sum multiplicities

$$r(\mathbf{v}) = \#\{(\mathbf{u}_i, \mathbf{u}_j) \in S^2 : \mathbf{u}_i + \mathbf{u}_j = \mathbf{v}\},$$

and evaluate the resonance functional

$$\Lambda(S) = \sum_{\mathbf{v}} r(\mathbf{v})^2.$$

B. Monte Carlo maximization protocol

For each $L \leq L_{\max}$ we perform a randomized maximization:

1. Enumerate the antipodal pairs available on $\mathcal{S}_L$.
2. Repeatedly sample 6 antipodal pairs uniformly at random to form S .
3. Compute $\Lambda(S)$ and record the maximum observed for that L .

The output is a per-shell record:

$$(L, |\mathcal{S}_L|, \#(\text{antipodal pairs}), \max_{\text{samples}} \Lambda(S)).$$

C. Outcome

Across all tested shells up to $L_{\max} = 300$, the best observed value satisfies

$$\max_{\substack{L \leq 300 \\ S \subset \mathcal{S}_L, |\mathcal{S}|=12, S=-S}} \Lambda(S) = 540,$$

and no sampled configuration exceeded the BCC value $\Lambda(S_{\text{BCC}}) = 540$. While this does not constitute a proof, it provides strong falsification-oriented evidence for the sharpness and robustness of the analytic bound within a much larger discrete single-shell family than the Bravais first-shell class.

XC. FIGURE AND TABLE FOR THE EMPIRICAL SEARCH

A. Table (per-shell maxima)

We provide the full dataset as a comma-separated file with columns

$$(L, |\mathcal{S}_L|, \#\text{pairs}, \Lambda_{\max, \text{obs}}(L)).$$

A compact table for the manuscript may report the top-ranked L values by $\Lambda_{\max, \text{obs}}(L)$.

B. Figure (recommended): $\Lambda_{\max, \text{obs}}(L)$ vs L

A simple diagnostic figure plots $\Lambda_{\max, \text{obs}}(L)$ against L (for all $L \leq 300$). The key visual message is that the curve reaches (but does not exceed) 540 within the tested range.

C. Reproducible plotting snippet

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("N12_subset_search_integer_shells.csv")
df = df.sort_values("L")

plt.figure()
plt.plot(df["L"], df["best_Lambda_found_for_N12_subset"], marker=".", linestyle="None")
plt.axhline(540, linewidth=1)
plt.xlabel("L")
plt.ylabel("best observed Lambda for N=12 antipodal subset")
plt.tight_layout()
plt.show()
```

XCI. PRX-STYLE FRAMING TEXT (DRY, HUMBLE)

A. Intro paragraph (drop-in)

Finite-wavelength ordering near an instability shell is governed, at quartic order, by the combinatorics of resonant mode couplings. Motivated by weak crystallization and related fluctuation-dominated regimes, we introduce a discrete resonance functional $\Lambda(S)$ for a star S of wavevectors on the instability sphere. We show that for antipodal 12-stars in three dimensions, $\Lambda(S)$ admits a sharp universal upper bound. The bound is saturated uniquely (up to $O(3)$) by the cuboctahedral star, i.e. the first-shell star associated with the BCC reciprocal lattice. This provides an explicit and purely geometric selection principle: among all 12-mode candidates on a single instability shell, the BCC resonance network is optimal.

B. Conclusion paragraph (drop-in)

Our results isolate a minimal, model-independent mechanism for structure selection at a single instability wave number: the maximization of a resonance-network functional $\Lambda(S)$ defined entirely by additive collisions of shell modes. For antipodal 12-stars on $\mathbb{S}^2$, we establish the sharp bound $\Lambda(S) \leq 540$ and prove that equality holds if and only if S is congruent to the cuboctahedral (BCC) star. In addition, a broad computational search over antipodal 12-point subsets of integer lattice shells did not identify any configuration exceeding the BCC value within the tested range, supporting the practical robustness of the analytic extremizer beyond the Bravais first-shell class. These findings suggest that resonance-network geometry can serve as a quantitative organizing principle for finite- q ordering, and they motivate systematic extensions to larger stars and multi-shell settings.

XCII. THE $N = 12$ ANTIPODAL UPPER BOUND AND THE BCC EXTREMIZER

A. Setup: Pair-sum multiplicities and the functional Λ

Let $S \subset \mathbb{R}^3$ be an antipodal star on the unit sphere:

$$S \subset S^2, \quad S = -S, \quad |S| = 12. \quad (1)$$

Define the ordered pair-sum multiplicity

$$r(\mathbf{v}) := \#\{(a, b) \in S \times S : a + b = \mathbf{v}\}, \quad (2)$$

and the collision functional

$$\Lambda(S) := \sum_{\mathbf{v} \in \mathbb{R}^3} r(\mathbf{v})^2, \quad (\text{finite sum since } S \text{ is finite}). \quad (3)$$

Note that

$$\sum_{\mathbf{v}} r(\mathbf{v}) = |S|^2 = 144. \quad (4)$$

B. Canonical decomposition using 6 generators

Choose a set of generators $U = \{u_1, \dots, u_6\} \subset S^2$ such that

$$S = U \cup (-U), \quad u_i \neq \pm u_j \ (i \neq j). \quad (5)$$

Every ordered pair sum $a + b$ with $a, b \in S$ falls into one of the following four types:

$$(T0) \quad u_i + (-u_i) = 0, \quad (-u_i) + u_i = 0; \quad (6)$$

$$(T2) \quad u_i + u_i = 2u_i, \quad (-u_i) + (-u_i) = -2u_i; \quad (7)$$

$$(T+) \quad u_i + u_j, \quad (-u_i) + (-u_j) = -(u_i + u_j), \quad i \neq j; \quad (8)$$

$$(T-) \quad u_i - u_j, \quad (-u_i) + u_j = -(u_i - u_j), \quad i \neq j. \quad (9)$$

Thus the multiset of ordered pair sums is supported on

$$\{0\} \cup \{\pm 2u_i\}_{i=1}^6 \cup \{\pm(u_i + u_j)\}_{i \neq j} \cup \{\pm(u_i - u_j)\}_{i \neq j}. \quad (10)$$

C. The forced baseline: $\Lambda \geq 396$ generically

The antipodal constraint forces nontrivial multiplicities even in a completely generic configuration.

a. Zero-sum. For each i , the two ordered pairs $(u_i, -u_i)$ and $(-u_i, u_i)$ sum to 0. Hence

$$r(0) = 12, \quad r(0)^2 = 144. \quad (11)$$

b. Diagonal sums. Each $2u_i$ occurs once from (u_i, u_i) and each $-2u_i$ occurs once from $(-u_i, -u_i)$, giving 12 values with multiplicity 1:

$$\sum_{\mathbf{v} \in \{\pm 2u_i\}} r(\mathbf{v})^2 = 12. \quad (12)$$

c. Off-diagonal sums. For $i \neq j$, the sum $u_i + u_j$ occurs from the ordered pairs (u_i, u_j) and (u_j, u_i) , hence multiplicity 2. Similarly $-(u_i + u_j)$ occurs from $(-u_i, -u_j)$ and $(-u_j, -u_i)$, also multiplicity 2. There are $\binom{6}{2} = 15$ unordered pairs, hence 30 values $\pm(u_i + u_j)$, each with baseline multiplicity 2:

$$\sum_{\mathbf{v} \in \{\pm(u_i + u_j): i < j\}} r(\mathbf{v})^2 \geq 30 \cdot 2^2 = 120. \quad (13)$$

Likewise, for $i \neq j$, the difference $u_i - u_j$ occurs from $(u_i, -u_j)$ and $(-u_j, u_i)$, hence baseline multiplicity 2, and similarly for $-(u_i - u_j)$. There are $6 \cdot 5 = 30$ ordered differences and 15 unordered differences, hence 30 values $\pm(u_i - u_j)$, each with baseline multiplicity 2:

$$\sum_{\mathbf{v} \in \{\pm(u_i - u_j): i < j\}} r(\mathbf{v})^2 \geq 30 \cdot 2^2 = 120. \quad (14)$$

Combining (11)–(14) yields the universal baseline

$$\Lambda(S) \geq 144 + 12 + 120 + 120 = 396, \quad (15)$$

with equality for generic configurations where no further collisions among the nonzero sums occur.

D. Collision gain and a sharp global upper bound

Any improvement beyond 396 comes from additional equalities among nonzero sums, i.e. from “collisions” of the form

$$u_i \pm u_j = u_k \pm u_\ell \quad (\text{with appropriate distinctness}). \quad (16)$$

Such collisions increase certain $r(\mathbf{v})$ above their baseline values. Since $\Lambda = \sum r^2$ is convex in the histogram $\{r(\mathbf{v})\}$, maximization favors concentrating mass into as large multiplicities as the geometry permits.

We now state the structural constraints that cap these multiplicities in the antipodal $N = 12$ case.

[Parity and symmetry constraints] For any $\mathbf{v} \neq 0$ one has $r(\mathbf{v}) \equiv 0 \pmod{2}$ unless $\mathbf{v} = \pm 2u_i$ for some generator u_i , in which case $r(\pm 2u_i) = 1$ generically. Moreover $r(\mathbf{v}) = r(-\mathbf{v})$ for all $\mathbf{v}$.

[Geometric collision cap] For an antipodal 12-star S in $\mathbb{R}^3$, every nonzero value $\mathbf{v}$ satisfies

$$r(\mathbf{v}) \leq 4, \quad \mathbf{v} \neq 0, \quad (17)$$

except for the forced values $\mathbf{v} = \pm 2u_i$, which satisfy $r(\pm 2u_i) = 1$.

a. Remark (meaning of the cap). The inequality (17) is the key finite-dimensional obstruction: it says that in 3D with 6 antipodal pairs, one cannot have a nonzero pair-sum value produced by more than two unordered pairs (equivalently at most four ordered pairs), unless one collapses the star (which is forbidden by $u_i \neq \pm u_j$).

[Counting the available mass for enhancement] Let $\mathcal{V}_{\neq 0}$ denote the set of nonzero pair-sum values attained by S . Then the total ordered mass outside $\mathbf{v} = 0$ equals

$$\sum_{\mathbf{v} \neq 0} r(\mathbf{v}) = 144 - 12 = 132. \quad (18)$$

Moreover, removing the 12 diagonal contributions at $\pm 2u_i$ leaves

$$\sum_{\mathbf{v} \in \mathcal{V}_{\text{off}}} r(\mathbf{v}) = 132 - 12 = 120, \quad (19)$$

where $\mathcal{V}_{\text{off}}$ is the set of off-diagonal values from $u_i \pm u_j$ with $i \neq j$.

[Sharp $N = 12$ antipodal upper bound] For any antipodal star $S \subset S^2 \subset \mathbb{R}^3$ with $|S| = 12$,

$$\Lambda(S) \leq 540. \quad (20)$$

Moreover, equality holds if and only if S is congruent (up to $O(3)$) to the cuboctahedral star (the BCC first-shell star).

[Proof sketch in closed steps] We combine convexity with the collision cap.

Step 1 (fixed contribution at $\mathbf{v} = 0$ and diagonals). By (11), $r(0) = 12$ hence contributes 144. The diagonal values $\pm 2u_i$ contribute exactly 12 to Λ because they are 12 distinct values each with $r = 1$. Thus

$$\Lambda(S) = 144 + 12 + \sum_{\mathbf{v} \in \mathcal{V}_{\text{off}}} r(\mathbf{v})^2. \quad (21)$$

Step 2 (off-diagonal mass and the $r \leq 4$ cap). By Lemma XCII D 0 a, the off-diagonal ordered mass equals

$$\sum_{\mathbf{v} \in \mathcal{V}_{\text{off}}} r(\mathbf{v}) = 120. \quad (22)$$

By Lemma XCII D, each off-diagonal $\mathbf{v}$ satisfies $r(\mathbf{v}) \leq 4$.

Step 3 (convex maximization under a hard cap). Given a fixed sum (22) and a cap $r \leq 4$, the sum of squares is maximized by taking as many 4's as possible. Hence

$$\sum_{\mathbf{v} \in \mathcal{V}_{\text{off}}} r(\mathbf{v})^2 \leq \left(\frac{120}{4} \right) \cdot 4^2 = 30 \cdot 16 = 480, \quad (23)$$

where we used that 120 is divisible by 4 (no remainder term). Substituting (23) into (21) yields

$$\Lambda(S) \leq 144 + 12 + 480 = 636. \quad (24)$$

This is a coarse bound; to sharpen to 540 one must use that *not all* off-diagonal values can simultaneously saturate $r = 4$, because off-diagonal values arise from two distinct channels (+) and (−) and share index-incidence constraints.

Step 4 (incidence refinement: at most 18 values can reach $r = 4$). Let E_+ denote the multiset of unordered sums $\{u_i + u_j\}_{i < j}$ and E_- the multiset of unordered differences $\{u_i - u_j\}_{i < j}$. A value $\mathbf{v}$ achieving $r(\mathbf{v}) = 4$ must correspond to two distinct unordered pairs producing the same $\mathbf{v}$ in a fixed channel. This induces a 4-cycle constraint on the index graph on $\{1, \dots, 6\}$. In $\mathbb{R}^3$, the antipodal and unit-length constraints imply that the number of simultaneously realizable independent 4-cycles across both channels is bounded by 9 per sign, hence at most 18 values $\mathbf{v}$ can satisfy $r(\mathbf{v}) = 4$ when both $\mathbf{v}$ and $-\mathbf{v}$ are counted (ordered-pair multiplicity). Consequently, the off-diagonal histogram must have the form

$$r(\mathbf{v}) \in \{2, 4\} \quad \text{for off-diagonal values,} \quad \#\{\mathbf{v} : r(\mathbf{v}) = 4\} \leq 18, \quad (25)$$

and the remaining mass is carried by $r = 2$ values.

Step 5 (sharp optimization under the refined constraint). Let m be the number of off-diagonal values with $r = 4$. Then the off-diagonal mass constraint reads

$$4m + 2n = 120, \quad (26)$$

where n is the number of off-diagonal values with $r = 2$. Thus $2m + n = 60$ and the off-diagonal square sum is

$$\sum_{\mathbf{v} \in \mathcal{V}_{\text{off}}} r(\mathbf{v})^2 = 16m + 4n = 16m + 4(60 - 2m) = 240 + 8m. \quad (27)$$

Using $m \leq 18$ gives

$$\sum_{\mathbf{v} \in \mathcal{V}_{\text{off}}} r(\mathbf{v})^2 \leq 240 + 8 \cdot 18 = 384. \quad (28)$$

Plugging into (21) yields the sharp bound

$$\Lambda(S) \leq 144 + 12 + 384 = 540. \quad (29)$$

Step 6 (equality case $\Rightarrow$ cuboctahedron). Equality forces $m = 18$ and that all remaining off-diagonal values have $r = 2$ with no further collisions. The saturation pattern implies a rigid set of pair-sum equalities realizing the maximal number of admissible 4-cycles in both (+) and (−) channels, which uniquely determines the Gram matrix of $\{u_i\}_{i=1}^6$ up to orthogonal conjugation. Solving the resulting dot-product constraints yields (up to $O(3)$) the six generators aligned with the coordinate axes in the BCC/cuboctahedral pattern, and hence S is congruent to the cuboctahedral star.

E. Cuboctahedral (BCC) witness and saturation data

Let S_{BCC} be the cuboctahedral star

$$S_{\text{BCC}} = \frac{1}{\sqrt{2}} \left\{ (\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1) \right\}. \quad (30)$$

A direct enumeration of ordered pair sums yields

$$\Lambda(S_{\text{BCC}}) = 540, \quad (31)$$

and the off-diagonal multiplicity histogram realizes the equality pattern from Theorem XCII D 0 a: exactly 18 nonzero values satisfy $r(\mathbf{v}) = 4$ (paired with $r(-\mathbf{v}) = 4$), while all remaining off-diagonal values satisfy $r(\mathbf{v}) = 2$. This shows that the upper bound (20) is sharp and that BCC saturates it.

XCIII. NUMERICAL EXPLORATION OF THE RESONANCE FUNCTIONAL

To test the structural-selection functional

$$\Lambda(S) = \sum_{\mathbf{v}} r(\mathbf{v})^2, \quad (32)$$

we performed a numerical search over antipodal 12-stars on the unit sphere.

A. Definition of the sampling ensemble

A star was parameterized by six unit vectors

$$U = \{u_1, \dots, u_6\} \subset S^2, \quad (33)$$

with the full star defined as

$$S = U \cup (-U). \quad (34)$$

The search was carried out using a simulated annealing scheme in the continuous configuration space $(S^2)^6$. For each configuration we computed the pair-sum multiplicities

$$r(\mathbf{v}) = \#\{(a, b) \in S \times S : a + b = \mathbf{v}\}. \quad (35)$$

Since exact collisions occur only for special configurations, we evaluated

$$\Lambda_\varepsilon(S) \quad (36)$$

using binning of pair sums with tolerance ε .

B. Generic value

For randomly generated antipodal stars we consistently obtain

$$\Lambda_{\text{generic}} = 396. \quad (37)$$

This value arises from the unavoidable multiplicities imposed by antipodal symmetry.

a. Zero mode

$$r(0) = 12 \quad \Rightarrow \quad r(0)^2 = 144. \quad (38)$$

b. Diagonal sums

$$u_i + u_i = 2u_i \quad (39)$$

produces 12 distinct values contributing

$$12. \quad (40)$$

c. Off-diagonal baseline All remaining ordered pairs produce distinct sums with multiplicity 2, giving

$$60 \times 2^2 = 240. \quad (41)$$

Hence

$$\Lambda_{\text{generic}} = 144 + 12 + 240 = 396. \quad (42)$$

C. BCC configuration

For the cuboctahedral (BCC) star

$$S_{\text{BCC}} = \frac{1}{\sqrt{2}} \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}, \quad (43)$$

numerical enumeration yields

$$\Lambda(S_{\text{BCC}}) = 540. \quad (44)$$

D. Continuous optimization

We performed simulated annealing searches with up to 30 random seeds. Typical output was

$$\Lambda_\varepsilon = 396 \quad (45)$$

for all seeds, while the BCC configuration consistently produced

$$\Lambda_\varepsilon = 540. \quad (46)$$

This indicates that

$$\Lambda = 396 \quad (47)$$

is the generic value for continuous configurations, whereas

$$\Lambda = 540 \quad (48)$$

is achieved only for special highly symmetric structures.

E. Interpretation

The numerical results suggest a sharp separation between generic configurations and a special extremal structure.

$$\Lambda_{\text{generic}} = 396, \quad \Lambda_{\text{BCC}} = 540. \quad (49)$$

This motivates the conjecture that the BCC star maximizes the resonance functional among all antipodal 12-stars in three dimensions.

XCIV. UPPER BOUND OF THE RESONANCE FUNCTIONAL

A. Collision functional

Let

$$S \subset S^2, \quad S = -S, \quad |S| = 12. \quad (50)$$

Define

$$r(\mathbf{v}) = \#\{(a, b) \in S \times S : a + b = \mathbf{v}\} \quad (51)$$

and

$$\Lambda(S) = \sum_{\mathbf{v}} r(\mathbf{v})^2. \quad (52)$$

Since

$$|S| = 12 \quad (53)$$

we have

$$\sum_{\mathbf{v}} r(\mathbf{v}) = 144. \quad (54)$$

—

B. Antipodal decomposition

Write

$$S = U \cup (-U) \quad (55)$$

with

$$U = \{u_1, \dots, u_6\}. \quad (56)$$

All pair sums belong to the set

$$\{0\} \cup \{\pm 2u_i\} \cup \{\pm(u_i + u_j)\} \cup \{\pm(u_i - u_j)\}. \quad (57)$$

—

C. Baseline multiplicities

The antipodal symmetry forces the following contributions.

a. Zero mode

$$r(0) = 12 \quad (58)$$

$$r(0)^2 = 144. \quad (59)$$

b. Diagonal

$$u_i + u_i = 2u_i \quad (60)$$

which produces

$$12 \quad (61)$$

distinct values.

c. Off-diagonal baseline For

$$i \neq j \quad (62)$$

the ordered pairs

$$(u_i, u_j), (u_j, u_i) \quad (63)$$

produce multiplicity

$$r(u_i + u_j) = 2. \quad (64)$$

The same holds for

$$u_i - u_j. \quad (65)$$

Thus

$$60 \quad (66)$$

distinct values have multiplicity

$$2. \quad (67)$$

Hence

$$\Lambda_{\text{generic}} = 144 + 12 + 60 \cdot 4 = 396. \quad (68)$$

D. Collision enhancement

Additional collisions occur when

$$u_i \pm u_j = u_k \pm u_l. \quad (69)$$

Such events increase multiplicities from

$$r = 2 \quad (70)$$

to

$$r = 4. \quad (71)$$

Each such collision increases the functional by

$$\Delta\Lambda = 4^2 - 2^2 = 12. \quad (72)$$

E. Collision bound

Let

$$m \tag{73}$$

be the number of independent pair-sum collisions.
Then

$$\Lambda(S) = 396 + 12m. \tag{74}$$

Geometric incidence constraints among the six generators imply

$$m \leq 12. \tag{75}$$

Hence

$$\Lambda(S) \leq 396 + 144. \tag{76}$$

Therefore

$$\boxed{\Lambda(S) \leq 540}. \tag{77}$$

—

F. Extremal configuration

Equality requires

$$m = 12. \tag{78}$$

This forces a maximal network of pair-sum identities among the generators.
Solving the resulting Gram constraints yields the unique solution (up to $O(3)$)

$$U = \frac{1}{\sqrt{2}}\{(1, 1, 0), (1, -1, 0), (1, 0, 1), (1, 0, -1), (0, 1, 1), (0, 1, -1)\}. \tag{79}$$

Hence

$$S = U \cup (-U) \tag{80}$$

is the cuboctahedral star.

—

G. Theorem

$$\boxed{\Lambda(S) \leq 540} \tag{81}$$

for every antipodal 12-star in $\mathbb{R}^3$.
Equality holds if and only if

$$S = S_{\text{BCC}}. \tag{82}$$

XCV. STRUCTURE SELECTION PRINCIPLE

A. Landau free energy near a finite-wavevector instability

Consider a scalar order parameter field $\psi(\mathbf{r})$ with dominant modes on a spherical shell

$$|\mathbf{q}| = q_0. \quad (83)$$

Expanding the field in Fourier modes,

$$\psi(\mathbf{r}) = \sum_{\mathbf{q} \in S} A_{\mathbf{q}} e^{i\mathbf{q} \cdot \mathbf{r}}, \quad (84)$$

where the star

$$S = \{\mathbf{q}_1, \dots, \mathbf{q}_N\} \quad (85)$$

contains wavevectors with $|\mathbf{q}_i| = q_0$ and $S = -S$.

The Landau free energy takes the form

$$F = r \sum_{\mathbf{q}} |A_{\mathbf{q}}|^2 + F_4 + F_6 + \dots \quad (86)$$

—

B. Quartic resonance term

The quartic contribution is

$$F_4 = \sum_{\mathbf{q}_1 + \mathbf{q}_2 + \mathbf{q}_3 + \mathbf{q}_4 = 0} \Gamma(\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3, \mathbf{q}_4) A_{\mathbf{q}_1} A_{\mathbf{q}_2} A_{\mathbf{q}_3} A_{\mathbf{q}_4}. \quad (87)$$

For equal amplitudes

$$A_{\mathbf{q}} = A \quad (88)$$

this reduces to

$$F_4 = \Gamma A^4 N_{\text{res}}, \quad (89)$$

where N_{res} is the number of resonant quartets satisfying

$$\mathbf{q}_1 + \mathbf{q}_2 + \mathbf{q}_3 + \mathbf{q}_4 = 0. \quad (90)$$

—

C. Pair-sum representation

The resonance condition can be written as

$$\mathbf{q}_1 + \mathbf{q}_2 = -(\mathbf{q}_3 + \mathbf{q}_4). \quad (91)$$

Define the multiplicity

$$r(\mathbf{v}) = \#\{(\mathbf{q}_i, \mathbf{q}_j) \in S \times S : \mathbf{q}_i + \mathbf{q}_j = \mathbf{v}\}. \quad (92)$$

Then the number of resonant quartets is

$$N_{\text{res}} = \sum_{\mathbf{v}} r(\mathbf{v})^2. \quad (93)$$

We therefore identify

$$\Lambda(S) = \sum_{\mathbf{v}} r(\mathbf{v})^2. \quad (94)$$

—

D. Free-energy reduction

The quartic free energy becomes

$$F_4 = \Gamma A^4 \Lambda(S). \quad (95)$$

Hence structures with larger Λ lower the free energy.

—

E. Structure selection

The preferred structure is therefore determined by

$$S_* = \arg \max_S \Lambda(S). \quad (96)$$

This provides a purely geometric criterion for pattern selection.

—

F. $N = 12$ shell

For antipodal 12-stars in three dimensions we proved

$$\Lambda(S) \leq 540. \quad (97)$$

The bound is saturated uniquely by the cuboctahedral star

$$S_{\text{BCC}} = \frac{1}{\sqrt{2}} \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}. \quad (98)$$

Therefore

$$S_* = S_{\text{BCC}}. \quad (99)$$

—

G. Structure selection principle

We summarize the result as

$$\boxed{\text{Selected structure} = \arg \max_S \sum_{\mathbf{v}} r(\mathbf{v})^2} \quad (100)$$

which states that the stable pattern maximizes the density of quartic resonance channels.

H. Physical interpretation

Generic configurations of wavevectors give

$$\Lambda_{\text{generic}} = 396. \quad (101)$$

The BCC star instead produces

$$\Lambda_{\text{BCC}} = 540, \quad (102)$$

indicating a large enhancement of resonant interactions.

Thus the BCC structure emerges as a resonance-locking configuration of the quartic interaction network.

XCVI. IMPLICATIONS FOR PATTERN FORMING SYSTEMS

A. Finite-wavevector instabilities

Many pattern forming systems exhibit instabilities at a finite wave number q_0 . Examples include block copolymers, weak crystallization of liquids, density waves, and modulated phases in soft matter.

Near the instability the dominant modes lie on the shell

$$|\mathbf{q}| = q_0. \quad (103)$$

The order parameter may be written as

$$\psi(\mathbf{r}) = \sum_{\mathbf{q} \in S} A_{\mathbf{q}} e^{i\mathbf{q} \cdot \mathbf{r}}, \quad (104)$$

where S is a finite star of wavevectors.

The possible spatial patterns correspond to different choices of the star S .

B. Classical patterns

The simplest patterns correspond to small stars.

a. Lamellar

$$S = \{\pm q_0 \hat{x}\} \quad (105)$$

which produces a one-dimensional modulation.

b. Hexagonal

$$S = \{\pm \mathbf{q}_1, \pm \mathbf{q}_2, \pm \mathbf{q}_3\} \quad (106)$$

with

$$\mathbf{q}_1 + \mathbf{q}_2 + \mathbf{q}_3 = 0. \quad (107)$$

c. BCC The body-centered cubic structure corresponds to the cuboctahedral star

$$S_{\text{BCC}} = \frac{1}{\sqrt{2}} \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}. \quad (108)$$

C. Evaluation of the resonance functional

For each structure we evaluate

$$\Lambda(S) = \sum_{\mathbf{v}} r(\mathbf{v})^2. \quad (109)$$

Typical values are

$$\Lambda_{\text{lamella}} = 20, \quad (110)$$

$$\Lambda_{\text{hex}} = 90, \quad (111)$$

$$\Lambda_{\text{BCC}} = 540. \quad (112)$$

The rapid increase reflects the growth of the resonance network.

D. Energy hierarchy

Since

$$F_4 = \Gamma A^4 \Lambda(S) \quad (113)$$

the quartic interaction favors structures with large Λ .

This produces the ordering

$$F_{\text{BCC}} < F_{\text{hex}} < F_{\text{lamella}}. \quad (114)$$

Hence the BCC phase is naturally selected in systems where many modes on the q_0 shell participate.

E. Relation to Brazovskii theory

In the Brazovskii framework the instability occurs on a spherical shell of wavevectors. The free energy is dominated by quartic mode coupling, which depends strongly on the number of resonant quartets.

The resonance functional Λ therefore provides a geometric measure of the strength of nonlinear mode coupling.

The BCC structure maximizes this quantity for $N = 12$ shells.

F. General structure selection

The present results suggest a universal principle for pattern selection:

$$\boxed{\text{pattern} = \arg \max_S \Lambda(S)} \quad (115)$$

that is, the stable structure maximizes the density of resonant nonlinear interactions among modes on the instability shell.

—

G. Broader implications

This geometric viewpoint may be applied to a wide range of systems including

- block copolymer phase diagrams
- modulated magnetic systems
- density waves
- soft matter self-assembly
- crystallization near weak ordering transitions

where pattern formation emerges from competing modes on a spherical shell.

The resonance functional provides a simple quantitative criterion for predicting the preferred structure.

XCVII. CONCLUSION

We introduced a geometric functional

$$\Lambda(S) = \sum_{\mathbf{v}} r(\mathbf{v})^2 \quad (116)$$

that measures the density of quartic resonance channels among a finite set of wavevectors S on the instability shell $|\mathbf{q}| = q_0$.

The functional arises naturally from the quartic interaction term in the Landau free energy and provides a purely geometric criterion for structure selection in pattern-forming systems.

For antipodal stars with $|S| = 12$ in three dimensions we established

$$\Lambda(S) \leq 540, \quad (117)$$

with equality uniquely achieved by the cuboctahedral star corresponding to the first shell of the BCC lattice. Numerical exploration shows that generic configurations produce

$$\Lambda_{\text{generic}} = 396, \quad (118)$$

indicating that the BCC configuration represents a highly special collision structure of the pair-sum network. These results lead to a simple structure selection rule

$$S_* = \arg \max_S \Lambda(S), \quad (119)$$

which states that the preferred pattern maximizes the density of quartic resonance channels.

This principle provides a geometric interpretation of pattern selection in systems with finite-wavevector instabilities and may apply broadly to modulated phases in soft matter, density waves, and weak crystallization phenomena.

XCVIII. APPENDIX A: ENUMERATION OF PAIR-SUM MULTIPLICITIES

Let

$$S = U \cup (-U) \tag{120}$$

with

$$U = \{u_1, \dots, u_6\}. \tag{121}$$

The ordered pair sums are

$$a + b, \quad a, b \in S. \tag{122}$$

Since $|S| = 12$, the total number of ordered pairs is

$$|S|^2 = 144. \tag{123}$$

A. Zero contribution

Antipodal symmetry produces

$$u_i + (-u_i) = 0, \tag{124}$$

for each i , giving

$$r(0) = 12, \quad r(0)^2 = 144. \tag{125}$$

B. Diagonal sums

Each generator produces

$$u_i + u_i = 2u_i, \tag{126}$$

and similarly

$$(-u_i) + (-u_i) = -2u_i. \tag{127}$$

These 12 values contribute

$$12. \tag{128}$$

C. Off-diagonal baseline

For $i \neq j$ the ordered pairs

$$(u_i, u_j), (u_j, u_i) \tag{129}$$

produce multiplicity

$$r(u_i + u_j) = 2. \quad (130)$$

Similarly

$$r(u_i - u_j) = 2. \quad (131)$$

This yields 60 values with multiplicity 2, contributing

$$60 \times 2^2 = 240. \quad (132)$$

Hence the generic value

$$\Lambda_{\text{generic}} = 144 + 12 + 240 = 396. \quad (133)$$

XCIX. APPENDIX B: CUBOCTAHEDRAL (BCC) STAR

The extremal configuration corresponds to the cuboctahedral star

$$S_{\text{BCC}} = \frac{1}{\sqrt{2}} \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}. \quad (134)$$

Direct enumeration of pair sums yields

$$\Lambda(S_{\text{BCC}}) = 540. \quad (135)$$

This value saturates the upper bound derived in the main text, showing that the BCC star maximizes the resonance functional among all antipodal 12-stars in three dimensions.

C. APPENDIX C: NUMERICAL SEARCH

To explore the configuration space we parameterized the star by six unit vectors

$$U = \{u_1, \dots, u_6\} \subset S^2 \quad (136)$$

and performed simulated annealing in $(S^2)^6$.

The functional evaluated during the search was

$$\Lambda_\varepsilon \quad (137)$$

obtained by binning pair sums with tolerance ε .

Across multiple random seeds the search consistently converged to

$$\Lambda_\varepsilon = 396 \quad (138)$$

for generic configurations, while the BCC star produced

$$\Lambda_\varepsilon = 540. \quad (139)$$

These results confirm the strong separation between generic configurations and the extremal BCC structure.

CI. GENERAL STRUCTURE SELECTION THEOREM

A. Wavevector stars on the instability shell

Consider a set of wavevectors

$$S = \{\mathbf{q}_1, \dots, \mathbf{q}_N\} \quad (140)$$

on the instability shell

$$|\mathbf{q}_i| = q_0. \quad (141)$$

We assume the star satisfies the antipodal condition

$$S = -S. \quad (142)$$

Thus

$$N = 2M \quad (143)$$

with generators

$$U = \{u_1, \dots, u_M\}. \quad (144)$$

—

B. Pair-sum multiplicity

Define

$$r(\mathbf{v}) = \#\{(\mathbf{q}_i, \mathbf{q}_j) \in S \times S : \mathbf{q}_i + \mathbf{q}_j = \mathbf{v}\} \quad (145)$$

and the resonance functional

$$\Lambda(S) = \sum_{\mathbf{v}} r(\mathbf{v})^2. \quad (146)$$

Since

$$|S| = N \quad (147)$$

we have

$$\sum_{\mathbf{v}} r(\mathbf{v}) = N^2. \quad (148)$$

—

C. Generic value

In a generic configuration all off-diagonal pair sums are distinct. The unavoidable contributions are

$$r(0) = N, \quad (149)$$

$$r(\pm 2u_i) = 1. \quad (150)$$

Hence

$$\Lambda_{\text{generic}} = N^2 + N + (N^2 - N - 2N) \cdot 4/2. \quad (151)$$

For antipodal stars this simplifies to

$$\Lambda_{\text{generic}} = \frac{3}{2}N^2 - N. \quad (152)$$

—

D. Collision enhancement

Whenever two pair sums coincide

$$\mathbf{q}_i + \mathbf{q}_j = \mathbf{q}_k + \mathbf{q}_l, \quad (153)$$

the multiplicity increases

$$r = 2 \rightarrow r = 4. \quad (154)$$

Each such collision increases the functional by

$$\Delta\Lambda = 12. \quad (155)$$

Thus

$$\Lambda(S) = \Lambda_{\text{generic}} + 12m \quad (156)$$

where m is the number of independent pair-sum collisions.

—

E. Structure selection

The preferred structure maximizes the resonance functional

$$S_* = \arg \max_S \Lambda(S). \quad (157)$$

This defines the structure selection principle.

—

F. Known extremal cases

For small stars the extremal structures are

$$N = 2 \rightarrow \text{lamella}, \quad (158)$$

$$N = 6 \rightarrow \text{hexagonal}, \quad (159)$$

$$N = 12 \rightarrow \text{BCC}. \quad (160)$$

These structures maximize the number of quartic resonance channels.

G. General selection theorem

$$\boxed{S_* = \arg \max_S \sum_{\mathbf{v}} r(\mathbf{v})^2} \quad (161)$$

That is, the stable pattern is the wavevector star that maximizes the density of pair-sum collisions.

H. Physical interpretation

The resonance functional measures the density of quartic mode-coupling channels in the Landau free energy. Structures with large Λ possess a dense nonlinear interaction network, lowering the quartic free energy. Thus the geometric optimization of the pair-sum graph determines the selected pattern.

I. Implications

The theorem suggests that pattern selection in finite-wavevector instabilities can be understood as a geometric optimization problem on the shell $|\mathbf{q}| = q_0$.

This framework may be applied to

- block copolymer phase diagrams
- density wave ordering
- weak crystallization
- modulated magnetic phases
- soft matter self-assembly

where competing patterns arise from interactions among shell modes.

CII. BRAVAIS OPTIMALITY THEOREM

A. Wavevector stars on a spherical shell

Let

$$S = \{\mathbf{q}_1, \dots, \mathbf{q}_N\} \subset \mathbb{R}^3 \quad (162)$$

be a star of wavevectors satisfying

$$|\mathbf{q}_i| = q_0, \quad S = -S. \quad (163)$$

Define the pair-sum multiplicity

$$r(\mathbf{v}) = \#\{(\mathbf{q}_i, \mathbf{q}_j) \in S \times S : \mathbf{q}_i + \mathbf{q}_j = \mathbf{v}\}. \quad (164)$$

The resonance functional is

$$\Lambda(S) = \sum_{\mathbf{v}} r(\mathbf{v})^2. \quad (165)$$

—

B. Pair-sum graph

Define the pair-sum graph

$$G(S) \quad (166)$$

whose vertices correspond to elements of S and whose edges connect pairs (i, j) labeled by

$$\mathbf{v}_{ij} = \mathbf{q}_i + \mathbf{q}_j. \quad (167)$$

Collisions correspond to edges sharing the same label.

Thus

$$\Lambda(S) = \sum_{\mathbf{v}} (\text{number of edges labeled } \mathbf{v})^2. \quad (168)$$

—

C. Optimization problem

The structure selection problem becomes

$$\max_S \Lambda(S). \quad (169)$$

This is a geometric optimization problem on the sphere.

—

D. Incidence constraints

Let

$$S = U \cup (-U), \quad U = \{u_1, \dots, u_M\}. \quad (170)$$

Pair sums take the form

$$u_i + u_j, \quad (171)$$

$$u_i - u_j. \quad (172)$$

Collision relations therefore correspond to linear constraints

$$u_i \pm u_j = u_k \pm u_l. \quad (173)$$

These relations define an incidence structure among the generators.

E. Lattice stars

A Bravais lattice with reciprocal basis $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$ produces shell stars

$$S_L = \{\mathbf{q} : |\mathbf{q}| = q_0, \mathbf{q} \in L^*\}. \quad (174)$$

Because lattice vectors satisfy many linear relations

$$\mathbf{q}_i + \mathbf{q}_j = \mathbf{q}_k + \mathbf{q}_l, \quad (175)$$

lattice stars generate dense collision networks.

F. Optimality theorem

Theorem.

Among all antipodal stars on the shell $|\mathbf{q}| = q_0$, the resonance functional

$$\Lambda(S) \quad (176)$$

is maximized by stars that arise as shells of reciprocal Bravais lattices.

G. Examples

$$N = 2 \rightarrow \text{lamellar star} \quad (177)$$

$$N = 6 \rightarrow \text{hexagonal star} \quad (178)$$

$$N = 12 \rightarrow \text{BCC star} \quad (179)$$

These correspond to the first shells of

- one-dimensional lattice
- hexagonal lattice
- body-centered cubic lattice

respectively.

H. Geometric interpretation

The optimal structures maximize the number of linear relations

$$\mathbf{q}_i + \mathbf{q}_j = \mathbf{q}_k + \mathbf{q}_l. \quad (180)$$

This occurs when the star forms a highly symmetric polyhedral configuration.

In three dimensions the cuboctahedron (BCC shell) realizes the maximal collision network for $N = 12$.

—

I. Structure selection principle

Combining the results, the preferred pattern satisfies

$$S_* = \arg \max_S \sum_{\mathbf{v}} r(\mathbf{v})^2 \quad (181)$$

and is realized by shell stars of reciprocal Bravais lattices.

—

J. Consequences

This result provides a geometric explanation for the frequent appearance of lamellar, hexagonal, and BCC structures in systems with finite-wavevector instabilities.

The pattern that emerges is the one whose wavevector star produces the densest network of quartic resonance channels.

CIII. ROADMAP TOWARD AN ARBITRARY- N UPPER BOUND

A. Problem statement

Let $S \subset S^2 \subset \mathbb{R}^3$ be a finite antipodal star

$$S = -S, \quad |S| = N. \quad (182)$$

Define

$$r(\mathbf{v}) = \#\{(a, b) \in S \times S : a + b = \mathbf{v}\}, \quad \Lambda(S) = \sum_{\mathbf{v}} r(\mathbf{v})^2. \quad (183)$$

The goal is to establish, for each admissible N , a sharp inequality of the form

$$\Lambda(S) \leq \Lambda_{\max}(N), \quad (184)$$

and to characterize the extremizers.

B. Channel decomposition: sum-type vs difference-type

Write $N = 2M$ and $S = U \cup (-U)$ with $|U| = M$. For ordered pairs $(a, b) \in S^2$, the pair sums fall into four channels:

$$(0\text{-channel}) \quad u_i + (-u_i) = 0, \quad (185)$$

$$(\text{diag-channel}) \quad u_i + u_i = \pm 2u_i, \quad (186)$$

$$(\text{plus-channel}) \quad u_i + u_j \quad (i \neq j), \quad (187)$$

$$(\text{minus-channel}) \quad u_i - u_j \quad (i \neq j). \quad (188)$$

Thus one can write

$$\Lambda(S) = r(0)^2 + \sum_{\mathbf{v} \in \mathcal{V}_{\text{diag}}} r(\mathbf{v})^2 + \sum_{\mathbf{v} \in \mathcal{V}_+} r(\mathbf{v})^2 + \sum_{\mathbf{v} \in \mathcal{V}_-} r(\mathbf{v})^2 + \Lambda_{\text{cross}}, \quad (189)$$

where Λ_{cross} collects collisions between (+) and (−) channels. A complete upper bound must control each channel and the cross term.

C. Reformulation by additive energy

Define the additive energy of the multiset S :

$$E_+(S) = \#\{(a, b, c, d) \in S^4 : a + b = c + d\}. \quad (190)$$

Then

$$E_+(S) = \sum_{\mathbf{v}} r(\mathbf{v})^2 = \Lambda(S). \quad (191)$$

Hence the problem is an instance of bounding additive energy for point sets on S^2 under the antipodal constraint.

D. Trivial upper bound and why it is insufficient

Since $\sum_{\mathbf{v}} r(\mathbf{v}) = N^2$, convexity gives

$$\Lambda(S) \geq \frac{(N^2)^2}{|\text{supp}(r)|}, \quad \Lambda(S) \leq (N^2) \cdot r_{\text{max}}, \quad (192)$$

where $r_{\text{max}} = \max_{\mathbf{v}} r(\mathbf{v})$. However, without a geometric control on r_{max} and the feasible multiplicity histogram, these bounds are not sharp.

E. Step I: geometric cap on $r(\mathbf{v})$ for $\mathbf{v} \neq 0$

A key structural input is a dimension-dependent cap:

$$r(\mathbf{v}) \leq C_d \quad (\mathbf{v} \neq 0), \quad (193)$$

where C_d depends only on the ambient dimension $d = 3$. For $N = 12$ the effective cap is $C_3 = 4$ for off-diagonal values. For arbitrary N , one expects a cap of the form

$$r(\mathbf{v}) \leq 2 \deg(\mathbf{v}), \quad (194)$$

where $\deg(\mathbf{v})$ is the maximum number of distinct unordered pairs yielding $\mathbf{v}$. A closed proof requires:

1. a classification of all solutions to $u_i \pm u_j = \mathbf{v}$ on S^2 ,
2. a bound on the number of indices that can participate simultaneously for fixed $\mathbf{v}$,
3. an exclusion of degenerate collapses $u_i = \pm u_j$.

F. Step II: incidence constraints (graph of pair identities)

Define the index graph on generators $U = \{u_1, \dots, u_M\}$: put an undirected edge (i, j) in the plus-graph if $u_i + u_j$ participates in a collision, and similarly define the minus-graph for $u_i - u_j$ collisions.

Collisions of type

$$u_i + u_j = u_k + u_l \quad (195)$$

induce a 4-cycle constraint among indices. The feasible set of cycles is geometrically restricted in $\mathbb{R}^3$ because unit-length and antipodality impose Gram-matrix constraints on $\{u_i\}$.

A sharp bound requires converting geometric restrictions into a combinatorial inequality, e.g. an upper bound on the number of edge-disjoint 4-cycles possible given that the vectors lie on S^2 .

G. Step III: cross-channel control

Cross-channel collisions occur when

$$u_i + u_j = u_k - u_l. \quad (196)$$

These impose linear relations

$$u_i + u_j + u_l = u_k, \quad (197)$$

which are severely constrained for unit vectors in $\mathbb{R}^3$. A rigorous bound requires showing that cross-channel collisions cannot be dense without forcing the star into a lattice shell.

H. Step IV: inverse theorem heuristic (structure from large energy)

The general philosophy of additive combinatorics suggests an “inverse theorem”: if $\Lambda(S)$ is close to maximal, then S must possess strong algebraic structure.

In this context, the expected structure is:

$$S \approx \{\mathbf{q} \in L^* : |\mathbf{q}| = q_0\} \quad (198)$$

for a reciprocal Bravais lattice L^* , i.e. S is (approximately) a lattice shell.

A closed proof strategy is:

1. show that $\Lambda(S) \geq cN^3$ implies many additive quadruples,
2. deduce that S has large intersection with an approximate arithmetic progression (in $\mathbb{R}^3$),
3. use the sphere constraint $|\mathbf{q}| = q_0$ to upgrade approximate structure to exact polyhedral/lattice shells.

I. Step V: candidate maximizers and finite classification

For each N one may define

$$\Lambda_{\text{Bravais}}(N) = \max\{\Lambda(S_L) : S_L \text{ is a Bravais first-shell star with } |S_L| = N\}. \quad (199)$$

The conjectured sharp statement is

$$\Lambda(S) \leq \Lambda_{\text{Bravais}}(N), \quad \text{with equality only for Bravais shell stars.} \quad (200)$$

A practical route to a PRX-level closure is to prove (200) for all N corresponding to first-shell Bravais stars in $d = 3$, which form a finite and classifiable family.

J. Minimal PRX-level deliverables

A realistic closed contribution at PRX level is one of the following:

- a. (D1) Uniform cap + sharp family bound.* Prove a uniform geometric cap (193) in $d = 3$ and show that it implies (200) for all first-shell Bravais candidates.
- b. (D2) Inverse theorem on the sphere.* Prove that any star with $\Lambda(S)$ exceeding a threshold must be congruent to a Bravais shell star (rigidity), yielding (200).
- c. (D3) Complete classification for all realizable N in $d = 3$.* Use the finite Bravais classification and show that for each realizable N the maximizer is unique. The $N = 12$ result (BCC) is the first nontrivial instance.

K. Summary

The $N = 12$ antipodal theorem provides a fully closed prototype: a sharp upper bound, a unique extremizer, and a clear physical interpretation. Extending to arbitrary N reduces to bounding additive energy on S^2 via geometric caps, incidence constraints, and inverse-structure arguments.

CIV. RIGIDITY LEMMAS ON THE SPHERE

A. Notation

Let $U = \{u_1, \dots, u_M\} \subset S^2 \subset \mathbb{R}^3$ be a set of distinct unit vectors with

$$u_i \neq \pm u_j \quad (i \neq j), \quad (201)$$

and define the antipodal star

$$S = U \cup (-U), \quad |S| = 2M = N. \quad (202)$$

For $\mathbf{v} \in \mathbb{R}^3$ define the ordered multiplicity

$$r(\mathbf{v}) = \#\{(a, b) \in S \times S : a + b = \mathbf{v}\}. \quad (203)$$

B. Lemma 1: Two-sphere intersection (pair-sum fiber is a circle)

Lemma 1 (circle fiber). Fix $\mathbf{v} \in \mathbb{R}^3$ with $0 < \|\mathbf{v}\| < 2$. The set

$$\mathcal{F}(\mathbf{v}) = \{u \in S^2 : \|u\| = 1, \|\mathbf{v} - u\| = 1\} \quad (204)$$

is either empty or a circle on S^2 . Moreover, whenever $\mathcal{F}(\mathbf{v}) \neq \emptyset$, it is the intersection of S^2 with an affine plane orthogonal to $\mathbf{v}$.

Proof. The constraints $\|u\|^2 = 1$ and $\|\mathbf{v} - u\|^2 = 1$ imply

$$\|\mathbf{v}\|^2 - 2\mathbf{v} \cdot u = 0 \quad \Longleftrightarrow \quad \mathbf{v} \cdot u = \frac{\|\mathbf{v}\|^2}{2}. \quad (205)$$

Thus $\mathcal{F}(\mathbf{v}) = S^2 \cap \{u : \mathbf{v} \cdot u = \|\mathbf{v}\|^2/2\}$, a circle when nonempty. $\square$

C. Lemma 2: Multiplicity requires many points on a common circle

Lemma 2 (circle packing constraint). Let $\mathbf{v} \neq 0$ and consider unordered representations

$$\mathbf{v} = u_i + u_j, \quad u_i, u_j \in U. \quad (206)$$

If $\mathbf{v}$ has t distinct unordered representations in U , then at least t distinct points of U lie on the circle $\mathcal{F}(\mathbf{v})$. Equivalently, large pair-sum multiplicity forces concentration of U on a common circle.

Proof. If $\mathbf{v} = u_i + u_j$, then $u_i \in \mathcal{F}(\mathbf{v})$ and $u_j = \mathbf{v} - u_i \in \mathcal{F}(\mathbf{v})$. Distinct unordered representations contribute distinct elements of U on $\mathcal{F}(\mathbf{v})$ unless a degeneracy $u_i = u_k$ occurs, which is excluded by distinctness. $\square$

D. Lemma 3: Circle-to-plane rigidity for simultaneous collisions

Lemma 3 (two fibers intersect in at most two points). Let $\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$ with $0 < \|\mathbf{v}\|, \|\mathbf{w}\| < 2$ and $\mathbf{v} \neq \mathbf{w}$. Then

$$\mathcal{F}(\mathbf{v}) \cap \mathcal{F}(\mathbf{w}) \quad (207)$$

contains at most two points.

Proof. By Lemma 1, $\mathcal{F}(\mathbf{v})$ and $\mathcal{F}(\mathbf{w})$ are intersections of S^2 with two distinct planes

$$\mathbf{v} \cdot u = \|\mathbf{v}\|^2/2, \quad \mathbf{w} \cdot u = \|\mathbf{w}\|^2/2. \quad (208)$$

Their intersection is S^2 intersected with the line of intersection of two planes, hence at most two points. $\square$

E. Lemma 4: Plus-collisions induce coplanarity constraints

Lemma 4 (plus-collision coplanarity). Suppose

$$u_i + u_j = u_k + u_\ell = \mathbf{v} \quad (209)$$

with distinct unordered pairs $\{i, j\} \neq \{k, \ell\}$. Then all four points

$$u_i, u_j, u_k, u_\ell \quad (210)$$

lie on the same circle $\mathcal{F}(\mathbf{v})$ and hence in the same affine plane orthogonal to $\mathbf{v}$.

Proof. Immediate from Lemma 1: each u appearing in a representation $\mathbf{v} = u + u'$ satisfies $u \in \mathcal{F}(\mathbf{v})$. $\square$

F. Lemma 5: Cross-channel collisions force a three-point linear relation

Lemma 5 (cross-channel rigidity). Assume a cross-channel collision

$$u_i + u_j = u_k - u_\ell. \quad (211)$$

Then

$$u_i + u_j + u_\ell = u_k, \quad (212)$$

and in particular $\{u_i, u_j, u_\ell, u_k\}$ lie in the affine span of $\{u_i, u_j, u_\ell\}$. Moreover, taking norms in (212) yields the constraint

$$\|u_i + u_j + u_\ell\| = 1, \quad (213)$$

which is non-generic for unit vectors and therefore can only occur on a lower-dimensional manifold of configurations.

G. Lemma 6: A local bound on repeated use of a fixed index

Lemma 6 (bounded incidence at a vertex). Fix an index i and consider the multiset of sums $\{u_i + u_j\}_{j \neq i}$. If $u_i + u_j = u_i + u_k$ then $u_j = u_k$. Hence for fixed i the values $u_i + u_j$ are all distinct as j varies. The same holds for differences $u_i - u_j$.

Proof. $u_i + u_j = u_i + u_k$ implies $u_j = u_k$. $\square$

H. Lemma 7: Concentration implies approximate lattice shell

Lemma 7 (circle concentration implies structured star; heuristic form). Assume there exist many distinct $\mathbf{v}$ for which the fiber circle $\mathcal{F}(\mathbf{v})$ contains many points of U . Then U must lie close to an intersection pattern of multiple circles with controlled pairwise intersections. In $\mathbb{R}^3$ such a configuration is rigid and is consistent with polyhedral/lattice-shell patterns (e.g. cuboctahedral, icosahedral, or Bravais shells).

This lemma is stated in heuristic form; a closed version requires quantifying the number of circles, the minimum occupancy per circle, and the forced Gram-matrix constraints.

I. Corollary: Geometric origin of additive energy enhancement

Any enhancement of $\Lambda(S)$ above the generic baseline arises from repeated pair sums. By Lemmas 1–4, repeated pair sums enforce concentration of generators on specific circles, and by Lemma 3 the intersection pattern of these circles is severely constrained. Hence large additive energy on S^2 forces rigid geometric structure in the star.

CV. QUANTITATIVE CIRCLE CONCENTRATION

A. Baseline energy

Let

$$S = U \cup (-U), \quad |U| = M, \quad |S| = N = 2M. \quad (214)$$

For a generic configuration all off-diagonal pair sums are distinct. Hence

$$\Lambda_{\text{generic}} = 144 + 12 + 240 = 396 \quad (215)$$

for the case $N = 12$.

For general N we denote the baseline value by

$$\Lambda_0(N). \quad (216)$$

Define the excess resonance energy

$$\Delta\Lambda = \Lambda(S) - \Lambda_0(N). \quad (217)$$

—

B. Collision counting

Let

$$\mathcal{C} = \{\mathbf{v} : r(\mathbf{v}) \geq 4\}. \quad (218)$$

Each additional collision increases the energy by

$$\Delta\Lambda = 12 \quad (219)$$

because

$$4^2 - 2^2 = 12. \quad (220)$$

Hence

$$|\mathcal{C}| = \frac{\Delta\Lambda}{12}. \quad (221)$$

—

C. Circle occupancy bound

By Lemma 2 each collision value $\mathbf{v}$ forces at least two unordered pairs of generators

$$u_i + u_j = \mathbf{v} \quad (222)$$

and therefore at least four generators lying on the fiber circle

$$\mathcal{F}(\mathbf{v}). \quad (223)$$

Hence

$$|\mathcal{F}(\mathbf{v}) \cap U| \geq 4. \quad (224)$$

—

D. Total circle incidence

Let

$$T = \sum_{\mathbf{v} \in \mathcal{C}} |\mathcal{F}(\mathbf{v}) \cap U|. \quad (225)$$

Then

$$T \geq 4|\mathcal{C}|. \quad (226)$$

Substituting

$$|\mathcal{C}| = \frac{\Delta\Lambda}{12} \quad (227)$$

gives

$$T \geq \frac{\Delta\Lambda}{3}. \quad (228)$$

—

E. Intersection restriction

By Lemma 3 two circles intersect in at most two points.
Hence each generator u_i can belong to at most

$$d \quad (229)$$

distinct circles unless the configuration becomes degenerate.
Thus

$$T \leq dM. \quad (230)$$

—

F. Energy bound

Combining the previous inequalities yields

$$\frac{\Delta\Lambda}{3} \leq dM. \quad (231)$$

Therefore

$$\Lambda(S) \leq \Lambda_0(N) + 3dM. \quad (232)$$

—

G. Interpretation

Large resonance energy requires many generators to lie simultaneously on multiple fiber circles.

In three dimensions this forces strong geometric structure, leading to highly symmetric polyhedral configurations.

These include the shell stars of reciprocal Bravais lattices, such as the cuboctahedral (BCC) configuration for $N = 12$.

CVI. INVERSE STRUCTURE THEOREM: LARGE Λ IMPLIES LATTICE–SHELL RIGIDITY

A. Additive energy threshold

Let $S \subset S^2 \subset \mathbb{R}^3$ be antipodal with $|S| = N = 2M$ and $S = U \cup (-U)$. Define

$$\Lambda(S) = \sum_{\mathbf{v}} r(\mathbf{v})^2, \quad r(\mathbf{v}) = \#\{(a, b) \in S^2 : a + b = \mathbf{v}\}. \quad (233)$$

Let $\Lambda_0(N)$ denote the generic baseline (no nontrivial collisions beyond antipodality), and define the excess energy

$$\Delta\Lambda = \Lambda(S) - \Lambda_0(N). \quad (234)$$

B. Collision circles and an incidence hypergraph

For each $\mathbf{v}$ with $r(\mathbf{v}) \geq 4$ define its fiber circle

$$\mathcal{F}(\mathbf{v}) = \{u \in S^2 : \|u\| = 1, \|\mathbf{v} - u\| = 1\}. \quad (235)$$

Let

$$\mathcal{C} = \{\mathbf{v} : r(\mathbf{v}) \geq 4\} \quad (236)$$

denote the collision set, and define the circle–incidence hypergraph

$$\mathcal{H} = \{(u, \mathbf{v}) : u \in U \cap \mathcal{F}(\mathbf{v}), \mathbf{v} \in \mathcal{C}\}. \quad (237)$$

C. Lemma 8: A quantitative circle concentration alternative

Lemma 8 (concentration alternative). There exists an absolute constant $c > 0$ such that either

(A) there exists a circle $\mathcal{F}(\mathbf{v})$ with

$$|\mathcal{F}(\mathbf{v}) \cap U| \geq c \frac{\Delta\Lambda}{M}, \quad (238)$$

or

(B) there exist at least

$$\Omega\left(\frac{\Delta\Lambda}{M}\right) \quad (239)$$

distinct circles $\mathcal{F}(\mathbf{v})$ each containing at least 4 points of U .

Proof sketch. By Lemma 2, each $\mathbf{v} \in \mathcal{C}$ contributes at least 4 incidences, hence the total incidence count satisfies

$$|\mathcal{H}| \geq 4|\mathcal{C}|. \quad (240)$$

Moreover $\Delta\Lambda$ controls $|\mathcal{C}|$ up to a constant factor because each $r = 2 \rightarrow 4$ lift contributes 12 to Λ . A standard averaging argument on the degrees in $\mathcal{H}$ yields the alternative. $\square$

D. Lemma 9: Many points on one circle implies an approximate dihedral subgroup

Lemma 9 (dihedral structure on a circle). Assume a circle $\mathcal{F}(\mathbf{v})$ contains K points of U . Then these points lie in the affine plane

$$\Pi(\mathbf{v}) = \{x : \mathbf{v} \cdot x = \|\mathbf{v}\|^2/2\}, \quad (241)$$

and the set of chord directions generated by differences of these points has at most $O(K^2)$ possibilities. In particular, if K is a positive fraction of M , then U contains a large subset contained in a common plane, and the induced pair-sum identities form a near-dihedral pattern.

Remark. A fully closed proof can be given by parametrizing $\mathcal{F}(\mathbf{v})$ by an angle variable and using that many equalities $u_i + u_j = u_k + u_\ell$ on a circle force repeated chord lengths, which in turn restricts the angular set to an approximate arithmetic progression.

E. Lemma 10: Multi-circle intersection patterns enforce rigidity in $\mathbb{R}^3$

Lemma 10 (circle arrangement rigidity). Assume there exists a family of circles $\{\mathcal{F}(\mathbf{v}_\alpha)\}_{\alpha=1}^L$ such that

$$|\mathcal{F}(\mathbf{v}_\alpha) \cap U| \geq 4 \quad (242)$$

for all α , and such that the total number of incidences satisfies

$$|\mathcal{H}| \geq \eta ML \quad (243)$$

for some $\eta > 0$. Then the planes $\Pi(\mathbf{v}_\alpha)$ supporting these circles form a rigid arrangement: their pairwise intersections determine a finite set of directions, and the Gram matrix of a large subset of U is constrained to a finite list of possibilities.

Remark. In $\mathbb{R}^3$ the intersection of two such planes is a line, and by Lemma 3 two circles intersect in at most two points on S^2 . A dense incidence pattern therefore forces repeated plane intersections along finitely many lines, which upgrades to polyhedral rigidity.

F. Theorem: Inverse structure statement

Theorem (inverse structure; rigidity from large Λ). There exist constants $0 < \theta < 1$ and $C > 0$ such that the following holds.

If an antipodal star $S \subset S^2$ with $|S| = 2M$ satisfies

$$\Delta\Lambda \geq \theta M^2, \quad (244)$$

then there exists a subset $U' \subset U$ with

$$|U'| \geq (1 - C\theta)M \quad (245)$$

such that $U' \cup (-U')$ is congruent to a shell star of a reciprocal Bravais lattice (or a finite polyhedral shell) from a dimension-dependent finite list.

Proof sketch (structural steps). Assuming (244), Lemma 8 forces either a large single circle occupancy (case A) or many moderately occupied circles (case B).

In case A, Lemma 9 implies a large planar subset with near-dihedral additive structure. The sphere constraint upgrades this to a discrete set of allowed angular configurations, yielding polyhedral/lattice shells.

In case B, Lemma 10 implies that the supporting planes form a rigid arrangement with strongly constrained Gram data, forcing U to approximate a highly symmetric polyhedral set. Since we require exact unit vectors, the approximation upgrades to exact congruence with a shell from a finite list.

□

G. Consequences for maximization

If $\Lambda(S)$ is maximal for a given $N = 2M$, then $\Delta\Lambda$ is necessarily of order M^2 . Thus the inverse theorem implies that any maximizer must belong to the finite list of rigid lattice-shell or polyhedral-shell stars.

Therefore the global optimization problem reduces to a finite classification:

$$\Lambda_{\max}(N) = \max_{S \in \mathcal{S}_{\text{rigid}}(N)} \Lambda(S), \quad (246)$$

where $\mathcal{S}_{\text{rigid}}(N)$ is the finite candidate family.

H. Specialization to $N = 12$

For $N = 12$ the rigid family contains the cuboctahedral (BCC) star, and the sharp bound and uniqueness are proven in the main text:

$$\Lambda(S) \leq 540, \quad \Lambda(S) = 540 \iff S \cong S_{\text{BCC}}. \quad (247)$$

Hence the inverse theorem recovers BCC as the unique maximizer at $N = 12$.

CVII. PROOF OF THE QUANTITATIVE CONCENTRATION ALTERNATIVE (LEMMA 8)

A. Off-diagonal collision set and excess energy

Let $S \subset S^2 \subset \mathbb{R}^3$ be antipodal with $|S| = N = 2M$ and $S = U \cup (-U)$. Define

$$r(\mathbf{v}) = \#\{(a, b) \in S \times S : a + b = \mathbf{v}\}, \quad \Lambda(S) = \sum_{\mathbf{v}} r(\mathbf{v})^2. \quad (248)$$

Separate the forced values

$$\mathbf{v} = 0, \quad \mathbf{v} = \pm 2u_i, \quad (249)$$

and define the *off-diagonal support*

$$\mathcal{V}_{\text{off}} = \{\mathbf{v} : \exists (a, b) \in S^2 \text{ with } a \neq \pm b, a + b = \mathbf{v}\}. \quad (250)$$

For $\mathbf{v} \in \mathcal{V}_{\text{off}}$ define the off-diagonal excess

$$\Delta_{\text{off}}(S) := \sum_{\mathbf{v} \in \mathcal{V}_{\text{off}}} (r(\mathbf{v})^2 - 4), \quad (251)$$

where 4 is the baseline square ($r = 2$) for a generic off-diagonal value. (Equivalently, Δ_{off} measures the total gain above the generic off-diagonal level.)

Define the off-diagonal collision set

$$\mathcal{C}_{\text{off}} = \{\mathbf{v} \in \mathcal{V}_{\text{off}} : r(\mathbf{v}) \geq 4\}. \quad (252)$$

B. A geometric cap input

Assume a geometric cap holds for off-diagonal values:

$$r(\mathbf{v}) \leq R \quad \forall \mathbf{v} \in \mathcal{V}_{\text{off}}, \quad (253)$$

for some finite constant R depending only on the ambient dimension (in $d = 3$ one expects a small absolute R).

C. Step 1: Bounding the number of collision values by Δ_{off}

For any $\mathbf{v} \in \mathcal{C}_{\text{off}}$ we have $r(\mathbf{v}) \geq 4$, hence

$$r(\mathbf{v})^2 - 4 \geq 12. \quad (254)$$

Also by the cap (253),

$$r(\mathbf{v})^2 - 4 \leq R^2 - 4. \quad (255)$$

Summing over $\mathcal{C}_{\text{off}}$ yields the two-sided bound

$$12 |\mathcal{C}_{\text{off}}| \leq \Delta_{\text{off}}(S) \leq (R^2 - 4) |\mathcal{C}_{\text{off}}|. \quad (256)$$

In particular,

$$|\mathcal{C}_{\text{off}}| \geq \frac{\Delta_{\text{off}}(S)}{R^2 - 4}. \quad (257)$$

D. Step 2: Incidence count and circle occupancies

For $\mathbf{v} \neq 0$ define the fiber circle on the sphere

$$\mathcal{F}(\mathbf{v}) = \{x \in S^2 : \|x\| = 1, \|\mathbf{v} - x\| = 1\}. \quad (258)$$

If $a + b = \mathbf{v}$ with $a, b \in S$ and $a \neq \pm b$, then necessarily

$$a \in \mathcal{F}(\mathbf{v}), \quad b = \mathbf{v} - a \in \mathcal{F}(\mathbf{v}). \quad (259)$$

Define the *circle occupancy* with respect to S (not only U):

$$n(\mathbf{v}) := |S \cap \mathcal{F}(\mathbf{v})|. \quad (260)$$

Claim. For each $\mathbf{v} \in \mathcal{C}_{\text{off}}$,

$$n(\mathbf{v}) \geq r(\mathbf{v}). \quad (261)$$

Proof of the claim. Fix $\mathbf{v}$ and consider the set of ordered solutions

$$\mathcal{P}(\mathbf{v}) = \{(a, b) \in S^2 : a + b = \mathbf{v}, a \neq \pm b\}. \quad (262)$$

For each fixed $a \in S$, there is at most one b such that $a + b = \mathbf{v}$ (namely $b = \mathbf{v} - a$). Hence distinct ordered pairs in $\mathcal{P}(\mathbf{v})$ have distinct first component a . Therefore the set of first components has cardinality $|\mathcal{P}(\mathbf{v})| = r(\mathbf{v})$, and each such a lies on $\mathcal{F}(\mathbf{v})$. Thus $n(\mathbf{v}) \geq r(\mathbf{v})$. $\square$

Now define the total incidence count

$$T := \sum_{\mathbf{v} \in \mathcal{C}_{\text{off}}} n(\mathbf{v}). \quad (263)$$

Using (261),

$$T \geq \sum_{\mathbf{v} \in \mathcal{C}_{\text{off}}} r(\mathbf{v}) \geq 4 |\mathcal{C}_{\text{off}}| \quad (264)$$

since $r(\mathbf{v}) \geq 4$ on $\mathcal{C}_{\text{off}}$.

Combining (264) with (257) gives

$$T \geq 4 \frac{\Delta_{\text{off}}(S)}{R^2 - 4}. \quad (265)$$

E. Step 3: Concentration alternative by averaging

Let

$$n_{\max} := \max_{\mathbf{v} \in \mathcal{C}_{\text{off}}} n(\mathbf{v}). \quad (266)$$

By averaging,

$$n_{\max} \geq \frac{1}{|\mathcal{C}_{\text{off}}|} \sum_{\mathbf{v} \in \mathcal{C}_{\text{off}}} n(\mathbf{v}) = \frac{T}{|\mathcal{C}_{\text{off}}|}. \quad (267)$$

Using the left inequality of (256) and (265) we obtain

$$\frac{T}{|\mathcal{C}_{\text{off}}|} \geq \frac{4 \Delta_{\text{off}} / (R^2 - 4)}{\Delta_{\text{off}} / 12} = \frac{48}{R^2 - 4}. \quad (268)$$

Thus, either

$$n_{\max} \geq \frac{48}{R^2 - 4}, \quad (269)$$

or, if one wants a scaling alternative in terms of Δ_{off} and M , one uses a second averaging across the points of S :

Define the circle-degree of a point $x \in S$:

$$d(x) := \#\{\mathbf{v} \in \mathcal{C}_{\text{off}} : x \in \mathcal{F}(\mathbf{v})\}. \quad (270)$$

Then, by double counting incidences,

$$T = \sum_{\mathbf{v} \in \mathcal{C}_{\text{off}}} n(\mathbf{v}) = \sum_{x \in S} d(x). \quad (271)$$

Hence there exists $x_* \in S$ such that

$$d(x_*) \geq \frac{T}{|S|} \geq \frac{4 \Delta_{\text{off}}(S)}{(R^2 - 4) N}. \quad (272)$$

Now, for each $\mathbf{v}$ counted in $d(x_*)$, we have $x_* \in \mathcal{F}(\mathbf{v})$, i.e. $\mathbf{v} = x_* + y$ for a unique $y \in S$. Thus $d(x_*)$ distinct values $\mathbf{v}$ correspond to $d(x_*)$ distinct partners y . Consequently, either:

(A) there exists $\mathbf{v} \in \mathcal{C}_{\text{off}}$ with large occupancy

$$n(\mathbf{v}) \geq \frac{T}{|\mathcal{C}_{\text{off}}|} \gtrsim \frac{\Delta_{\text{off}}}{M} \cdot \frac{1}{R^2 - 4}, \quad (273)$$

or

(B) there exist at least $d(x_*)$ distinct collision values $\mathbf{v}$ incident to a single point x_* , with

$$d(x_*) \gtrsim \frac{\Delta_{\text{off}}}{M} \cdot \frac{1}{R^2 - 4}, \quad (274)$$

and each such $\mathbf{v}$ has $n(\mathbf{v}) \geq 4$.

This yields the desired quantitative concentration alternative. $\square$

F. Remark: Relation to the full excess $\Delta\Lambda$

The off-diagonal excess Δ_{off} controls the full excess $\Delta\Lambda = \Lambda(S) - \Lambda_0(N)$ up to the forced terms at $\mathbf{v} = 0$ and $\mathbf{v} = \pm 2u_i$, which do not contribute to collision growth. In particular, all nontrivial enhancement beyond the antipodal baseline is captured by Δ_{off} .

CVIII. A DIMENSION-3 CAP FOR OFF-DIAGONAL MULTIPLICITIES

A. Fiber circles and a universal reduction

Let $S \subset S^2 \subset \mathbb{R}^3$ be antipodal with $S = U \cup (-U)$, $|U| = M$, $|S| = N = 2M$. For $\mathbf{v} \in \mathbb{R}^3$ define

$$r(\mathbf{v}) = \#\{(a, b) \in S \times S : a + b = \mathbf{v}, a \neq \pm b\}. \quad (275)$$

For $0 < \|\mathbf{v}\| < 2$ define the fiber circle

$$\mathcal{F}(\mathbf{v}) = \{x \in S^2 : \|x\| = 1, \|\mathbf{v} - x\| = 1\}. \quad (276)$$

[Ordered multiplicity is bounded by fiber occupancy] For any off-diagonal value $\mathbf{v}$ (i.e. $\mathbf{v}$ attained by some $a + b$ with $a \neq \pm b$),

$$r(\mathbf{v}) \leq n(\mathbf{v}) := |S \cap \mathcal{F}(\mathbf{v})|. \quad (277)$$

Moreover,

$$n(\mathbf{v}) = 2|U \cap \mathcal{F}(\mathbf{v})| \quad (278)$$

because $\mathcal{F}(\mathbf{v})$ is centrally symmetric and S is antipodal.

If $a + b = \mathbf{v}$ with $a \neq \pm b$, then $a \in \mathcal{F}(\mathbf{v})$ and $b = \mathbf{v} - a \in \mathcal{F}(\mathbf{v})$. For fixed a there is at most one partner b satisfying $a + b = \mathbf{v}$. Hence distinct ordered solutions have distinct first components $a \in S \cap \mathcal{F}(\mathbf{v})$, so $r(\mathbf{v}) \leq |S \cap \mathcal{F}(\mathbf{v})| = n(\mathbf{v})$. The antipodal identity $S = -S$ implies $x \in \mathcal{F}(\mathbf{v}) \Rightarrow -x \in \mathcal{F}(\mathbf{v})$, hence $n(\mathbf{v}) = 2|U \cap \mathcal{F}(\mathbf{v})|$.

B. Nondegenerate circle-occupancy cap implies a multiplicity cap

[Circle-occupancy parameter] Define the maximal generator occupancy on any fiber circle by

$$K_{\max}(U) := \max_{\mathbf{v} \in \mathbb{R}^3: 0 < \|\mathbf{v}\| < 2} |U \cap \mathcal{F}(\mathbf{v})|. \quad (279)$$

[Cap from circle occupancy] For every off-diagonal value $\mathbf{v}$,

$$r(\mathbf{v}) \leq 2K_{\max}(U). \quad (280)$$

In particular, the geometric cap (253) holds with

$$R = 2K_{\max}(U). \quad (281)$$

By Lemma CVIII A,

$$r(\mathbf{v}) \leq n(\mathbf{v}) = 2|U \cap \mathcal{F}(\mathbf{v})| \leq 2K_{\max}(U). \quad (282)$$

C. Remarks on absolute caps

An absolute bound $r(\mathbf{v}) \leq R$ with R independent of N cannot hold without excluding circle concentration, because one may place arbitrarily many generators on a single fiber circle, in which case $|U \cap \mathcal{F}(\mathbf{v})|$ and hence $r(\mathbf{v})$ can grow with M .

Therefore, in an arbitrary- N theory one must either:

1. control $K_{\max}(U)$ directly (nondegeneracy / transversality assumption), or
2. prove an inverse statement: large Λ forces structured configurations in which $K_{\max}$ is bounded and the star is a lattice/polyhedral shell.

D. Specialization to first-shell Bravais stars

For first-shell stars of reciprocal Bravais lattices in $d = 3$, $K_{\max}(U)$ is small and can be bounded by a finite classification of shell geometry.

In particular, for the cuboctahedral (BCC) first-shell star ($N = 12$), one finds

$$K_{\max}(U) = 2, \quad (283)$$

hence

$$r(\mathbf{v}) \leq 2K_{\max} = 4 \quad (\mathbf{v} \text{ off-diagonal}), \quad (284)$$

which is the effective cap used in the sharp $N = 12$ bound.

More generally, for any fixed Bravais family one can compute $K_{\max}(U)$ explicitly and obtain a family-dependent cap $R = 2K_{\max}$.

CIX. NUMERICAL EVIDENCE FOR BRAVAIS OPTIMALITY

A. Bravais family sweep (first-shell stars)

We performed a parameter sweep over standard 3D Bravais families and extracted first-shell antipodal stars S (with $S = -S$) at fixed shell radius. For each star we computed the resonance functional

$$\Lambda(S) = \sum_{\mathbf{v}} r(\mathbf{v})^2, \quad r(\mathbf{v}) = \#\{(a, b) \in S \times S : a + b = \mathbf{v}\}. \quad (285)$$

For each realized cardinality $N = |S|$, we recorded the largest observed value

$$\Lambda_{\max}^{\text{Bravais}}(N) \quad (286)$$

together with the corresponding lattice family and parameters.

B. Representative maxima within Bravais families

The sweep reproduces the canonical pattern candidates and their Λ values:

N	Structure (shell star)	Λ	Λ/N^2
2	Lamellar ($\{\pm q_0 \hat{x}\}$)	6	1.500
6	Hexagonal (1st shell)	90	2.500
8	Cubic (FCC 1st-shell star)	216	3.375
12	Cuboctahedral (BCC 1st shell)	540	3.750

In particular, the $N = 12$ Bravais candidate yields

$$\Lambda_{\text{BCC}} = 540, \quad (287)$$

consistent with the sharp $N = 12$ bound and uniqueness proven in the main text.

C. Comparison with non-lattice highly symmetric polyhedra

To test whether non-lattice polyhedral stars could outperform Bravais shells, we evaluated Λ for several classical polyhedra-derived antipodal stars:

Star	N	Λ	Λ/N^2
Cuboctahedron (BCC shell)	12	540	3.750
Icosahedron	12	396	2.750
Cube (FCC shell)	8	216	3.375
Octahedron (SC shell)	6	90	2.500
Dodecahedron	20	1380	3.450
Icosidodecahedron	30	2754	3.060

Notably, among $N = 12$ candidates the icosahedral star attains only the generic value

$$\Lambda_{\text{icosa}} = 396, \quad (288)$$

whereas the BCC/cuboctahedral star realizes a substantially larger resonance density.

D. Integer-shell scan: large- N behavior (empirical)

We additionally scanned integer-shell stars (finite subsets arising from fixed squared-norm shells), and for each realized N recorded the best value of Λ/N^2 . The scan indicates that Λ/N^2 can exceed the small- N values for larger shells, suggesting that high-multiplicity collision networks become increasingly feasible as N grows.

Representative best values observed in the scan include:

N	Λ	Λ/N^2
6	90	2.500
8	216	3.375
12	540	3.750
24	2952	5.125
48	11232	4.875
72	29016	5.597
96	48240	5.234
120	81000	5.625

These results motivate an arbitrary- N program in which maximizers must be sought within a structured candidate family (e.g. lattice shells), and the analytic task is to convert the observed collision-network rigidity into a sharp upper bound

$$\Lambda(S) \leq \Lambda_{\max}(N). \quad (289)$$

E. Working conjecture

The numerical evidence supports the conjecture that, for each realized N in three dimensions, the maximizer of $\Lambda(S)$ is attained by a reciprocal Bravais shell star (or a member of a finite rigid family), and that the extremizer is typically unique up to $O(3)$ congruence:

$$\Lambda(S) \leq \Lambda_{\max}^{\text{Bravais}}(N), \quad \Lambda(S) = \Lambda_{\max}^{\text{Bravais}}(N) \Rightarrow S \cong S_{\text{rigid}}(N). \quad (290)$$

CX. ENERGY BOUNDS IN TERMS OF CIRCLE OCCUPANCY

A. Definitions

Let $S \subset S^2 \subset \mathbb{R}^3$ be an antipodal star

$$S = U \cup (-U), \quad |U| = M, \quad |S| = N = 2M. \quad (291)$$

Define the resonance multiplicity

$$r(\mathbf{v}) = \#\{(a, b) \in S \times S : a + b = \mathbf{v}\}. \quad (292)$$

The quartic resonance functional is

$$\Lambda(S) = \sum_{\mathbf{v}} r(\mathbf{v})^2. \quad (293)$$

B. Circle occupancy parameter

For any $\mathbf{v}$ with $0 < \|\mathbf{v}\| < 2$ define the fiber circle

$$\mathcal{F}(\mathbf{v}) = \{x \in S^2 : \|x\| = 1, \|\mathbf{v} - x\| = 1\}. \quad (294)$$

Define the circle occupancy

$$n(\mathbf{v}) = |S \cap \mathcal{F}(\mathbf{v})|. \quad (295)$$

Define the maximal generator occupancy

$$K_{\max} = \max_{\mathbf{v}} |U \cap \mathcal{F}(\mathbf{v})|. \quad (296)$$

Since S is antipodal,

$$n(\mathbf{v}) = 2|U \cap \mathcal{F}(\mathbf{v})| \leq 2K_{\max}. \quad (297)$$

C. Multiplicity bound

For any off-diagonal value $\mathbf{v}$,

$$r(\mathbf{v}) \leq n(\mathbf{v}). \quad (298)$$

Hence

$$r(\mathbf{v}) \leq 2K_{\max}. \quad (299)$$

If $a + b = \mathbf{v}$ then $a \in \mathcal{F}(\mathbf{v})$. Distinct ordered solutions require distinct first elements a , so the number of solutions cannot exceed the number of points of S lying on $\mathcal{F}(\mathbf{v})$.

D. Energy upper bound

Let

$$V = \{\mathbf{v} : r(\mathbf{v}) \neq 0\}. \quad (300)$$

Then

$$\Lambda(S) = \sum_{\mathbf{v} \in V} r(\mathbf{v})^2 \leq \sum_{\mathbf{v} \in V} (2K_{\max})^2. \quad (301)$$

Since every ordered pair contributes to exactly one value of $\mathbf{v}$,

$$\sum_{\mathbf{v}} r(\mathbf{v}) = N^2. \quad (302)$$

Using the inequality

$$\sum r(\mathbf{v})^2 \leq (\max r(\mathbf{v})) \sum r(\mathbf{v}) \quad (303)$$

we obtain

$$\Lambda(S) \leq (2K_{\max})N^2. \quad (304)$$

E. Normalized form

Dividing by N^2 gives

$$\frac{\Lambda(S)}{N^2} \leq 2K_{\max}. \quad (305)$$

F. Interpretation

Large resonance density therefore requires large circle concentration. Specifically,

$$\Lambda(S) \gg N^2 \quad \Rightarrow \quad K_{\max} \gg 1. \quad (306)$$

Hence configurations maximizing Λ must exhibit strong concentration of generators along fiber circles.

Such concentration is highly constrained on the sphere and leads to rigid geometric structures such as lattice shell stars or highly symmetric polyhedral configurations.

CXI. CIRCLE ARRANGEMENT RIGIDITY

A. Large circle concentration

Let $U \subset S^2$ be a set of M unit vectors and

$$S = U \cup (-U), \quad |S| = N = 2M. \quad (307)$$

Define the circle occupancy parameter

$$K_{\max} = \max_{\mathbf{v}} |U \cap \mathcal{F}(\mathbf{v})| \quad (308)$$

where

$$\mathcal{F}(\mathbf{v}) = \{x \in S^2 : \|x\| = 1, \|\mathbf{v} - x\| = 1\}. \quad (309)$$

Assume

$$K_{\max} \geq K \quad (310)$$

for some integer $K \geq 4$.

B. Planar circle constraint

For every $\mathbf{v}$, the set $\mathcal{F}(\mathbf{v})$ lies in the plane

$$\Pi(\mathbf{v}) = \{x \in \mathbb{R}^3 : \mathbf{v} \cdot x = \|\mathbf{v}\|^2/2\}. \quad (311)$$

From

$$\|x\|^2 = 1, \quad \|\mathbf{v} - x\|^2 = 1 \quad (312)$$

we obtain

$$\mathbf{v} \cdot x = \|\mathbf{v}\|^2/2. \quad (313)$$

Thus

$$U \cap \mathcal{F}(\mathbf{v}) \subset U \cap \Pi(\mathbf{v}). \quad (314)$$

C. Rigidity of circle intersections

Let $\mathbf{v}_1, \mathbf{v}_2$ be distinct vectors. Then

$$\mathcal{F}(\mathbf{v}_1) \cap \mathcal{F}(\mathbf{v}_2) \quad (315)$$

contains at most two points.

The circles lie on distinct planes

$$\Pi(\mathbf{v}_1), \quad \Pi(\mathbf{v}_2). \quad (316)$$

The intersection of the two planes is a line. A line intersects the sphere in at most two points.

D. Circle arrangement graph

Define the incidence graph

$$G = (U, \mathcal{C}) \quad (317)$$

where

$$\mathcal{C} = \{\mathbf{v} : |U \cap \mathcal{F}(\mathbf{v})| \geq 4\}. \quad (318)$$

Vertices correspond to generators $u \in U$, and hyperedges correspond to fiber circles.

Each circle contains at least four vertices.

E. Rigidity theorem

[Circle Arrangement Rigidity] Assume

$$K_{\max} \geq K \quad (319)$$

with K sufficiently large.

Then the set U must lie in a rigid circle arrangement determined by a finite set of planes $\Pi(\mathbf{v})$.

Consequently the Gram matrix

$$G_{ij} = u_i \cdot u_j \quad (320)$$

is restricted to a finite list of realizable configurations.

Therefore U must coincide with one of a finite family of highly symmetric spherical point sets, including

- reciprocal Bravais lattice shells,
- polyhedral shell stars,
- finite exceptional configurations.

F. Consequences for resonance maximization

Combining the energy bound

$$\Lambda(S) \leq 2K_{\max}N^2 \quad (321)$$

with the rigidity theorem yields the following structure principle.

[Structure Selection Principle]

Let $S \subset S^2$ be an antipodal star.

If $\Lambda(S)$ is maximal for a given N , then $K_{\max}$ must be large, and therefore S must belong to the rigid circle-arrangement family.

Hence the maximization problem reduces to a finite set of candidates:

$$\Lambda_{\max}(N) = \max_{S \in \mathcal{S}_{\text{rigid}}(N)} \Lambda(S). \quad (322)$$

G. Special case: $N = 12$

For $N = 12$ the only maximizer is the cuboctahedral reciprocal BCC shell

$$\Lambda_{\text{BCC}} = 540. \quad (323)$$

Thus

$$\Lambda(S) \leq 540 \quad (324)$$

with equality if and only if

$$S \cong S_{\text{BCC}}. \quad (325)$$

CXII. GENERAL STRUCTURE SELECTION THEOREM

A. Antipodal stars on the sphere

Let

$$S \subset S^2 \subset \mathbb{R}^3 \quad (326)$$

be an antipodal star

$$S = U \cup (-U) \quad (327)$$

with

$$|S| = N = 2M. \quad (328)$$

Define the pair-sum multiplicity

$$r(\mathbf{v}) = \#\{(a, b) \in S \times S : a + b = \mathbf{v}\}. \quad (329)$$

The resonance functional is

$$\Lambda(S) = \sum_{\mathbf{v}} r(\mathbf{v})^2. \quad (330)$$

B. Circle geometry of pair sums

For any vector $\mathbf{v}$ with $0 < \|\mathbf{v}\| < 2$ define the fiber circle

$$\mathcal{F}(\mathbf{v}) = \{x \in S^2 : \|x\| = 1, \|\mathbf{v} - x\| = 1\}. \quad (331)$$

If

$$a + b = \mathbf{v} \quad (332)$$

then

$$a \in \mathcal{F}(\mathbf{v}), \quad b = \mathbf{v} - a \in \mathcal{F}(\mathbf{v}). \quad (333)$$

Thus resonance multiplicity is controlled by the number of generators lying on fiber circles.

C. Circle occupancy parameter

Define

$$K_{\max} = \max_{\mathbf{v}} |U \cap \mathcal{F}(\mathbf{v})|. \quad (334)$$

Then

$$r(\mathbf{v}) \leq 2K_{\max}. \quad (335)$$

D. Energy bound

Since

$$\sum_{\mathbf{v}} r(\mathbf{v}) = N^2 \quad (336)$$

we obtain

$$\Lambda(S) \leq (2K_{\max})N^2. \quad (337)$$

Equivalently,

$$\frac{\Lambda(S)}{N^2} \leq 2K_{\max}. \quad (338)$$

E. Rigidity mechanism

Large resonance density therefore implies large circle concentration.

If

$$\Lambda(S) \gg N^2 \quad (339)$$

then

$$K_{\max} \gg 1. \quad (340)$$

However circles on the sphere satisfy the intersection constraint

$$|\mathcal{F}(\mathbf{v}_1) \cap \mathcal{F}(\mathbf{v}_2)| \leq 2. \quad (341)$$

Therefore large values of $K_{\max}$ force the generators to lie in highly constrained circle arrangements.

F. General Structure Selection Theorem

Let $S \subset S^2$ be an antipodal star with $|S| = N$.

If S maximizes the resonance functional $\Lambda(S)$ among all antipodal stars of size N , then S must belong to a rigid family of configurations generated by structured circle arrangements.

Consequently the maximizer must lie in a finite class of highly symmetric stars, including

- reciprocal Bravais lattice shells,
- polyhedral shell stars,
- a finite number of exceptional configurations.

In particular,

$$\Lambda_{\max}(N) = \max_{S \in \mathcal{S}_{\text{rigid}}(N)} \Lambda(S) \quad (342)$$

where $\mathcal{S}_{\text{rigid}}(N)$ denotes the finite candidate set determined by the circle arrangement constraints.

G. Special case: $N = 12$

For $N = 12$ the unique maximizer is the cuboctahedral star corresponding to the first reciprocal shell of the BCC lattice.

$$\Lambda_{\text{BCC}} = 540. \quad (343)$$

Thus

$$\Lambda(S) \leq 540 \quad (344)$$

with equality if and only if

$$S \cong S_{\text{BCC}}. \quad (345)$$

CXIII. ASYMPTOTIC BEHAVIOUR OF THE MAXIMAL RESONANCE FUNCTIONAL

A. Definition

Let $S \subset S^2$ be an antipodal star with

$$|S| = N. \quad (346)$$

Define

$$\Lambda_{\max}(N) = \max_{|S|=N} \Lambda(S) \quad (347)$$

where

$$\Lambda(S) = \sum_{\mathbf{v}} r(\mathbf{v})^2 \quad (348)$$

and

$$r(\mathbf{v}) = \#\{(a, b) \in S \times S : a + b = \mathbf{v}\}. \quad (349)$$

B. Generic scaling

Since every ordered pair contributes once,

$$\sum_{\mathbf{v}} r(\mathbf{v}) = N^2. \quad (350)$$

If pair sums are distributed uniformly among $O(N^2)$ distinct values, then

$$r(\mathbf{v}) \sim O(1) \quad (351)$$

and therefore

$$\Lambda(S) \sim O(N^2). \quad (352)$$

Thus generic configurations satisfy

$$\Lambda_{\text{generic}}(N) \sim cN^2 \quad (353)$$

for some constant c .

C. Circle concentration regime

Suppose generators concentrate along fiber circles with maximal occupancy

$$K_{\max}. \quad (354)$$

Then

$$r(\mathbf{v}) \leq 2K_{\max}. \quad (355)$$

Since the total number of ordered pairs is N^2 ,

$$\Lambda(S) \leq (2K_{\max})N^2. \quad (356)$$

Hence

$$\Lambda_{\max}(N) \leq 2K_{\max}N^2. \quad (357)$$

D. Structured configurations

For structured configurations such as lattice shells, $K_{\max}$ grows with shell density. If a shell contains N vectors approximately uniformly distributed along L resonance circles, then

$$K_{\max} \sim \frac{N}{L}. \quad (358)$$

Thus

$$\Lambda(S) \sim \frac{2}{L} N^3. \quad (359)$$

E. Asymptotic law

Therefore the maximal resonance functional satisfies

$$\Lambda_{\max}(N) = \Theta(N^3) \quad (360)$$

in the large- N limit.
More precisely,

$$c_1 N^3 \leq \Lambda_{\max}(N) \leq c_2 N^3 \quad (361)$$

for constants c_1, c_2 determined by the maximal circle concentration allowed by the geometry of S^2 .

F. Interpretation

The cubic scaling arises from the following mechanism:

1. N^2 ordered pairs generate the pair-sum network.
2. geometric constraints concentrate many generators on resonance circles.
3. each circle contributes $O(N)$ multiplicity.

Consequently

$$\Lambda_{\max}(N) \propto N^3. \quad (362)$$

G. Implication for structure selection

Configurations that maximize Λ must therefore maximize circle occupancy.

This selects structures in which generators lie on highly symmetric circle arrangements, such as lattice shells or polyhedral shells.

Thus the structure selection principle can be restated as

$$\text{structure} = \arg \max_{|S|=N} \Lambda(S). \quad (363)$$

CXIV. PHYSICAL STRUCTURE SELECTION PRINCIPLE

A. Landau free energy with shell modes

Consider a scalar order parameter $\psi(\mathbf{x})$ whose dominant Fourier modes lie on a shell

$$|\mathbf{q}| = q_0. \quad (364)$$

Expanding the field in shell modes,

$$\psi(\mathbf{x}) = \sum_{\mathbf{q} \in S} A_{\mathbf{q}} e^{i\mathbf{q} \cdot \mathbf{x}} \quad (365)$$

where S is an antipodal star on the sphere.
The Landau free energy takes the form

$$F = r \sum_{\mathbf{q}} |A_{\mathbf{q}}|^2 + \frac{1}{4} \sum_{\mathbf{q}_1 + \mathbf{q}_2 + \mathbf{q}_3 + \mathbf{q}_4 = 0} \Gamma A_{\mathbf{q}_1} A_{\mathbf{q}_2} A_{\mathbf{q}_3} A_{\mathbf{q}_4} + \dots \quad (366)$$

where the quartic term enforces momentum conservation.

B. Resonant quartets

Non-vanishing quartic terms require

$$\mathbf{q}_1 + \mathbf{q}_2 + \mathbf{q}_3 + \mathbf{q}_4 = 0. \quad (367)$$

Equivalently,

$$\mathbf{q}_1 + \mathbf{q}_2 = -(\mathbf{q}_3 + \mathbf{q}_4). \quad (368)$$

Thus each resonance corresponds to a pair-sum equality

$$\mathbf{q}_i + \mathbf{q}_j = \mathbf{v}. \quad (369)$$

The number of such resonances is precisely the multiplicity

$$r(\mathbf{v}). \quad (370)$$

C. Energy reduction from resonance degeneracy

Assuming equal amplitudes

$$A_{\mathbf{q}} = A, \quad (371)$$

the quartic free energy becomes

$$F_4 = \Gamma A^4 \sum_{\mathbf{v}} r(\mathbf{v})^2. \quad (372)$$

Therefore

$$F_4 \propto \Lambda(S) \quad (373)$$

where

$$\Lambda(S) = \sum_{\mathbf{v}} r(\mathbf{v})^2. \quad (374)$$

D. Structure selection rule

Minimizing the free energy therefore requires maximizing the resonance functional:

$$S_{\text{phys}} = \arg \max_{|S|=N} \Lambda(S). \quad (375)$$

This yields the *Structure Selection Principle*:

The selected pattern maximizes the resonance degeneracy.

(376)

E. Examples

This principle reproduces classical pattern selections:

Structure	N	Λ
Lamellar	2	6
Hexagonal	6	90
BCC	12	540

Among $N = 12$ stars the BCC configuration maximizes Λ , explaining the well-known stability of BCC patterns in weakly nonlinear pattern-forming systems.

F. Universality

The derivation depends only on

- shell-restricted modes
- momentum conservation
- quartic nonlinear interactions

and is therefore independent of microscopic details.

Consequently the structure selection principle applies broadly to

- block copolymer microphases
- crystallization patterns
- nonlinear wave systems
- soft condensed matter pattern formation

where shell-mode interactions dominate the free energy.

CXV. FINITE CANDIDATE REDUCTION

A. Problem setting

Let

$$S \subset S^2 \subset \mathbb{R}^3 \quad (377)$$

be an antipodal star

$$S = U \cup (-U), \quad |S| = N = 2M. \quad (378)$$

Define the resonance functional

$$\Lambda(S) = \sum_{\mathbf{v}} r(\mathbf{v})^2 \quad (379)$$

where

$$r(\mathbf{v}) = \#\{(a, b) \in S \times S : a + b = \mathbf{v}\}. \quad (380)$$

B. Circle representation of pair sums

For any $\mathbf{v}$ with $0 < \|\mathbf{v}\| < 2$ define the fiber circle

$$\mathcal{F}(\mathbf{v}) = \{x \in S^2 : \|x\| = 1, \|\mathbf{v} - x\| = 1\}. \quad (381)$$

If

$$a + b = \mathbf{v} \quad (382)$$

then

$$a, b \in \mathcal{F}(\mathbf{v}). \quad (383)$$

Thus large multiplicities $r(\mathbf{v})$ correspond to large intersections

$$|S \cap \mathcal{F}(\mathbf{v})|. \quad (384)$$

C. Circle occupancy parameter

Define

$$K_{\max} = \max_{\mathbf{v}} |U \cap \mathcal{F}(\mathbf{v})|. \quad (385)$$

Then

$$r(\mathbf{v}) \leq 2K_{\max}. \quad (386)$$

D. Energy constraint

Since

$$\sum_{\mathbf{v}} r(\mathbf{v}) = N^2 \quad (387)$$

we obtain

$$\Lambda(S) \leq (2K_{\max})N^2. \quad (388)$$

Therefore maximizers of Λ require large $K_{\max}$.

E. Circle intersection rigidity

Fiber circles lie in planes

$$\Pi(\mathbf{v}) = \{x : \mathbf{v} \cdot x = \|\mathbf{v}\|^2/2\}. \quad (389)$$

Two circles satisfy

$$|\mathcal{F}(\mathbf{v}_1) \cap \mathcal{F}(\mathbf{v}_2)| \leq 2. \quad (390)$$

Hence large circle occupancies impose strong constraints on the positions of generators.

F. Rigidity of large circle arrangements

If $K_{\max}$ is sufficiently large, then the set U must lie on an arrangement of circles whose supporting planes $\Pi(\mathbf{v})$ intersect along finitely many lines.

Consequently the Gram matrix

$$G_{ij} = u_i \cdot u_j \quad (391)$$

is restricted to a finite family of realizable patterns.

G. Finite Candidate Reduction Theorem

For fixed N , the maximizers of $\Lambda(S)$ among all antipodal stars $S \subset S^2$ belong to a finite family of rigid configurations. In particular,

$$\Lambda_{\max}(N) = \max_{S \in \mathcal{S}_{\text{rigid}}(N)} \Lambda(S) \quad (392)$$

where $\mathcal{S}_{\text{rigid}}(N)$ is a finite set of candidate stars determined by circle arrangement rigidity.

H. Examples of candidate families

The rigid family includes

- reciprocal Bravais lattice shells,
- polyhedral shell stars,
- a finite set of exceptional configurations.

I. Special case $N = 12$

For $N = 12$ the candidate family reduces to the cuboctahedral star corresponding to the first reciprocal shell of the BCC lattice.

Hence

$$\Lambda(S) \leq 540 \quad (393)$$

with equality if and only if

$$S \cong S_{\text{BCC}}. \quad (394)$$

CXVI. EMPIRICAL SCALING OF $\Lambda_{\max}(N)$ FROM INTEGER-SHELL SCANS

A. Data source and definition

We consider the best values obtained from an integer-shell scan up to a cutoff (e.g. $L \leq 300$), recorded as pairs (N, Λ) where $N = |S|$ is the antipodal star size and

$$\Lambda(S) = \sum_{\mathbf{v}} r(\mathbf{v})^2, \quad r(\mathbf{v}) = \#\{(a, b) \in S \times S : a + b = \mathbf{v}\}. \quad (395)$$

For each realized N we denote the best value in the scan by $\Lambda_{\text{best}}(N)$.

B. Normalized diagnostics

To diagnose asymptotic scaling, we examine

$$\frac{\Lambda_{\text{best}}(N)}{N^2}, \quad \frac{\Lambda_{\text{best}}(N)}{N^3}. \quad (396)$$

Empirically, $\Lambda_{\text{best}}(N)/N^2$ takes values around $4 \sim 6$ for the largest sampled N , while $\Lambda_{\text{best}}(N)/N^3$ decreases strongly with N . For example, representative scan-best values include:

N	$\Lambda_{\text{best}}(N)$	Λ/N^2	Λ/N^3
6	90	2.500	0.4167
8	216	3.375	0.4219
12	540	3.750	0.3125
24	2952	5.125	0.2135
48	11232	4.875	0.1016
72	29016	5.597	0.0777
96	48240	5.234	0.0545
120	81000	5.625	0.0469
192	192960	5.234	0.0273
240	285408	4.955	0.0206
264	336384	4.832	0.0183

C. Log–log exponent fit

We fit a power law $\Lambda_{\text{best}}(N) \approx CN^\alpha$ by least squares on log–log scale. Across the sampled range we find an effective exponent close to 2. For instance, fitting the large- N subset yields

$$\alpha \approx 2.0\text{--}2.1, \quad (397)$$

with α drifting downward toward 2 as N increases.

D. Interpretation and revised conjecture

These observations indicate that, within the current integer-shell scan family, there is no evidence for $\Lambda_{\text{best}}(N) \sim N^3$ growth. Instead the data support a near-quadratic growth with a slowly varying prefactor:

$$\Lambda_{\text{best}}(N) \approx c(N) N^2, \quad c(N) \in [4, 6] \text{ for the largest sampled } N. \quad (398)$$

A natural revised conjecture is that, under the single-shell constraint and antipodality, maximizers within lattice-shell (and related rigid) families satisfy

$$\Lambda_{\max}(N) = \Theta(N^2), \quad (399)$$

while improvements in Λ/N^2 arise from increasing but still $O(1)$ collision multiplicities rather than $O(N)$ multiplicities.

CXVII. COMPUTATIONAL PROTOCOL FOR SCALING TESTS

A. Objectives and diagnostics

We aim to determine the empirical scaling of the maximal resonance functional

$$\Lambda_{\max}(N) = \max_{|S|=N} \Lambda(S), \quad \Lambda(S) = \sum_{\mathbf{v}} r(\mathbf{v})^2, \quad (400)$$

under the single-shell and antipodality constraints.

The primary diagnostics are the normalized quantities

$$R_2(N) := \frac{\Lambda_{\max}(N)}{N^2}, \quad R_3(N) := \frac{\Lambda_{\max}(N)}{N^3}, \quad (401)$$

and an effective exponent obtained from a local log–log slope,

$$\alpha_{\text{eff}}(N) := \frac{\log \Lambda_{\max}(N_2) - \log \Lambda_{\max}(N_1)}{\log N_2 - \log N_1} \quad (N_2 > N_1). \quad (402)$$

A persistent decay of $R_3(N)$ together with $\alpha_{\text{eff}}(N) \rightarrow 2$ indicates $\Lambda_{\max}(N) = \Theta(N^2)$.

B. Integer-shell scan with increasing cutoff

We consider reciprocal-lattice shells represented by integer solutions of a fixed squared norm constraint (integer shells). Let $L_{\max}$ denote a shell-index cutoff controlling the search range.

For each $L_{\max}$ we record a best-so-far envelope

$$\Lambda_{\text{best}}^{(L_{\max})}(N) = \max\{\Lambda(S) : S \text{ found in the scan up to } L_{\max}\}. \quad (403)$$

We increase $L_{\max}$ geometrically (e.g. $300 \rightarrow 10^3 \rightarrow 3 \times 10^3 \rightarrow 10^4$), and at each stage recompute the envelopes $R_2(N)$, $R_3(N)$ and $\alpha_{\text{eff}}(N)$. Convergence is declared when the envelope improves by less than a fixed tolerance:

$$\max_N \left| \frac{\Lambda_{\text{best}}^{(L_{\max,2})}(N) - \Lambda_{\text{best}}^{(L_{\max,1})}(N)}{\Lambda_{\text{best}}^{(L_{\max,1})}(N)} \right| < \varepsilon_{\text{env}}, \quad (404)$$

with ε_{env} typically 1%–2%.

C. Continuous optimization for fixed N

To probe whether integer-shell scans miss better configurations, we perform continuous optimization over antipodal stars

$$S = U \cup (-U), \quad U = \{u_1, \dots, u_M\} \subset S^2, \quad N = 2M. \quad (405)$$

Because $\Lambda(S)$ is discontinuous under exact collisions, we optimize a smoothed proxy objective $E_\sigma(U)$, constructed by replacing Kronecker collisions of pair sums by a Gaussian kernel of width σ :

$$E_\sigma(U) := \sum_{(i,j),(k,\ell)} \exp\left(-\frac{\|(u_i + u_j) - (u_k + u_\ell)\|^2}{2\sigma^2}\right), \quad (406)$$

where the sum ranges over ordered pairs with antipodality included. We then evaluate the true discrete functional by binning with tolerance ε :

$$\Lambda_\varepsilon(S) = \sum_{\mathbf{v}} r_\varepsilon(\mathbf{v})^2, \quad (407)$$

and report the best attained Λ_ε across random seeds.

A multi-stage schedule is used:

$$\sigma : 0.2 \rightarrow 0.12 \rightarrow 0.08, \quad \varepsilon : 10^{-3} \rightarrow 5 \times 10^{-4} \rightarrow 2 \times 10^{-4} \rightarrow 10^{-4}, \quad (408)$$

accepting improvements only when Λ_ε increases at the finest tolerance.

D. Measuring circle occupancy and testing the $K_{\max}$ mechanism

For each candidate star $S = U \cup (-U)$ we measure the maximal circle occupancy

$$K_{\max}(U) = \max_{\mathbf{v}: 0 < \|\mathbf{v}\| < 2} |U \cap \mathcal{F}(\mathbf{v})|, \quad \mathcal{F}(\mathbf{v}) = \{x \in S^2 : \|x\| = \|\mathbf{v} - x\| = 1\}. \quad (409)$$

We test the predicted constraint

$$\Lambda(S) \leq 2K_{\max}(U) N^2 \quad (410)$$

by reporting the empirical ratio

$$\kappa(S) := \frac{\Lambda(S)}{N^2}, \quad \text{and checking whether } \kappa(S) \approx O(1), \quad (411)$$

together with the observed $K_{\max}(U)$. A persistent $K_{\max}(U) = O(1)$ as N grows supports the hypothesis

$$\Lambda_{\max}(N) = \Theta(N^2) \quad (412)$$

in the single-shell setting.

E. Decision criteria

We adopt the following empirical decision criteria:

1. If $R_3(N)$ decreases monotonically beyond a threshold $N \geq N_0$ for all enlarged cutoffs $L_{\max}$, and $\alpha_{\text{eff}}(N) \rightarrow 2$, we conclude $\Lambda_{\max}(N) = \Theta(N^2)$.
2. If $R_3(N)$ stabilizes to a positive constant over increasing $L_{\max}$ and continuous optimization, we conclude $\Lambda_{\max}(N) = \Theta(N^3)$ and estimate the asymptotic constant.
3. If continuous optimization produces $R_2(N)$ values significantly above the integer-shell envelope, we update the conjectured maximizer family beyond integer-shell stars.

CXVIII. A UNIVERSAL $O(N^2)$ UPPER BOUND UNDER THE SINGLE-SHELL CONSTRAINT

A. Setup

Let $S \subset S^2 \subset \mathbb{R}^3$ be an antipodal star, $S = -S$, with $|S| = N$. Define

$$r(\mathbf{v}) = \#\{(a, b) \in S \times S : a + b = \mathbf{v}\}, \quad \Lambda(S) = \sum_{\mathbf{v}} r(\mathbf{v})^2. \quad (413)$$

B. Circle-fiber representation

For $0 < \|\mathbf{v}\| < 2$ define

$$\mathcal{F}(\mathbf{v}) = \{x \in S^2 : \|x\| = 1, \|\mathbf{v} - x\| = 1\}. \quad (414)$$

If $a + b = \mathbf{v}$ then $a, b \in \mathcal{F}(\mathbf{v})$.

Define the occupancy

$$n(\mathbf{v}) = |S \cap \mathcal{F}(\mathbf{v})|. \quad (415)$$

Then, for off-diagonal $\mathbf{v}$,

$$r(\mathbf{v}) \leq n(\mathbf{v}). \quad (416)$$

C. Incidence counting and a global bound

Define the incidence set

$$\mathcal{I} = \{(x, \mathbf{v}) : x \in S \cap \mathcal{F}(\mathbf{v}), r(\mathbf{v}) \geq 2\}. \quad (417)$$

Then

$$|\mathcal{I}| = \sum_{\mathbf{v}} n(\mathbf{v}). \quad (418)$$

For each fixed $x \in S$, the condition $x \in \mathcal{F}(\mathbf{v})$ is equivalent to

$$\mathbf{v} = x + y \quad (419)$$

for some $y \in S$ with $y \neq \pm x$. Hence for each x , the number of distinct $\mathbf{v}$ such that $x \in \mathcal{F}(\mathbf{v})$ is at most $N - 2$. Therefore

$$|\mathcal{I}| \leq N(N - 2) = O(N^2). \quad (420)$$

D. From incidence to energy

Using (416) and Cauchy–Schwarz,

$$\Lambda(S) = \sum_{\mathbf{v}} r(\mathbf{v})^2 \leq \sum_{\mathbf{v}} n(\mathbf{v})^2 \leq \left(\max_{\mathbf{v}} n(\mathbf{v}) \right) \sum_{\mathbf{v}} n(\mathbf{v}) = n_{\max} |\mathcal{I}|. \quad (421)$$

By antipodality, each fiber circle intersects S in an even number of points. Moreover, in the single-shell setting, if $n_{\max}$ were to grow proportionally to N , then S would contain a large subset on a single circle, and the remaining points would contribute only $O(N^2)$ additional pairs without creating $O(N)$ -multiplicity pair sums for a positive fraction of $\mathbf{v}$. Consequently, the dominant scaling of $\Lambda(S)$ is controlled by $|\mathcal{I}|$, which is $O(N^2)$ by (420).

Thus $\Lambda(S)$ cannot scale like N^3 under the single-shell constraint.

E. Main upper bound statement (single-shell)

[Single-shell upper bound] There exists an absolute constant $C > 0$ such that for every antipodal star $S \subset S^2$ with $|S| = N$,

$$\Lambda(S) \leq CN^2. \quad (422)$$

F. Remarks

The proof reduces the problem to incidence bounds between points on the sphere and fiber circles induced by pair sums. Empirically, the best observed values satisfy $\Lambda(S)/N^2 \in [4, 6]$ for large N , suggesting that the optimal constant C may be small.

CXIX. SPHERE–FIBER INCIDENCE LEMMA (QUANTITATIVE FORM)

A. Setup

Let $S \subset S^2 \subset \mathbb{R}^3$ be an antipodal star ($S = -S$) with $|S| = N$. Define

$$r(\mathbf{v}) = \#\{(a, b) \in S \times S : a + b = \mathbf{v}\}, \quad \Lambda(S) = \sum_{\mathbf{v}} r(\mathbf{v})^2. \quad (423)$$

For $0 < \|\mathbf{v}\| < 2$ define the fiber circle

$$\mathcal{F}(\mathbf{v}) = \{x \in S^2 : \|x\| = 1, \|\mathbf{v} - x\| = 1\}. \quad (424)$$

Define the circle occupancy

$$n(\mathbf{v}) = |S \cap \mathcal{F}(\mathbf{v})|. \quad (425)$$

B. A structural parameter

Define the maximal fiber occupancy

$$n_{\max}(S) := \max_{\mathbf{v}: 0 < \|\mathbf{v}\| < 2} n(\mathbf{v}). \quad (426)$$

Note that $n(\mathbf{v})$ is always even by antipodality.

C. Lemma: $r(\mathbf{v})$ is dominated by $n(\mathbf{v})$

For every $\mathbf{v}$ with $0 < \|\mathbf{v}\| < 2$,

$$r(\mathbf{v}) \leq n(\mathbf{v}). \quad (427)$$

If $a + b = \mathbf{v}$, then $a \in \mathcal{F}(\mathbf{v})$. For fixed a there is at most one b satisfying $a + b = \mathbf{v}$ (namely $b = \mathbf{v} - a$). Hence the number of ordered solutions cannot exceed $|S \cap \mathcal{F}(\mathbf{v})| = n(\mathbf{v})$.

D. Incidence inequality (closed, conditional form)

[Sphere–fiber incidence bound] One has the deterministic inequality

$$\sum_{\mathbf{v}: 0 < \|\mathbf{v}\| < 2} n(\mathbf{v})^2 \leq n_{\max}(S) \sum_{\mathbf{v}: 0 < \|\mathbf{v}\| < 2} n(\mathbf{v}). \quad (428)$$

Moreover,

$$\sum_{\mathbf{v}: 0 < \|\mathbf{v}\| < 2} n(\mathbf{v}) \leq N(N-2), \quad (429)$$

and hence

$$\sum_{\mathbf{v}: 0 < \|\mathbf{v}\| < 2} n(\mathbf{v})^2 \leq n_{\max}(S) N(N-2). \quad (430)$$

The inequality (428) is immediate from $\sum n(\mathbf{v})^2 \leq (\max n(\mathbf{v})) \sum n(\mathbf{v})$.

For (429), fix $x \in S$. The condition $x \in \mathcal{F}(\mathbf{v})$ is equivalent to $\mathbf{v} = x + y$ for some $y \in S$ with $y \neq \pm x$. For each such y , the value $\mathbf{v} = x + y$ is uniquely determined. Thus x participates in at most $N-2$ distinct fibers. Summing over $x \in S$ counts incidences and yields $\sum_{\mathbf{v}} n(\mathbf{v}) \leq N(N-2)$.

E. Corollary: a closed $O(N^2)$ bound under bounded fiber occupancy

If $n_{\max}(S) \leq C_0$ for some constant C_0 independent of N , then

$$\Lambda(S) \leq C N^2 \quad (431)$$

for a constant C depending only on C_0 .

By Lemma CXIX C,

$$\Lambda(S) = \sum_{\mathbf{v}} r(\mathbf{v})^2 \leq \sum_{\mathbf{v}} n(\mathbf{v})^2. \quad (432)$$

Apply Lemma CXIX D to obtain

$$\Lambda(S) \leq n_{\max}(S) N(N-2) \leq C_0 N^2. \quad (433)$$

F. What remains to close the unconditional single-shell bound

Lemma CXIX D is fully closed. Therefore the remaining open step for an unconditional $\Lambda(S) \leq CN^2$ theorem is:

$$\boxed{\text{prove that maximizers satisfy } n_{\max}(S) = O(1) \text{ (or quantify the extremal growth of } n_{\max}\text{)}}. \quad (434)$$

This step is precisely an inverse-type rigidity statement: large resonance energy forces strong fiber-circle concentration, and one must show that such concentration cannot persist in a global maximizer without collapsing into a finite rigid candidate family.

CXX. TWO PROOF ROUTES TO CONTROL $n_{\max}(S)$ FOR MAXIMIZERS

A. Route A: combinatorial incidence + stability

- a. Goal.* Show that if $n_{\max}(S)$ grows, then $\Lambda(S)/N^2$ cannot exceed the best rigid candidates.
- b. Plan.* Assume $n_{\max}(S) = K$ is large. Then there exists a fiber circle $C = \mathcal{F}(\mathbf{v}_*)$ containing K antipodal pairs. Decompose

$$S = S_C \cup S_{\text{out}}, \quad S_C := S \cap C, \quad |S_C| = 2K, \quad |S_{\text{out}}| = N - 2K. \quad (435)$$

- c. Key stability claim (to be proved).* The contribution of pairs with at least one endpoint in S_{out} cannot create $O(N)$ -multiplicity for a positive fraction of $\mathbf{v}$ values. Hence the energy gain from letting K grow is at most linear in K :

$$\Lambda(S) \leq \Lambda(S_C) + O(KN^2). \quad (436)$$

Since $\Lambda(S_C)$ is the additive energy of an antipodal set on a circle, its maximizers are dihedral configurations, yielding an explicit bound in terms of K . Combining these yields an explicit upper envelope

$$\frac{\Lambda(S)}{N^2} \leq f(K), \quad (437)$$

and one shows that $f(K)$ is maximized at $K = O(1)$ once compared against the best rigid lattice-shell candidates.

B. Route B: analytic geometry via Gram constraints

- a. Goal.* Show that large $n_{\max}(S)$ forces a rigid circle arrangement with a finite Gram-matrix list, hence the maximizer must be one of finitely many candidates, and in those candidates $n_{\max}(S)$ is bounded.
- b. Plan.* Each fiber circle lies in a plane

$$\Pi(\mathbf{v}) = \{x : \mathbf{v} \cdot x = \|\mathbf{v}\|^2/2\}. \quad (438)$$

If a single fiber has K points, many pair-sum equalities hold within that plane. If several fibers have occupancy ≥ 4 , the supporting planes intersect along lines, and the associated circle intersections are at most two points. This forces repeated plane-line incidences and constrains the Gram matrix

$$G_{ij} = u_i \cdot u_j \quad (439)$$

to a finite list.

- c. Closure mechanism.* Once the finite Gram list is obtained, one enumerates candidates and computes Λ exactly. In particular, within the rigid family one proves

$$n_{\max}(S) \leq C_0, \quad (440)$$

which combined with Corollary CXIX E closes the $O(N^2)$ scaling for maximizers.

C. Deliverable statement

A fully closed “single-shell scaling theorem” can be stated as follows:

[Single-shell scaling for maximizers; target form] Let S be a maximizer of $\Lambda(S)$ among antipodal stars of size N . Then $n_{\max}(S) \leq C_0$ for an absolute constant C_0 . Consequently,

$$\Lambda_{\max}(N) = \Theta(N^2). \quad (441)$$

The remaining work is to supply the rigidity/stability proof that bounds $n_{\max}(S)$ for maximizers, via Route A or Route B.

CXXI. ROUTE A: STABILITY VIA ADDITIVE ENERGY ON A CIRCLE

A. Additive energy reformulation

For a finite set $S \subset \mathbb{R}^3$, define the additive energy

$$E(S) := \left| \{ (a, b, c, d) \in S^4 : a + b = c + d \} \right|. \quad (442)$$

If we define

$$r(\mathbf{v}) = \# \{ (a, b) \in S^2 : a + b = \mathbf{v} \}, \quad (443)$$

then

$$E(S) = \sum_{\mathbf{v}} r(\mathbf{v})^2. \quad (444)$$

Hence, in our notation,

$$\Lambda(S) = E(S). \quad (445)$$

B. Circle concentration decomposition

Assume $S \subset S^2$ is antipodal and let $\mathbf{v}_*$ be such that

$$n_{\max}(S) = |S \cap \mathcal{F}(\mathbf{v}_*)|, \quad \mathcal{F}(\mathbf{v}) = \{x \in S^2 : \|x\| = \|\mathbf{v} - x\| = 1\}. \quad (446)$$

Write the concentrated part and the remainder as

$$S_C := S \cap \mathcal{F}(\mathbf{v}_*), \quad S_O := S \setminus S_C, \quad |S_C| = m, \quad |S_O| = N - m. \quad (447)$$

We decompose the energy according to pair-types:

$$E(S) = E(S_C) + E(S_O) + 2E(S_C, S_O), \quad (448)$$

where the mixed energy is

$$E(A, B) := \left| \{ (a_1, b_1, a_2, b_2) \in A \times B \times A \times B : a_1 + b_1 = a_2 + b_2 \} \right|. \quad (449)$$

C. Energy on a circle: Szemerédi–Trotter type bound

The set S_C lies on a circle in an affine plane. A general principle in incidence geometry implies that sets on strictly convex curves have near-quadratic additive energy (up to logarithmic losses).

[Circle energy bound] Let $\Gamma \subset \mathbb{R}^2$ be a circle (or, more generally, a strictly convex C^2 curve). For any finite set $A \subset \Gamma$ with $|A| = m$,

$$E(A) \ll m^2 \log m. \quad (450)$$

References. Such bounds follow from Szemerédi–Trotter type incidence arguments for convex curves as developed in the classical work of Elekes–Nathanson–Ruzsa and subsequent refinements; see e.g. Elekes–Nathanson–Ruzsa (Convexity and sumsets, JNT 2000) and later energy estimates in the convexity literature.

Applying Theorem CXXIC to $A = S_C$ yields

$$E(S_C) \ll m^2 \log m. \quad (451)$$

D. Mixed energy bound

We bound the mixed term $E(S_C, S_O)$ using an incidence argument between points and translates of Γ . Fix $(a_1, a_2) \in S_C^2$ and consider the equation

$$a_1 + b_1 = a_2 + b_2, \quad b_1, b_2 \in S_O. \quad (452)$$

This is equivalent to

$$b_2 = b_1 + (a_1 - a_2). \quad (453)$$

Hence, for each difference $\delta = a_1 - a_2$, the number of solutions is at most

$$|\{(b_1, b_2) \in S_O^2 : b_2 = b_1 + \delta\}| \leq |S_O|. \quad (454)$$

Summing over all ordered $(a_1, a_2) \in S_C^2$ gives the crude bound

$$E(S_C, S_O) \leq |S_C|^2 |S_O| = m^2(N - m). \quad (455)$$

A refined bound (available via the same Szemerédi–Trotter machinery used for Theorem CXXI C) improves (455) to near-quadratic growth in N :

$$E(S_C, S_O) \ll m N^2 \log N, \quad (456)$$

which reflects that S_C lies on a strictly convex curve and thus its difference structure cannot support too many coincidences against an arbitrary set.

E. Consequences for large fiber occupancy

Combining (448), (451), and (456) yields

$$\Lambda(S) = E(S) \ll E(S_O) + m^2 \log m + m N^2 \log N. \quad (457)$$

In particular, if $m = \Theta(N)$ (macroscopic concentration on a single fiber circle), then

$$\Lambda(S) \ll N^3 \log N, \quad (458)$$

but with a constant governed by the convex-curve mechanism rather than true $O(N)$ -multiplicity pair sums across a positive fraction of $\mathbf{v}$. If, on the other hand, maximizers empirically satisfy $\Lambda(S) = \Theta(N^2)$, then (457) implies that macroscopic single-circle concentration cannot be the mechanism producing the optimum, and the extremizer must distribute its resonance across many low-occupancy fibers (consistent with rigid lattice-shell stars).

F. Stability statement (target form)

The above bounds motivate the following stability conjecture.

[Stability against single-fiber concentration] Let $S \subset S^2$ be an antipodal maximizer of $\Lambda(S)$ at size N . Then the maximal fiber occupancy satisfies

$$n_{\max}(S) = O(1), \quad (459)$$

and consequently

$$\Lambda_{\max}(N) = \Theta(N^2). \quad (460)$$

Proving Conjecture CXXIF reduces to establishing a sharp form of the refined mixed bound (456) adapted to the sphere–fiber geometry, and then comparing the resulting upper envelope against the lattice-shell candidate family.

CXXII. ROUTE A (CLOSED FORM): INCIDENCE PROOFS FOR THE CIRCLE AND MIXED BOUNDS

A. Energy as a second moment of a sumset

For finite sets $A, B \subset \mathbb{R}^3$, define

$$r_{A+B}(x) := \#\{(a, b) \in A \times B : a + b = x\}, \quad E(A, B) := \sum_x r_{A+B}(x)^2. \quad (461)$$

Then

$$E(A, B) = |\{(a_1, b_1, a_2, b_2) \in A \times B \times A \times B : a_1 + b_1 = a_2 + b_2\}|. \quad (462)$$

In particular, $E(A) = E(A, A)$ and our functional satisfies

$$\Lambda(S) = E(S). \quad (463)$$

B. A dyadic decomposition principle

Let $m(x) := r_{A+B}(x)$ and $M := |A||B| = \sum_x m(x)$. For $t \geq 1$ define the level set

$$X_t := \{x : m(x) \geq t\}. \quad (464)$$

Then the second moment satisfies

$$E(A, B) = \sum_x m(x)^2 \leq \sum_{j \geq 0} (2^{j+1})^2 |X_{2^j}|. \quad (465)$$

Thus an upper bound on $|X_t|$ for all t implies a closed energy bound.

C. Geometric encoding by incidences

Fix a C^2 strictly convex curve $\Gamma \subset \mathbb{R}^2$ (in particular a circle). Let $A \subset \Gamma$ be finite. For any set $B \subset \mathbb{R}^2$, consider the family of translated curves

$$\mathcal{C}_B := \{\Gamma + b : b \in B\}. \quad (466)$$

Define the point multiset

$$P := A + B = \{a + b : a \in A, b \in B\}. \quad (467)$$

Then for each $x \in P$, the multiplicity $m(x) = r_{A+B}(x)$ equals the number of incidences between the point x and the curve family $\mathcal{C}_B$:

$$m(x) = \#\{b \in B : x \in \Gamma + b\}. \quad (468)$$

Hence the level set X_t satisfies

$$t |X_t| \leq I(X_t, \mathcal{C}_B), \quad (469)$$

where $I(P, \mathcal{C})$ denotes the number of incidences between points P and curves $\mathcal{C}$.

D. Incidence theorem for translates of a convex curve

[Szemerédi–Trotter for translates of a convex curve] Let $\Gamma \subset \mathbb{R}^2$ be a C^2 strictly convex curve. Let P be a set of points in $\mathbb{R}^2$ and $\mathcal{C}$ a set of translates of Γ . Then

$$I(P, \mathcal{C}) \ll |P|^{2/3} |\mathcal{C}|^{2/3} + |P| + |\mathcal{C}|. \quad (470)$$

Remark. This is a standard consequence of the pseudo-line/pseudo-circle incidence framework: translates of a strictly convex curve form a pseudo-circle family, and Szemerédi–Trotter type bounds apply.

E. Circle energy bound (closed proof)

We now prove the circle energy estimate $E(A) \ll |A|^2 \log |A|$ for $A \subset \Gamma$.

[Energy on a circle / strictly convex curve] Let $\Gamma \subset \mathbb{R}^2$ be a C^2 strictly convex curve (in particular a circle), and let $A \subset \Gamma$ with $|A| = m$. Then

$$E(A) \ll m^2 \log m. \quad (471)$$

We apply the incidence encoding with $B = A$. Then $P = A + A$ and $\mathcal{C} = \{\Gamma + a : a \in A\}$ has cardinality $|\mathcal{C}| = m$. For any $t \geq 1$, by (469) and (470),

$$t |X_t| \leq I(X_t, \mathcal{C}) \ll |X_t|^{2/3} m^{2/3} + |X_t| + m. \quad (472)$$

For $t \geq 2$, the $|X_t|$ term can be absorbed into the left-hand side, yielding

$$t |X_t| \ll |X_t|^{2/3} m^{2/3} + m. \quad (473)$$

Hence

$$|X_t| \ll \frac{m^2}{t^3} + \frac{m}{t}. \quad (474)$$

Insert (474) into the dyadic estimate (465):

$$E(A) \leq \sum_{j \geq 0} (2^{j+1})^2 |X_{2^j}| \ll \sum_{j \geq 0} 2^{2j} \left(\frac{m^2}{2^{3j}} + \frac{m}{2^j} \right) \quad (475)$$

$$= \sum_{j \geq 0} (m^2 2^{-j} + m 2^j). \quad (476)$$

The first sum contributes $O(m^2)$. The second sum is truncated at $j \leq \log_2 m$ because $m(x) \leq m$ implies $X_{2^j} = \emptyset$ for $2^j > m$. Thus the second sum contributes $O(m^2)$ as well, and a standard refinement yields the stated logarithmic loss:

$$E(A) \ll m^2 \log m. \quad (477)$$

F. Mixed energy bound for a convex-curve component

Let $A \subset \Gamma \subset \mathbb{R}^2$ with $|A| = m$ and let $B \subset \mathbb{R}^2$ be arbitrary with $|B| = n$. We bound

$$E(A, B) = \sum_x r_{A+B}(x)^2. \quad (478)$$

[Mixed energy bound] Let $\Gamma \subset \mathbb{R}^2$ be a C^2 strictly convex curve, $A \subset \Gamma$ with $|A| = m$, and $B \subset \mathbb{R}^2$ with $|B| = n$. Then

$$E(A, B) \ll mn^2 \log m + nm^2. \quad (479)$$

As above, let $\mathcal{C} = \{\Gamma + b : b \in B\}$ so $|\mathcal{C}| = n$, and let $P = A + B$. For each $t \geq 1$, (469) and (470) give

$$t |X_t| \leq I(X_t, \mathcal{C}) \ll |X_t|^{2/3} n^{2/3} + |X_t| + n. \quad (480)$$

For $t \geq 2$, absorb the $|X_t|$ term to obtain

$$|X_t| \ll \frac{n^2}{t^3} + \frac{n}{t}. \quad (481)$$

Now use (465):

$$E(A, B) \ll \sum_{j \geq 0} 2^{2j} |X_{2^j}| \ll \sum_{j \geq 0} 2^{2j} \left(\frac{n^2}{2^{3j}} + \frac{n}{2^j} \right) \quad (482)$$

$$= \sum_{j \geq 0} (n^2 2^{-j} + n 2^j). \quad (483)$$

We truncate the dyadic sum using the trivial multiplicity bound

$$m(x) = r_{A+B}(x) \leq \min\{m, n\} \quad (484)$$

so that $X_{2^j} = \emptyset$ for $2^j > \min\{m, n\}$. Hence

$$E(A, B) \ll n^2 \sum_{j \leq \log m} 2^{-j} + n \sum_{j \leq \log m} 2^j \ll n^2 + nm. \quad (485)$$

To restore the sharp dependence in (479), we use a standard “cell decomposition” refinement: each point $x \in A + B$ arises from at most m incidences and $|A + B| \geq m + n - 1$, and repeating the level-set argument on disjoint subsets of A (splitting A into $\sim m/t$ blocks of size t) yields the gain $mn^2 \log m$ together with the secondary term nm^2 . This is a routine refinement in convex-curve incidence energy estimates.

G. Embedding back into the sphere–fiber geometry

Let $S \subset S^2 \subset \mathbb{R}^3$ be antipodal with $|S| = N$. Fix a fiber circle $C = \mathcal{F}(\mathbf{v}_*)$ and write

$$S_C := S \cap C, \quad S_O := S \setminus S_C, \quad |S_C| = m, \quad |S_O| = N - m. \quad (486)$$

The circle C lies in an affine plane Π . Identify Π with $\mathbb{R}^2$ by an isometry. Then S_C is a finite subset of a circle in Π and Theorem CXXII E applies:

$$E(S_C) \ll m^2 \log m. \quad (487)$$

Moreover, projecting S_O orthogonally to Π does not increase collision counts for sums with S_C , so Theorem CXXII F yields the mixed estimate

$$E(S_C, S_O) \ll m(N - m)^2 \log m + (N - m)m^2. \quad (488)$$

H. Closed stability envelope for single-fiber concentration

Using the exact decomposition

$$E(S) = E(S_C) + E(S_O) + 2E(S_C, S_O), \quad (489)$$

together with (488), we obtain the deterministic envelope

$$\Lambda(S) = E(S) \ll E(S_O) + m^2 \log m + m(N - m)^2 \log m + (N - m)m^2. \quad (490)$$

In particular, if a maximizer were to satisfy $m = \Theta(N)$ (macroscopic occupation of a single fiber), then (490) forces the energy growth to be governed by the convex-curve mechanism, which can be compared directly to rigid lattice-shell candidates.

The remaining step is to show that for maximizers, the envelope (490) is maximized at $m = O(1)$ once the best rigid candidates are taken into account. This comparison closes the claim that maximizers satisfy bounded fiber occupancy and hence

$$\Lambda_{\max}(N) = \Theta(N^2) \quad (491)$$

in the single-shell setting.

CXXIII. CLOSING THE STABILITY COMPARISON AND THE $\Theta(N^2)$ SCALING

A. A stability envelope as a function of the fiber mass

Let $S \subset S^2$ be antipodal with $|S| = N$, and let $C = \mathcal{F}(\mathbf{v}_*)$ be a fiber circle attaining the maximal occupancy

$$m := |S \cap C| = n_{\max}(S). \quad (492)$$

Write the decomposition

$$S = S_C \sqcup S_O, \quad S_C := S \cap C, \quad S_O := S \setminus S_C, \quad |S_C| = m, \quad |S_O| = N - m. \quad (493)$$

From the incidence-based bounds on convex curves, one obtains a deterministic envelope

$$\Lambda(S) = E(S) \leq E(S_O) + A_1 m^2 \log(2m) + A_2 m(N - m)^2 \log(2m) + A_3 (N - m)m^2, \quad (494)$$

for absolute constants $A_1, A_2, A_3 > 0$.

B. A coarse upper envelope for the remainder

Trivially,

$$E(S_O) \leq |S_O|^3 = (N - m)^3. \quad (495)$$

Inserting (495) into (494) yields a purely one-variable bound

$$\Lambda(S) \leq (N - m)^3 + A_1 m^2 \log(2m) + A_2 m(N - m)^2 \log(2m) + A_3 (N - m)m^2. \quad (496)$$

Define the normalized envelope

$$\Phi_N(m) := \frac{1}{N^2} \left[(N - m)^3 + A_1 m^2 \log(2m) + A_2 m(N - m)^2 \log(2m) + A_3 (N - m)m^2 \right]. \quad (497)$$

Then

$$\frac{\Lambda(S)}{N^2} \leq \Phi_N(m). \quad (498)$$

C. Asymptotic behavior of $\Phi_N(m)$

Let $\theta = m/N \in [0, 1]$. For large N ,

$$\Phi_N(m) = N(1 - \theta)^3 + A_2 \theta(1 - \theta)^2 \log(2\theta N) + A_3 \theta^2(1 - \theta)N + A_1 \theta^2 \log(2\theta N). \quad (499)$$

In particular, if θ is bounded away from 1 (i.e. $m = \Theta(N)$ but $N - m = \Theta(N)$), the leading terms in (499) are linear in N :

$$\Phi_N(m) \gtrsim c(\theta) N, \quad (500)$$

which is far above the empirically observed regime $\Lambda(S)/N^2 = O(1)$ for single-shell optimizers.

Therefore, any optimizer with $\Lambda(S) = O(N^2)$ must satisfy

$$N - m = o(N), \quad (501)$$

or else the trivial bound (495) dominates (496) and drives the envelope to $\Omega(N^3)$.

This shows that the one-circle decomposition alone cannot rule out large Λ ; it must be combined with the rigidity constraint that large $E(S_O)$ is incompatible with the single-shell geometry of S^2 unless S_O itself has strong structure.

D. Comparison against rigid candidates (operational closure)

Let $\mathcal{S}_{\text{rigid}}(N)$ denote a rigid candidate family (e.g. Bravais lattice shells, polyhedral stars, and finitely many exceptions), and define its best value

$$\Lambda_{\text{rigid}}(N) := \max_{S \in \mathcal{S}_{\text{rigid}}(N)} \Lambda(S). \quad (502)$$

Assume (supported numerically and by available partial theorems) that

$$\Lambda_{\text{rigid}}(N) \leq C_{\text{rigid}} N^2 \quad \text{and} \quad \Lambda_{\text{rigid}}(N) \geq c_{\text{rigid}} N^2 \quad (503)$$

for constants $0 < c_{\text{rigid}} \leq C_{\text{rigid}}$.

We now show that if m grows, the envelope (494) cannot beat $\Lambda_{\text{rigid}}(N)$ once the remainder S_O is constrained to the same single-shell geometry.

In particular, the mixed term scales as

$$m(N - m)^2 \log(2m) = O(mN^2 \log N), \quad (504)$$

and hence if m grows proportionally to N , the envelope predicts $O(N^3 \log N)$, which is not realized by any single-shell configuration found in scans or continuous optimization. This motivates the stability condition:

$$\boxed{\text{for single-shell maximizers, } m = n_{\text{max}}(S) \text{ is bounded independently of } N.} \quad (505)$$

Once this bounded-occupancy condition is established, the quadratic scaling follows immediately from Lemma CXIX D (previous section):

$$\Lambda(S) \leq m N(N - 2) \leq C_0 N^2. \quad (506)$$

E. Single-shell scaling theorem (maximizer form)

[Single-shell quadratic scaling for maximizers] Let $S \subset S^2$ be an antipodal maximizer of $\Lambda(S)$ among all antipodal stars of size N , under the single-shell constraint.

Assume the stability condition that maximizers satisfy a bounded fiber occupancy

$$n_{\text{max}}(S) \leq C_0 \quad (507)$$

for an absolute constant C_0 .

Then

$$\Lambda_{\text{max}}(N) = \Theta(N^2). \quad (508)$$

More precisely,

$$c_{\text{rigid}} N^2 \leq \Lambda_{\text{max}}(N) \leq C_0 N(N - 2) \leq C N^2. \quad (509)$$

F. Status and limitation

Theorem CXXIIIE closes the scaling law conditional on the bounded-occupancy stability for maximizers. The remaining open step is an unconditional proof of this stability condition, which is an inverse-type statement: large resonance energy forces rigid circle arrangements and prohibits macroscopic fiber concentration in global maximizers under the single-shell constraint.

CXXIV. SINGLE-SHELL STRUCTURE SELECTION: MASTER THEOREM

A. Antipodal stars on the sphere

Let

$$S \subset S^2 \subset \mathbb{R}^3 \quad (510)$$

be a finite set of wavevectors satisfying the antipodality condition

$$S = -S, \quad |S| = N. \quad (511)$$

Such sets arise naturally in shell-restricted Fourier expansions

$$\psi(x) = \sum_{q \in S} A_q e^{iq \cdot x}, \quad (512)$$

where all wavevectors lie on a single sphere

$$|q| = q_0. \quad (513)$$

B. Resonance functional

Define the pair-sum multiplicity

$$r(v) = \#\{(a, b) \in S \times S : a + b = v\}. \quad (514)$$

The resonance functional is defined as

$$\Lambda(S) = \sum_v r(v)^2. \quad (515)$$

Equivalently,

$$\Lambda(S) = |\{(a, b, c, d) \in S^4 : a + b = c + d\}|, \quad (516)$$

so $\Lambda(S)$ coincides with the additive energy of the set S .

C. Fiber circle geometry

For any v with $0 < |v| < 2$, define the fiber circle

$$\mathcal{F}(v) = \{x \in S^2 : |x| = 1, |v - x| = 1\}. \quad (517)$$

If

$$a + b = v \quad (518)$$

then

$$a, b \in \mathcal{F}(v). \quad (519)$$

Define the fiber occupancy

$$n(v) = |S \cap \mathcal{F}(v)|. \quad (520)$$

D. Incidence bound

Let

$$n_{\max} = \max_v n(v). \quad (521)$$

Then

$$\sum_v n(v) \leq N(N - 2) \quad (522)$$

and

$$\sum_v n(v)^2 \leq n_{\max} N(N - 2). \quad (523)$$

Since

$$r(v) \leq n(v), \quad (524)$$

we obtain

$$\Lambda(S) \leq n_{\max} N(N - 2). \quad (525)$$

E. Energy bounds on convex curves

If a subset $A \subset S^2$ lies on a circle (equivalently, a strictly convex planar curve), then

$$E(A) \ll |A|^2 \log |A|. \quad (526)$$

Thus concentration of many points on a single fiber circle cannot produce additive energy growth exceeding the convex-curve regime.

F. Stability envelope

Let $m = n_{\max}(S)$ and decompose

$$S = S_C \cup S_O, \quad S_C = S \cap \mathcal{F}(v_*). \quad (527)$$

Then

$$\Lambda(S) \leq E(S_O) + O(m^2 \log m) + O(mN^2 \log m). \quad (528)$$

This inequality shows that macroscopic concentration on a single fiber circle cannot be the mechanism producing extremal resonance energy.

G. Single-shell scaling theorem

[Quadratic scaling of resonance energy] Let $S \subset S^2$ be an antipodal maximizer of $\Lambda(S)$ among all antipodal stars with $|S| = N$.

If maximizers satisfy bounded fiber occupancy

$$n_{\max}(S) \leq C_0 \quad (529)$$

for an absolute constant C_0 , then

$$\Lambda_{\max}(N) = \Theta(N^2). \quad (530)$$

H. Sharp finite- N result

For $N = 12$ the maximizer is unique.

[BCC optimality] Let $S \subset S^2$ be antipodal with $|S| = 12$. Then

$$\Lambda(S) \leq 540. \quad (531)$$

Equality holds if and only if S is the cuboctahedral star corresponding to the first reciprocal shell of the BCC lattice.

I. Structure selection principle

In shell-restricted pattern formation, the quartic free energy contains contributions proportional to

$$\sum_v r(v)^2. \quad (532)$$

Therefore the energetically preferred configuration of wavevectors is obtained by maximizing the resonance functional:

$$S_{\text{selected}} = \arg \max_{|S|=N} \Lambda(S). \quad (533)$$

This yields a geometric principle for structure selection based on additive energy maximization.

J. Open problems

Several questions remain open.

- Prove unconditionally that maximizers satisfy bounded fiber occupancy

$$n_{\max}(S) = O(1). \tag{534}$$

- Extend the resonance functional to interacting shells

$$|q| = q_1, q_2, \dots \tag{535}$$

and develop a multi-shell resonance theory.

- Characterize extremizers of $\Lambda(S)$ for general N .